

HERMITE INTERPOLATION AT THE PRIME POWERS: A POSITIVITY CALCULUS, A POLYNOMIAL MODEL, AND A FAMILY OF TEST FUNCTIONS VANISHING AT THE ZEROS OF $\zeta(s)$

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ABSTRACT. We study nested Hermite interpolation with double nodes at the points $y_j = 2\pi n_j$, where n_j runs through the prime powers, for three kernels: the Fourier transform of Ξ^2 , which defines an explicit family of test functions $\Xi(t)^2 P_J(t^2)$ for the explicit formula of the Riemann zeta function, all vanishing at every non-trivial zero; its archimedean part; and the kernel e^{-y} , which gives a polynomial model, equivalently a radial Fourier interpolation problem on the circles $|v| = \sqrt{2n}$ in $\mathbb{R}^2$. A kernel-free calculus (Christoffel, ladder, insertion and half-step identities) reduces positivity of the interpolant beyond the nodes, in the model, to one likelihood-ratio law per level. A computer-assisted certificate verifies this law at every level up to 160, which proves the positivity of the front of the model for every $J \leq 80$. We prove far-node asymptotics for the model and for the true kernel, show that sign regularity of the associated five-function systems cannot give the tail of the model, and show that a natural sign property fails at the critical node density. For the true family we reduce the open hypotheses of a criterion for uniqueness and sharpness, proved in a companion paper, to explicit step laws, which remain open. Numerical observations are collected in one section and are never used in proofs.

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1. INTRODUCTION

1.1. The interpolation problem. The companion paper [15] studies all positive solutions of Weil’s explicit formula with the archimedean data of the Riemann zeta function: pairs of positive measures, one on the zero side and one on the prime side beyond $\log 2/2\pi$, that satisfy the formula on a natural test class. By linear-programming duality such pairs exist if and only if a slack κ^* is non-negative, and κ^* is the optimal degree-one conductor bound of the Odlyzko–Poitou–Serre method in logarithmic form. It is conjectured there that every such solution, if one exists, is ζ ’s own pair (Conjecture U), and that $\kappa^* \leq 0$ (Conjecture S). A self-contained summary of the setting, with the results of [15] that we use, is in Section 2.

The main tool of [15] is a family of zero-killing functions. Every function $\Xi(t)^2 H(t)$, with H analytic on a strip containing $|\operatorname{Im} t| \leq \frac{1}{2}$, vanishes at every non-trivial zero of ζ , on or off the critical line; following [15] we call such functions *zero-killing*. For them the explicit formula has no zero side. The exact member $F_J = \Xi^2 P_J(t^2)$ is defined by asking that its Fourier transform $\widehat{F}_J$ vanish to second order at the first J prime powers, with $P_J(0) = 1$ (Fact 2.7); it is the error of Hermite interpolation of the kernel $\Psi = \widehat{\Xi}^2$ at the nodes $y_j = 2\pi n_j$. The same family certifies $\kappa^* \leq 1.1508391 \cdot 10^{-906}$ at $J = 100$ in [15], and a subsequential limit of it would be an exact magic function: by the criterion of [15, Theorem D] (Fact 2.10 below), four hypotheses on the family, (N), (G_a) , (Z_∞) and (Mg), imply both conjectures. These hypotheses are open; only their finite- J instances are certified, at finitely many J .

This paper studies the interpolation problem behind the family, for an arbitrary kernel and for three kernels in particular: the true kernel Ψ , its archimedean part $\Psi_1 = \widehat{\Xi_\infty}^2$, and the kernel e^{-y} . The last gives a polynomial model (Section 5): its members are polynomials P for which $Q_P := e^y P(-\theta^2) e^{-y}$, $\theta = y \frac{d}{dy}$, has a double zero at every node. At leading order the true family *is* such a problem (Proposition 5.5), and the model has a second reading: by Proposition 5.3 a double zero of Q_P at $2\pi n$ means that a radial function f on $\mathbb{R}^2$ and its Fourier transform both vanish on the circle $|v| = \sqrt{2n}$. Interpolation from all circles of radius $\sqrt{n}$ is the subject of Stoller’s theorem [21], which extends the one-dimensional interpolation formula of Radchenko and Viazovska [19] to every dimension $d \geq 2$; the general question for derivatives up to order k at the radii $\sqrt{kn}$ is [4, Open Problem 7.1]. Here the nodes are the prime powers only, a sparse subset, and the question is not uniqueness but *positivity* of the interpolant.

Thesis. For nested double-node Hermite interpolation at the prime powers, positivity of the interpolant beyond the nodes reduces to one likelihood-ratio law per level. In the model this law is certified at every level up to 160, and for the true kernel the hypotheses of the criterion of [15] reduce to explicit step laws of the same kind.

1.2. Results. *The calculus* (Sections 3 and 4). For a base kernel φ with linearly independent iterates $(-\theta^2)^k \varphi$ we set up nested families of Hermite interpolants on chains of points with multiplicity, and prove the identities that govern them: a Christoffel identity, which adds one point (Lemma 4.2); a ladder identity for moving a node (Lemma 4.3); an insertion calculus (Proposition 4.6), which computes the response of the coefficient p_1 to a new node as a ratio of Wronskians; and a half-step calculus (Proposition 4.7), in which the two-point interaction of the p_1 -problem is a divided difference of one function with respect to another. These are kernel-free and apply verbatim to all three kernels. Two tail bounds follow (Proposition 4.8 and Theorem 4.9).

The model (Sections 5–8). The top of the cofactor $R = Q / \prod_j (y - y_j)^2$ sees the node set only through the sum of the nodes (Lemma 5.7), which is where the arithmetic of the prime powers enters. Along a chain $z_1, z_2, \dots$ let $\mathbf{f}^{(K)}$ be the monic cofactor at level K and $\mathbf{g}^{(K)}$ the cofactor

of the chain augmented by the point 0. The Christoffel identity, read on coefficients, gives the following criterion.

Theorem 6.3. *Suppose that, for every level $K < K_0$, the ratios g_k/f_k of the coefficients of $\mathbf{g}^{(K)}$ and $\mathbf{f}^{(K)}$ are non-decreasing in k (the likelihood-ratio law $(\Pi')_K$) and $\mathbf{g}^{(K)}(z_{K+1}) > 0$. Then every cofactor up to level K_0 has positive coefficients; for a doubled chain this gives, for every J with $2J \leq K_0$, non-singularity, $p_{2J} > 0$, positivity of the front $Q_J > 0$ off the nodes, and a strict step law.*

Along the prime powers the law is verified by computer.

Theorem 6.5. *For the nodes $y_j = 2\pi n_j$ and every $J \leq 80$, the model solution exists, its cofactor R_J has positive coefficients, and in particular $Q_J > 0$ on $(0, \infty)$ off the nodes; the step law and positivity of the half-step cofactor hold as well, and the likelihood-ratio law holds at every level $K \leq 160$.*

The proof is computer-assisted, by two independent certificates in ball arithmetic. Without it, the front of the model is proved here only for $J \leq 4$ (Corollary 5.13). An induction on the level does not close (Remark 6.4), and on the computed data the law is a sparse-regime law, which fails at the critical density of the integer nodes (Numerical observation 10.14(b)). For the tail of the model we prove far-node asymptotics (Theorem 7.1), certify the tail laws on finite ranges (Propositions 7.6–7.8), and show that no family of open sign conditions on the five-function system that governs the tail can prove it (Proposition 7.11). Finally a natural sign property (W) of the derivative weights, which would give a comparison principle in the node set, fails at and strictly inside the critical density (Proposition 8.2), although it holds on the whole class for two and three nodes (Propositions 8.3 and 8.4).

The true family (Section 9). The hypotheses (G_a) and (Z_∞) of the criterion of [15] follow from root counts and from a coefficientwise step law (Theorem 9.10), and (Mg) near the front follows from a step law (TEL) together with a base certificate and a summability condition (Theorem 9.15). Combining them:

Corollary 9.21. *Assume $(N)_J$ for every $J \geq J_0$, the coefficient bound (POS_a) for all large J with some $a < \pi/2$, the base hypothesis $(B^+)_{J_0}$, the step law (TEL_J) for every $J > J_0$, and erosion summability (ES). Then Conjectures U and S of [15] hold, and the Riemann hypothesis is equivalent to the existence of an admissible pair.*

None of these five hypotheses is known for all J . Away from the front, (Mg) reduces to a finite certificate plus the summability of one scalar sequence (Corollary 9.25), and we prove the far-node asymptotics of the true kernel (Theorem 9.27): the frozen double root $(u + \frac{1}{4})^2$ of the Gamma factor acts like one extra node, so the constant $8J$ of the model becomes $8(J+1)$. We observe numerically that the model and the true family agree quantitatively at the front, to within a few hundredths in the step-law profile for $J \geq 40$ (Numerical observation 10.13); we have no transfer theorem between them, and a positive smoothing cannot provide one (Proposition 5.6).

1.3. Status of the results. This paper uses three kinds of statement. *Proved* statements (theorem, proposition, lemma, corollary) include the kernel-free calculus, the criterion of Theorem 6.3, the far-node asymptotics for a fixed node set, the obstruction of Proposition 7.11, and the conditional reductions of Section 9; the hypotheses of every conditional statement are named in it. *Computer-assisted* statements are finite certificates, each with its script and parameter file (Appendices B and C): the front of the model for $J \leq 80$, small cases, base data, counterexamples to (W), tail laws on finite ranges, and base data for the true family at $J = 10, 60, 110$. *Numerical observations* are computed but not certified; they are all in Section 10 and are never

used in a proof. The open statements are in Section 11: five conjectures, each with what it would imply, the evidence and what a finite prefix of the computed family can and cannot decide, and four open problems.

Nothing infinite in J is proved for the families themselves. No result here bears on the Riemann hypothesis beyond the criterion of [15], and that criterion, even with all its hypotheses, gives Conjecture U, which does not imply the Riemann hypothesis. The model results say nothing about the true family.

1.4. Related work. Christoffel’s formula for orthogonal polynomials [22, Theorem 2.5] is the prototype of the identity of Lemma 4.2, here for a nested biorthogonal system with a test functional; the pointwise forms of Proposition 4.5 are statements about Chebyshev pairs in the sense of Karlin and Studden [16]. The model is built from Touchard polynomials [25, 6], and the radial reading of Proposition 5.3 is expressed through Laguerre polynomials; Laguerre–Gaussian functions with prescribed double roots are the standard ansatz for sign-uncertainty problems [3], whose extremizers have infinitely many double roots [13]; as the dimension grows, polynomials of degree sublinear in the dimension times a Gaussian do asymptotically no better than those of degree at most three [2]. Fourier interpolation from the radii $\sqrt{n}$ is due to Radchenko and Viazovska in dimension one [19], to Stoller in every dimension $d \geq 2$ [21], and, with derivatives at the radii $\sqrt{2n}$ in dimensions 8 and 24, to Cohn, Kumar, Miller, Radchenko and Viazovska [4]; Feigenbaum, Grabner and Hardin [11] describe the Fourier eigenfunctions with prescribed zeros behind these magic functions, in every dimension divisible by 4. Bondarenko, Radchenko and Seip [1] interpolate at the zeros of ζ and at the points $\log m/4\pi$. Cohn and Miller [5] study sequences of functions defined by interpolation conditions and conjecture that they converge to functions with the positivity and growth properties of a magic function; the criterion of [15, Theorem D] and Section 9 pose the analogous question for the explicit formula of ζ . The Bessel form of Ψ used here is a differentiated form of the extended Koshliakov formula [17, 9]; its attribution is discussed in [15].

1.5. Organisation. Section 2 recalls the setting of [15]. Sections 3 and 4 develop the kernel-free calculus. Sections 5–8 treat the model: definition, small cases and the dictionary to the true kernel; the Christoffel chain and the front for $J \leq 80$; the tail and the obstruction; and the sign property (W). Section 9 treats the true family. Section 10 collects the numerical observations and Section 11 the conjectures and open problems. Long proofs are in Appendix A, the design of the certificates and the tables in Appendix B, and the index of the ancillary files in Appendix C.

Conventions. PP is the set of prime powers, n_j the j -th prime power, $\mathbf{y}_J^{\text{PP}} = (2\pi n_1, \dots, 2\pi n_J)$, and $\theta = y \frac{d}{dy}$ is the Euler operator; $\mathbb{R}_{>0}[y]$ is the set of polynomials with positive coefficients, and inequalities between polynomials are coefficientwise when so stated. Named conditions are written in parentheses, for example (N); their definitions are indexed in Table 2.

2. SETTING AND RESULTS FROM THE COMPANION PAPER

This section fixes the notation and recalls the definitions and results of [15] that are used below. Statements quoted from [15] are called *Facts*; their proofs are there and are not repeated. The only statement of this section that is proved here is Corollary 2.11.

2.1. Admissible pairs and the slack. We write $\hat{F}(\xi) = \int_{\mathbb{R}} F(t) e^{-2\pi i \xi t} dt$, $\xi_n = \log n/(2\pi)$ for real $n > 0$, and $\xi_2 = \log 2/(2\pi)$ for the *gap*. PP is the set of prime powers $n > 1$, n_j is the j -th prime power, $\text{PP}_J = \{n_1, \dots, n_J\}$, $\xi_{\text{PP}} = \{\xi_n : n \in \text{PP}\}$, Λ is von Mangoldt’s function, and $\mathcal{M} := \{\log m/(4\pi) : m \geq 1\}$; since $\xi_n = \log(n^2)/(4\pi)$, $\xi_{\text{PP}} \subset \mathcal{M}$. The archimedean data of ζ are

$$\Omega_{\infty}(t) := \frac{1}{2\pi} \left(\operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right), \quad \mathcal{A}(F) := F\left(\frac{i}{2}\right) + F\left(-\frac{i}{2}\right) + \int_{\mathbb{R}} F(t) \Omega_{\infty}(t) dt,$$

with $\psi = \Gamma'/\Gamma$. For $0 < \delta < \frac{1}{2}$, $\mathcal{T}_\delta$ is the set of even functions F , real on $\mathbb{R}$ and analytic in a neighbourhood of $\{|\operatorname{Im} z| \leq \frac{1}{2} + \delta\}$, with $\|F\|_\delta := \sup_{|\operatorname{Im} z| \leq \frac{1}{2} + \delta} (1 + |z|)^2 |F(z)| < \infty$, and $\mathcal{T} := \bigcup_{\delta > 0} \mathcal{T}_\delta$ is the *test class*.

Fact 2.1 (Explicit formula on $\mathcal{T}$; [15, Lemma 2.3], [26], [1, (1.1)]). *For every $F \in \mathcal{T}$, $\mathcal{A}(F) = \sum_\rho F(t_\rho) + \frac{1}{\pi} \sum_{n \geq 2} \Lambda(n) n^{-1/2} \widehat{F}(\xi_n)$, with $t_\rho := -i(\rho - \frac{1}{2})$, where ρ runs over the non-trivial zeros of ζ with multiplicity. Both series converge absolutely; no hypothesis on the zeros is made.*

Let $Z_\zeta = \{\gamma \in \mathbb{R} : \zeta(\frac{1}{2} + i\gamma) = 0\}$, $\mu_\zeta := \sum_{\gamma \in Z_\zeta} m_\gamma \delta_\gamma$ (m_γ the multiplicity), $\nu_\zeta := \sum_{n \geq 2} \Lambda(n) n^{-1/2} \delta_{\xi_n}$ and $p_\zeta := (\mu_\zeta, \nu_\zeta)$.

Definition 2.2 (Admissible pairs, cones, slack; [15, Definitions 2.4 and 2.11]). An *admissible pair* is a pair (μ, ν) of positive Radon measures, μ even on $\mathbb{R}$ and ν on $[\xi_2, \infty)$, such that for every $F \in \mathcal{T}$ the integrals converge absolutely and $\mathcal{A}(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$. $\mathcal{K}$ is the set of admissible pairs. Put

$\mathcal{C} := \{F \in \mathcal{T} : F \geq 0 \text{ on } \mathbb{R}, \widehat{F} \geq 0 \text{ on } [\xi_2, \infty)\}$, $\mathcal{C}_{\text{Ops}} := \{F \in \mathcal{T} : F \geq 0 \text{ and } \widehat{F} \geq 0 \text{ on } \mathbb{R}\}$, $\kappa^* := \inf\{\mathcal{A}(F) : F \in \mathcal{C}, \int F = 1\}$ (the *slack*), and consider the statements (E): $\mathcal{K} \neq \emptyset$; (U): $\mathcal{K} \subseteq \{p_\zeta\}$; (S): $\kappa^* \leq 0$. For F on $\mathbb{R}$ put $Z(F) := F^{-1}(0) \cap \mathbb{R}$ and $\widehat{Z}(F) := \widehat{F}^{-1}(0) \cap [\xi_2, \infty)$. An *exact magic function* is an $F \in \mathcal{C}$ with $\int F > 0$ and $\mathcal{A}(F) = 0$.

The companion paper conjectures (U) (Conjecture U) and (S) (Conjecture S); the slack κ^* governs the optimal degree-one conductor bound of the explicit-formula method, $q_{\min} = e^{-2\pi\kappa^*}$.

Fact 2.3 (Duality, logic and complementary slackness; [15, Lemma 2.5, Theorems 2.7 and 2.9, Lemma 2.13]). (a) (Weak duality.) *If $(\mu, \nu) \in \mathcal{K}$ and $F \in \mathcal{C}$, then $\mathcal{A}(F) \geq 0$.* (b) (No duality gap.) (E) $\Leftrightarrow \kappa^* \geq 0$. (c) *RH implies $p_\zeta \in \mathcal{K}$; (U) holds if and only if both [(E) $\Rightarrow$ RH] and [RH $\Rightarrow \mathcal{K} = \{p_\zeta\}$] hold. So (U) is the conjunction of a criterion for RH and a uniqueness statement under RH, and under (U), RH $\Leftrightarrow$ (E) $\Leftrightarrow \kappa^* \geq 0$; moreover (U) $\Rightarrow$ (S).* (d) (Complementary slackness.) *If $F \in \mathcal{C}$ and $\mathcal{A}(F) = 0$, every admissible pair (μ, ν) has μ carried by $Z(F)$ and ν by $\widehat{Z}(F)$.*

Fact 2.4 (Magic-function principle; [15, Corollary 3.8]). *Suppose some $F \in \mathcal{C}$ satisfies $\mathcal{A}(F) = 0$, has only finitely many real zeros outside Z_ζ , and has $\widehat{Z}(F) \subseteq \mathcal{M}$. Then $\mathcal{K} \subseteq \{p_\zeta\}$, i.e. (U) holds. If (E) holds as well, then RH holds and $\mathcal{K} = \{p_\zeta\}$.*

2.2. The zero-killing family. Let $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$, $\Xi(t) := \xi(\frac{1}{2} + it)$ with $\xi(s) = \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta(s)$ (Riemann's Ξ -function), and $\Xi_\infty(t) := \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)|_{s=1/2+it}$, its archimedean part. Every function $\Xi^2 H$, with H analytic on a strip containing $|\operatorname{Im} t| \leq \frac{1}{2}$, vanishes at every non-trivial zero of ζ , on or off the critical line; such functions are called *zero-killing*. Put

$$\Psi := \widehat{\Xi^2}, \quad \Psi_1 := \widehat{\Xi_\infty^2}.$$

Throughout, $x = e^{2\pi\xi}$ and $y = 2\pi x$, so that the prime power n sits at $\xi = \xi_n$, at $x = n$ and at $y = 2\pi n$, and $\theta = x \frac{d}{dx} = y \frac{d}{dy} = \frac{1}{2\pi} \frac{d}{d\xi}$. For $a < \pi/2$, $\mathcal{H}_a$ is the class of even functions H , real on $\mathbb{R}$ and analytic on some closed strip $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$ ($\delta > 0$), with $|H(t)| \leq C e^{a|\operatorname{Re} t|}$ there; $\mathcal{H} := \bigcup_{a < \pi/2} \mathcal{H}_a$. Stirling's formula and the convexity bound for ζ give, for every $b \in (0, 1]$, a constant C_b with

$$(2.1) \quad |\Xi(t)| \leq C_b (1 + |\operatorname{Re} t|)^3 e^{-\pi|\operatorname{Re} t|/4} \quad (|\operatorname{Im} t| \leq b)$$

[24]. (For $|\operatorname{Im} t| = b$ the exponent of $1 + |\operatorname{Re} t|$ given by Stirling's formula is $\frac{7}{4} + \frac{b}{2}$ once $b \geq \frac{1}{2}$, up to the size of ζ on the line $\operatorname{Re} s = \frac{1}{2} + b$; so the exponent 3 is valid for $b \leq 1$, which is all we

use, but not for every $b > 0$.) Hence $\Xi^2 H \in \mathcal{T}_\delta$ whenever $H \in \mathcal{H}_a$ is analytic near $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, and $\Xi^2 P(t^2) \in \mathcal{T}_\delta$ for every polynomial P and every $\delta \in (0, \frac{1}{2})$.

Fact 2.5 (Bessel form; [15, Lemma 4.1 and Proposition 4.2]). *Let $W_0(z) := z^2[(z^2 + 9)K_0(z) - 6zK_1(z)] = \theta^2(\theta + 1)^2 K_0(z)$, $\theta = z \frac{d}{dz}$, with K_0, K_1 the modified Bessel functions. For every real ξ ,*

$$\Psi(\xi) = 2\pi\sqrt{x} \sum_{N \geq 1} d(N) W_0(2\pi Nx), \quad \Psi_1(\xi) = 2\pi\sqrt{x} W_0(2\pi x),$$

the series converging absolutely and uniformly on compact sets with all its derivatives. Moreover $\Psi > 0$ on $\mathbb{R}$, and $\Psi(\xi_n) = (C_\infty + o(1))n^4 e^{-2\pi n}$ with $C_\infty = \pi(2\pi)^4$.

Fact 2.6 (The explicit formula for zero-killing functions; [15, Corollary 4.4]). *If $F = \Xi^2 H \in \mathcal{T}$ with H analytic on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$ (for instance $H \in \mathcal{H}$, or H a polynomial in t^2), then $\mathcal{A}(F) = \frac{1}{\pi} \sum_{n \in \text{PP}} \Lambda(n) n^{-1/2} \widehat{F}(\xi_n)$. No hypothesis on the zeros is used.*

Fact 2.7 (The exact family; [15, Proposition 4.5]). *Let $J \geq 1$ and $m_k := \widehat{t^{2k} \Xi^2} = (-\theta^2)^k \Psi$. If the matrix $M_J = [m_k(\xi_{n_j}); m'_k(\xi_{n_j})]_{j \leq J, 1 \leq k \leq 2J}$ is non-singular, we write $J \in \mathcal{J}$; then there is exactly one polynomial $P_J = 1 + \sum_{k=1}^{2J} p_k^{(J)} u^k$ such that $\widehat{F}_J := \widehat{\Xi^2 P_J(t^2)}$ has double zeros at $\xi_{n_1}, \dots, \xi_{n_J}$. Equivalently, $\widehat{F}_J = \Psi - \Pi_J \Psi$ is the error of Hermite interpolation of Ψ by $\text{span}\{m_1, \dots, m_{2J}\}$ at the nodes. We write $H_J(t) := P_J(t^2)$ and $F_J := \Xi^2 H_J \in \mathcal{T}$; then $\mathcal{A}(F_J) = \frac{1}{\pi} \sum_{n \in \text{PP}, n > n_J} \Lambda(n) n^{-1/2} \widehat{F}_J(\xi_n)$.*

In the language of Section 3, this is the true family at the node set $\mathbf{y}_J^{\text{PP}} = (2\pi n_1, \dots, 2\pi n_J)$.

Fact 2.8 (An integral representation; [15, Lemma 4.6]). *For a polynomial P put $\Omega_P(\theta) := P(-(\theta + \frac{1}{2})^2) \theta^2 (\theta + 1)^2$ and $Q_P^{\text{Ab}}(y) := e^y \Omega_P(\theta) e^{-y}$, with $\theta = y \frac{d}{dy}$. Then Q_P^{Ab} is a polynomial in y of degree $2 \deg P + 4$, with leading coefficient $(-1)^{\deg P}$ times that of P , and $Q_P^{\text{Ab}}(0) = 0$. For $z > 0$,*

$$g_P(z) := [P(-(\theta + \frac{1}{2})^2) W_0](z) = \int_z^\infty Q_P^{\text{Ab}}(y) e^{-y} (y^2 - z^2)^{-1/2} dy,$$

and $\widehat{\Xi^2 P(t^2)}(\xi) = 2\pi\sqrt{x} \sum_N d(N) g_P(2\pi Nx)$.

In [15] the polynomial Q_P^{Ab} is written Q_P ; here Q_P is reserved for the model (Section 5), and we call Q_P^{Ab} the *Abel preimage* of P . With $P^\Xi(u) := P(u)(u + \frac{1}{4})^2$ one has $\Omega_P(\theta) = P^\Xi(-(\theta + \frac{1}{2})^2)$, since $-(\theta + \frac{1}{2})^2 + \frac{1}{4} = -\theta(\theta + 1)$. Since $\widehat{t^{2k} f} = (-\theta^2)^k \widehat{f}$ and $\widehat{\theta(\sqrt{x} h)} = \sqrt{x}(\theta + \frac{1}{2}) \widehat{h}$, the same computation with $\Psi_1 = 2\pi\sqrt{x} W_0(2\pi x)$ in place of Ψ gives $\Xi_\infty^2 P(t^2)(\xi) = 2\pi\sqrt{x} g_P(2\pi x)$.

Write $S_b := \{t : |\operatorname{Im} t| < b\}$.

Fact 2.9 (Robust compactness; [15, Lemma 4.7]). *Let H_J , J in an infinite index set, be even, real on $\mathbb{R}$ and analytic on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, and put $F_J := \Xi^2 H_J$. Assume: (i) $|H_J(t)| \leq C e^{a|\operatorname{Re} t|}$ on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, with C and $a < \pi/2$ independent of J ; (ii) $H_J(0) = 1$; (iii) $\liminf_J H_J(t) \geq 0$ for every real t ; (iv) $\liminf_J \widehat{F}_J(\xi) \geq 0$ for every $\xi \geq \xi_2$; (v) $\widehat{F}_J(\xi_n) \rightarrow 0$ for every prime power n . Then a subsequence H_J converges locally uniformly on $S_{1/2+\delta}$ to some H_∞ , and $F_\infty := \Xi^2 H_\infty \in \mathcal{C} \cap \mathcal{T}$, with $\mathcal{A}(F_\infty) = 0$ and $\widehat{F}_\infty(0) = \int F_\infty > 0$. Hence $\kappa^* \leq 0$; moreover $\mathcal{A}(F) \geq 0$ for every $F \in \mathcal{C}$ of the form $\Xi^2 H$, so the infimum of $\mathcal{A}(F)/\int F$ over such F is 0 and is attained at F_∞ .*

Uniqueness then comes from Fact 2.4: if, in addition, H_∞ has finitely many real zeros and $\widehat{Z}(F_\infty) \subseteq \mathcal{M}$, then $\mathcal{K} \subseteq \{p_\zeta\}$.

For the exact family $(P_J)_{J \in \mathcal{J}}$ the companion paper considers the following hypotheses.

(N) $\mathcal{J}$ is infinite.

(G_a) There are $\delta \in (0, \frac{1}{2})$, $a < \pi/2$ and C with $|H_J(t)| \leq Ce^{a|\operatorname{Re} t|}$ on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$ for all $J \in \mathcal{J}$.

(Z_∞) No real point is a limit of zeros of H_{J_k} with $J_k \in \mathcal{J}$, $J_k \rightarrow \infty$.

(Mg) For every $\xi \in [\xi_2, \infty) \setminus \xi_{\text{PP}}$, $\liminf_{J \in \mathcal{J}} \hat{F}_J(\xi) > 0$.

Given (G_a), (Z_∞) is equivalent to: for every $T > 0$, $\liminf_{J \in \mathcal{J}} \min_{|t| \leq T} H_J(t) > 0$; it is implied by (R_η): there are $\eta > 0$ and J_0 such that every zero of H_J , $J \in \mathcal{J}$, $J \geq J_0$, has $|\operatorname{Im} t| \geq \eta$ [15].

Fact 2.10 (A criterion through the zero-killing family; [15, Theorem D]). *Assume (N), (G_a), (Z_∞) and (Mg) for the exact family. Then some subsequential limit $F_\infty = \Xi^2 H_\infty$, with $H_\infty \in \mathcal{H}$ and $H_\infty > 0$ on $\mathbb{R}$, satisfies $F_\infty \in \mathcal{C} \cap \mathcal{T}$, $\mathcal{A}(F_\infty) = 0$, $F_\infty^{-1}(0) \cap \mathbb{R} = Z_\zeta$ and $\hat{F}_\infty^{-1}(0) \cap [\xi_2, \infty) = \xi_{\text{PP}}$. Hence (S) and (U) hold, and $\text{RH} \Leftrightarrow (\text{E})$; under either, $\kappa^* = 0$ and $\mathcal{K} = \{p_\zeta\}$.*

None of (N), (G_a), (Z_∞), (Mg) is known for all J , and the hypotheses of Fact 2.10 amount to the existence of an exact magic function of the form $\Xi^2 H$, which is strictly more than (S). This paper reduces them further, to explicit step laws (Section 9); those laws are open as well. The following variants are used below; (M⁺) is also in [15].

Corollary 2.11 (Variants of the criterion). *The conclusions of Fact 2.10 ((S), (U), and $\text{RH} \Leftrightarrow (\text{E})$) also hold under each of the following replacements, all other hypotheses being unchanged.*

(M[#]) (Mg) replaced by (Mg_ℳ): $\liminf_{J \in \mathcal{J}} \hat{F}_J(\xi) > 0$ for $\xi \in [\xi_2, \infty) \setminus \mathcal{M}$, and $\liminf_{J \in \mathcal{J}} \hat{F}_J \geq 0$ on $\mathcal{M} \setminus \xi_{\text{PP}}$.

(M⁺) (Z_∞) replaced by (Z⁺): there is $h : \mathbb{R} \rightarrow (0, \infty)$ with $H_J \geq h$ on $\mathbb{R}$ for all large $J \in \mathcal{J}$ (the pointwise form $\liminf_{J \in \mathcal{J}} H_J(t) > 0$ for every t suffices); then $H_\infty \geq h > 0$.

(M_{fin}) (Z_∞) replaced by: $\liminf_{J \in \mathcal{J}} H_J \geq 0$ on $\mathbb{R}$, and either (R_k) there are $k \in \mathbb{N}$, $\eta > 0$ and J_0 such that H_J ($J \geq J_0$) has at most k zeros, counted with multiplicity, in S_η ; or (R_E) there is a finite $E_0 \subset \mathbb{R}$ with $\liminf_{J \in \mathcal{J}} \min_K H_J > 0$ for every compact $K \subset \mathbb{R} \setminus E_0$; and (Mg) replaced by (Mg_ℳ). (H_∞ may then have finitely many real zeros.)

Proof. In each case Fact 2.9 applies as in the proof of Fact 2.10 in [15]: (i) and (ii) are (G_a) and $P_J(0) = 1$; (v) holds because $\hat{F}_J(\xi_{n_j}) = 0$ for every $J \in \mathcal{J}$ with $J \geq j$; (iv) holds at $\xi \in \xi_{\text{PP}}$ by (v) and elsewhere by (Mg) or by (Mg_ℳ); and (iii) holds by (Z_∞) (through its equivalent form above) or by the hypothesis that replaces it. It remains to check the uniqueness clause. Under (Mg_ℳ), $\hat{F}_\infty > 0$ on $[\xi_2, \infty) \setminus \mathcal{M}$. Under (Z⁺), $H_\infty(t) = \lim H_J(t) \geq h(t) > 0$. Under (R_E), $H_\infty > 0$ on $\mathbb{R} \setminus E_0$. Under (R_k), H_∞ has at most k zeros in S_η : otherwise Hurwitz's theorem, applied on disjoint small discs around $k+1$ of them, would give more than k zeros of H_J in S_η for large J [8, VII.2.5]. In every case H_∞ has finitely many real zeros, and Fact 2.4 gives (U), and RH under (E). $\square$

Fact 2.12 (Coefficient bounds control the zero side; [15, Proposition 4.9]). *Suppose that for all $J \in \mathcal{J}$ with $J \geq J_0$,*

$$(\text{POS}_a) \quad 0 \leq p_k^{(J)} \leq C a^{2k} / (2k)! \quad \text{for all } k, \quad \text{with } 0 \leq a < \pi/2 \text{ and } C \text{ independent of } J.$$

Then for these J : (1) $H_J(t) \geq 1$ for all real t , so (Z_∞) holds; (2) $|H_J(t)| \leq C \cosh(a|t|)$ for all $t \in \mathbb{C}$, so (G_a) holds on every strip; (3) $\{H_J\}$ is normal on $\mathbb{C}$, and every limit H_∞ is entire of exponential type $\leq a$, with $0 \leq [t^{2k}]H_\infty \leq C a^{2k} / (2k)!$ and $H_\infty \geq 1$ on $\mathbb{R}$; (4) $|P_J(u)| \leq P_J(|u|)$, and every root u of P_J satisfies $|u| \geq \min_k p_{k-1}^{(J)} / p_k^{(J)}$ (minimum over k with $p_k^{(J)} > 0$).

By (1), (POS_a) gives (Z⁺) with $h \equiv 1$, so (N), (POS_a) and (Mg) imply (S) and (U), by Corollary 2.11(M⁺), as in [15, Remark 4.10(a)].

Fact 2.13 (Certified finite- J instances; [15, Proposition 5.5]). *For $J \in \{10, 60, 61, 110, 111\}$ the Hermite matrix M_J is non-singular, every coefficient of the exact P_J is positive, and $\max_{1 \leq k \leq 2J} ((2k)! p_k^{(J)})^{1/2k} \leq 0.71636$, that is, $0 < p_k^{(J)} \leq 0.71636^{2k}/(2k)!$. Hence $H_J \geq 1$ on $\mathbb{R}$ and $|\widehat{H}_J(t)| \leq \cosh(0.71636|t|)$ on $\mathbb{C}$.*

These are computer-assisted statements. The certified enclosures of P_J on which they rest (the files `exact/out/PJ_ball.txt`) are also the input of the certificates of Proposition 9.9 (root geometry) and of Section 9.3.

These are finite- J facts only: they are finite- J instances of (N) and of the coefficient bounds behind (G_a) and (Z_∞), and (Mg), a lim inf condition, has no finite- J instance. The companion paper states the natural conjecture behind Fact 2.10.

Conjecture 2.14 (Prime-power magic function; [15, Conjecture 6.1]). (a) M_J is non-singular for every $J \geq 1$.

(b) For every $J \geq 10$ all coefficients $p_k^{(J)}$, $0 \leq k \leq 2J$, are positive, and $a^* := \sup_{J \geq 10} \max_{1 \leq k \leq 2J} ((2k)! p_k^{(J)})^{1/(2k)} < \pi/2$.

(c) $P_J \rightarrow P_\infty$ coefficientwise, and $F_\infty := \Xi^2 P_\infty(t^2)$ satisfies $\widehat{F}_\infty(\xi) > 0$ for every $\xi \in [\xi_2, \infty) \setminus \xi_{PP}$.

The conjecture implies (S) and (U) [15, Section 6.1]. This paper also considers, as its own statement, the *front statement* (FR) at every J : $\widehat{F}_J > 0$ on $[\xi_2, \infty) \setminus \{\xi_{n_1}, \dots, \xi_{n_J}\}$ and $\widehat{F}_J'' > 0$ at $\xi_{n_1}, \dots, \xi_{n_J}$. It is not part of the conjecture of [15]; by Theorem 9.15 below it follows, for $J \geq J_0$, from the step law of Conjecture 11.1 together with the base hypothesis (B⁺)_{J₀}, which is certified only in part (Proposition 9.19).

2.3. Certificates. Every computer-assisted statement of this paper is a finite computation in exact integer or rational arithmetic, or in Arb ball arithmetic [14] with outward rounding, run by a standalone script of the ancillary files (Appendix C); the conventions are those of [15] and are recalled in Appendix B. In particular a sign is accepted only if the ball excludes 0, and a linear system is solved only by a ball solver that certifies invertibility. Every such statement names its script and parameter file. Numerical observations (Section 10) are never used in a proof.

3. NESTED HERMITE FAMILIES

The interpolation problems of this paper share one linear-algebraic structure, which we set up for an arbitrary kernel. All derivatives are taken in y .

Definition 3.1 (Nested Hermite families). A *base kernel* is a function φ , real-analytic on an interval $I = (y_-, \infty) \subseteq (0, \infty)$, such that the functions $\varphi_k := (-\theta^2)^k \varphi$ ($k \geq 0$) are linearly independent on I . Put $\sigma_k := \theta^{2k} \varphi = (-1)^k \varphi_k$ and $\mathcal{A}_K := \text{span}\{\varphi_0, \dots, \varphi_K\}$. For a polynomial $P(u) = \sum_k p_k u^k$ put $\mathcal{F}_P := P(-\theta^2) \varphi = \sum_k p_k \varphi_k$. For $F \in \mathcal{A}_K$ we write $[F]_r$ for its φ_r -coefficient (so $[\mathcal{F}_P]_r = [u^r]P$), and $\ell_0(F) := [F]_0$.

(i) *Chains.* A *chain of level K* is a list $Z = (z_1, \dots, z_K)$ of points of I , repetitions allowed; $\text{mult}_Z(z)$ is the number of occurrences of z . Its *data functionals* are $\lambda_r(f) := f^{(m_r)}(z_r)$, with $m_r := \#\{s < r : z_s = z_r\}$, and f *vanishes on Z* if $\lambda_r(f) = 0$ for all r . Put $\Delta(Z) := \det[\lambda_r(\sigma_b)]_{1 \leq r \leq K, 0 \leq b < K}$ ($\Delta(\emptyset) := 1$). Z is *regular* if $\Delta(Z) \neq 0$. Then there is a unique *top-normalised interpolant* $\mathcal{F}_Z \in \sigma_K + \mathcal{A}_{K-1}$ vanishing on Z ; its *cofactor* is $f_Z := \mathcal{F}_Z / \omega_Z$, $\omega_Z(y) := \prod_r (y - z_r)$, an analytic function on I . For $z \in I$, $Z + z := (z_1, \dots, z_K, z)$, and the *next data functional* at z is $\delta_z^Z(f) := f^{(\mu)}(z)$, $\mu := \text{mult}_Z(z)$; it is the functional that $Z + z$ adds to the data of Z .

- (ii) *Node sets.* A *node set* is $\mathbf{y} = (y_1 < \dots < y_J) \subset I$; its *doubled chain* $(y_1, y_1, \dots, y_J, y_J)$ has level $2J$, with data functionals $\delta_{y_j}, \delta'_{y_j}$. The *Hermite problem* at $\mathbf{y}$ asks for P with $P(0) = 1$, $\deg P \leq 2J$ and $\mathcal{F}_P(y_j) = \mathcal{F}'_P(y_j) = 0$ ($j \leq J$). Condition $(N)_{\mathbf{y}}$ states that the $2J \times 2J$ matrix $[\varphi_k(y_j); \varphi'_k(y_j)]$, $j \leq J$, $1 \leq k \leq 2J$, is non-singular; then the solution $P_{\mathbf{y}} = 1 + \sum_{k=1}^{2J} p_k(\mathbf{y})u^k$ is unique, and $\mathcal{F}_{\mathbf{y}} := \mathcal{F}_{P_{\mathbf{y}}}$.
- (iii) *The p_1 -rule.* Assume $(N)_{\mathbf{y}}$. The Hermite projection $\Pi_{\mathbf{y}}$ maps a polynomial G to the unique element of $\text{span}\{u, \dots, u^{2J}\}$ with the same data at $\mathbf{y}$ as G . The p_1 -rule is $\mathcal{L}_{\mathbf{y}}(G) := [u] \Pi_{\mathbf{y}} G$; its moments are $\ell_k := \mathcal{L}_{\mathbf{y}}(u^k)$ ($k \geq 1$), so that $\ell_k = \delta_{k1}$ for $k \leq 2J$, and $\ell_1^* := \ell_{2J+1}$, $\ell_2^* := \ell_{2J+2}$. The *extension elements* are $E_m := u^{2J+m} - \Pi_{\mathbf{y}} u^{2J+m}$ ($m \geq 1$), with zero data at $\mathbf{y}$; the *insertion element* is $E_* := \ell_1^* E_2 - \ell_2^* E_1$, and the *screening factor* is $S_{\mathbf{y}} := 1 - \mathcal{L}_{\mathbf{y}}(u P_{\mathbf{y}})$.
- (iv) *Insertion.* For $Y \in I \setminus \mathbf{y}$, $\mathbf{y} \cup \{Y\}$ is the node set with Y added. If (N) holds for $\mathbf{y}$ and for $\mathbf{y} \cup \{Y\}$, the *insertion response* is $\Delta_{\mathbf{y}}(Y) := p_1(\mathbf{y} \cup \{Y\}) - p_1(\mathbf{y})$.
- (v) *Instances.* The *true family* has $\varphi = \Psi$, viewed as a function of $y \in I = (0, \infty)$; then $\mathcal{F}_P = \widehat{\Xi^2 P(t^2)}$. The *Gamma-only family* has $\varphi = \Psi_1$, and the *model* (Section 5) has $\varphi = e^{-y}$.

The number ℓ_0 is a functional on $\mathcal{A}_K$; it is not one of the moments ℓ_k , $k \geq 1$. We record some elementary facts.

- (a) $P \mapsto \mathcal{F}_P$ is a linear isomorphism from the polynomials of degree $\leq K$ onto $\mathcal{A}_K$. For the true and Gamma-only kernels, linear independence holds because $\sum_k c_k \varphi_k$ is the Fourier transform of $\Xi^2 \sum_k c_k t^{2k}$ (resp. $\Xi_{\infty}^2 \sum_k c_k t^{2k}$), which is real-analytic in $\xi \in \mathbb{R}$; if it vanishes on an interval it vanishes identically, and then the polynomial is 0. For e^{-y} , $\varphi_k = e^{-y} \tau_k$ with τ_k a polynomial of exact degree $2k$.
- (b) A chain Z of level K is regular if and only if the only element of $\mathcal{A}_{K-1}$ vanishing on Z is 0. Reordering Z permutes the rows of $\Delta(Z)$, so regularity is a property of the multiset.
- (c) Under $(N)_{\mathbf{y}}$, $\mathcal{L}_{\mathbf{y}}(u P_{\mathbf{y}}) = \sum_k p_k \ell_{k+1} = 1 + p_{2J} \ell_1^*$, so $S_{\mathbf{y}} = -p_{2J}(\mathbf{y}) \ell_1^*$.
- (d) *Bottom and top normalisation.* If $(N)_{\mathbf{y}}$ holds, the elements of $\mathcal{A}_{2J}$ vanishing on the doubled chain Z of $\mathbf{y}$ form the line spanned by $\mathcal{F}_{\mathbf{y}}$, so Z is regular if and only if $p_{2J}(\mathbf{y}) \neq 0$, and then $\mathcal{F}_{\mathbf{y}} = p_{2J}(\mathbf{y}) \mathcal{F}_Z$. Conversely, if Z is regular, then $(N)_{\mathbf{y}}$ holds if and only if $\ell_0(\mathcal{F}_Z) \neq 0$, and then $P_{\mathbf{y}}$ corresponds to $\mathcal{F}_Z / \ell_0(\mathcal{F}_Z)$.
- (e) *Sign at $+\infty$.* For the three instances, $\varphi_k(y) = (-1)^k c_k y^{2k+d} e^{-y} (1 + O_k(y^{-1}))$ as $y \rightarrow \infty$, with $c_k > 0$ and $d = 4$ for the true and Gamma-only kernels (Lemma 9.26) and $d = 0$ for e^{-y} . So σ_K is positive near $+\infty$ and dominates every element of $\mathcal{A}_{K-1}$ there, and every top-normalised interpolant is positive near $+\infty$.
- (f) For the true family it is often convenient to use ξ -derivatives, as in Fact 2.7. Since $dy/d\xi = 2\pi y \neq 0$, the change of coordinate multiplies the derivative rows by non-zero factors. It does not affect (N) , regularity, $p_k(\mathbf{y})$, or any ratio of Wronskians with equal numbers of entries.

Coefficient positivity is not a property of the kernel alone: for the true family at the prime powers and $3 \leq J \leq 9$, the coefficients $p_{2J-1}^{(J)}$ (and, for some J , $p_{2J-3}^{(J)}$) are negative, while $p_{2J}^{(J)} > 0$ for every J computed (Numerical observation 10.1).

Lemma 3.2 (p_1 -rule and node derivative). *Let φ be a base kernel and $\mathbf{y} = (y_1 < \dots < y_J)$ with $(N)_{\mathbf{y}}$.*

- (a) *There are unique numbers λ_j, μ_j such that $\mathcal{L}_{\mathbf{y}}(G) = \sum_{j \leq J} (\lambda_j \mathcal{F}_G(y_j) + \mu_j \mathcal{F}'_G(y_j))$ for every polynomial G ; this functional is characterised by $\mathcal{L}_{\mathbf{y}}(u^k) = \delta_{k1}$ ($1 \leq k \leq 2J$). Moreover $p_1(\mathbf{y}) = -\mathcal{L}_{\mathbf{y}}(1)$.*

- (b) If (N) also holds for $\mathbf{y}' = \mathbf{y} \cup \{y_{J+1}\}$, then $p_1(\mathbf{y}') - p_1(\mathbf{y}) = -\mathcal{L}_{\mathbf{y}'}(P_{\mathbf{y}}) = -(\lambda'_{J+1}r + \mu'_{J+1}r')$, where $(r, r') := (\mathcal{F}_{\mathbf{y}}, \mathcal{F}'_{\mathbf{y}})(y_{J+1})$ and $\lambda'_{J+1}, \mu'_{J+1}$ are the weights of $\mathcal{L}_{\mathbf{y}'}$ at y_{J+1} .
- (c) If $\mathbf{y} \subset \mathbf{y}'$ with $|\mathbf{y}'| = J' > J$ and (N) $_{\mathbf{y}'}$, then $p_1(\mathbf{y}) - p_1(\mathbf{y}') = \sum_{k=2J+1}^{2J'} p_k(\mathbf{y}') \ell_k$, with $\ell_k = \mathcal{L}_{\mathbf{y}}(u^k)$.
- (d) (Node derivative.) If (N) holds near $\mathbf{y}$, then p_1 is real-analytic in the nodes and $\partial p_1 / \partial y_j = -\mathcal{F}_{\mathbf{y}}''(y_j) \mu_j$.

The proof is in Appendix A.2.

The weights λ_j, μ_j depend on the coordinate; the products in (b) and (d) do not.

Lemma 3.3 (Increment identity). *Let $\mathbf{y}_J = (y_1 < \dots < y_J)$ and $\mathbf{y}_{J+1} = \mathbf{y}_J \cup \{y_{J+1}\}$ be nested node sets for a base kernel, with (N) for both, and write $\mathcal{F}_J := \mathcal{F}_{\mathbf{y}_J}$. For a function f that is C^1 near the nodes, let $\Pi_J f$ be the element of $\text{span}\{\varphi_1, \dots, \varphi_{2J}\}$ with the data of f at $\mathbf{y}_J$, and let $A_{J+1}, B_{J+1} \in \text{span}\{\varphi_1, \dots, \varphi_{2J+2}\}$ be the cardinal functions at the newest node: zero data at $y_1, \dots, y_J$, and $(A, A')(y_{J+1}) = (1, 0)$, $(B, B')(y_{J+1}) = (0, 1)$. Then*

- (a) $\mathcal{F}_J = \varphi - \Pi_J \varphi$, and $D_J := \mathcal{F}_{J+1} - \mathcal{F}_J = -\Pi_{J+1} \mathcal{F}_J = -r_J A_{J+1} - r'_J B_{J+1}$ with $(r_J, r'_J) := (\mathcal{F}_J, \mathcal{F}'_J)(y_{J+1})$;
- (b) D_J lies in the two-dimensional space of elements of $\text{span}\{\varphi_1, \dots, \varphi_{2J+2}\}$ with zero data at $y_1, \dots, y_J$; $D_J(y_{J+1}) = -r_J$; $D_J = \mathcal{F}_{P_{J+1}-P_J}$ with $P_{J+1} - P_J$ vanishing at $u = 0$; and, telescoping, $\mathcal{F}_J = \mathcal{F}_{J_1} - \sum_{J_1 \leq K < J} (r_K A_{K+1} + r'_K B_{K+1})$ for nested node sets $\mathbf{y}_{J_1} \subset \dots \subset \mathbf{y}_J$;
- (c) (removable singularity) if $\mathcal{F}_K$ has a double zero at a node y_i with $\mathcal{F}_K''(y_i) \neq 0$, and G is C^3 with a double zero at y_i , then $G/\mathcal{F}_K$ extends continuously to y_i with value $G''(y_i)/\mathcal{F}_K''(y_i)$.

For the true family at the prime powers $(J, J+1 \in \mathcal{J})$ these are statements about $\hat{F}_J$.

Proof. (a) $\mathcal{F}_J$ is the element of $\varphi + \text{span}\{\varphi_1, \dots, \varphi_{2J}\}$ with zero data, so $\mathcal{F}_J = \varphi - \Pi_J \varphi$. Π_{J+1} is the identity on $\text{span}\{\varphi_1, \dots, \varphi_{2J+2}\}$, hence $\Pi_{J+1} \Pi_J = \Pi_J$ and $D_J = \Pi_J \varphi - \Pi_{J+1} \varphi = -\Pi_{J+1}(\varphi - \Pi_J \varphi)$. As $\mathcal{F}_J$ has zero data at $y_1, \dots, y_J$, $\Pi_{J+1} \mathcal{F}_J = r_J A_{J+1} + r'_J B_{J+1}$. (b) follows from (a). (c) is Taylor's formula with remainder at y_i for G and $\mathcal{F}_K$. $\square$

Lemma 3.4 (Node addition). *Let φ be a base kernel, $\mathbf{y} = (y_1 < \dots < y_J)$, $Y \in I \setminus \mathbf{y}$, with (N) for $\mathbf{y}$ and $\mathbf{y} \cup \{Y\}$, and put $\mathcal{W} := \text{Wr}(\mathcal{F}_{E_1}, \mathcal{F}_{E_2})$.*

- (a) (Additive form.) $P_{\mathbf{y} \cup \{Y\}} = P_{\mathbf{y}} + aE_1 + bE_2$, with

$$a = p_{2J+1}(\mathbf{y} \cup \{Y\}) = -\frac{\text{Wr}(\mathcal{F}_{P_{\mathbf{y}}}, \mathcal{F}_{E_2})(Y)}{\mathcal{W}(Y)}, \quad b = p_{2J+2}(\mathbf{y} \cup \{Y\}) = \frac{\text{Wr}(\mathcal{F}_{P_{\mathbf{y}}}, \mathcal{F}_{E_1})(Y)}{\mathcal{W}(Y)}.$$

Hence $p_k(\mathbf{y} \cup \{Y\}) - p_k(\mathbf{y}) = a[u^k]E_1 + b[u^k]E_2$ for $1 \leq k \leq 2J$.

- (b) (Multiplicative form.) Let $\mathfrak{D}_k(\mathbf{y})$ be the maximal minor, with column k deleted, of the $2J \times (2J+1)$ data matrix of $(\varphi_0, \dots, \varphi_{2J})$ at $\mathbf{y}$, so that $p_k = (-1)^k \mathfrak{D}_k / \mathfrak{D}_0$. For $0 \leq k \leq 2J$ with $p_k(\mathbf{y}) \neq 0$,

$$\mathfrak{D}_k(\mathbf{y} \cup \{Y\}) = \mathfrak{D}_k(\mathbf{y}) \mathfrak{W}_k(Y), \quad \frac{p_k(\mathbf{y} \cup \{Y\})}{p_k(\mathbf{y})} = \frac{\mathfrak{W}_k(Y)}{\mathfrak{W}_0(Y)},$$

where $\mathfrak{W}_k := \text{Wr}(\mathcal{F}_{G_{k,1}}, \mathcal{F}_{G_{k,2}})$, $G_{k,m} := u^{2J+m} - I_k(u^{2J+m})$, and $I_k(G)$ is the interpolant of the data of G at $\mathbf{y}$ in $\text{span}\{u^j : j \leq 2J, j \neq k\}$. Moreover $G_{k,m} = E_m + (\pi_k^{(m)} / p_k) P_{\mathbf{y}}$ with $\pi_k^{(m)} := [u^k] \Pi_{\mathbf{y}} u^{2J+m}$, and $\mathfrak{W}_0 = \mathcal{W}$.

- (c) (Cofactor form.) With $\mathcal{G}_{\mathbf{y}} := \mathcal{F}_{\mathbf{y}} / \prod_{j \leq J} (1 - y/y_j)^2$, $Y := y_J$ and $\mathbf{y}^- := \mathbf{y} \setminus \{Y\}$, $\mathcal{G}_{\mathbf{y}} = \mathcal{G}_{\mathbf{y}^-} (1 - y/Y)^{-2} + (\mathcal{F}_{\mathbf{y}} - \mathcal{F}_{\mathbf{y}^-}) / \prod_{j \leq J} (1 - y/y_j)^2$.
- (d) If all coefficients of $P_{\mathbf{y}}$ are positive, then all coefficients of $P_{\mathbf{y} \cup \{Y\}}$ are positive if and only if $\mathfrak{W}_k(Y) / \mathcal{W}(Y) > 0$ for $1 \leq k \leq 2J$ and $a, b > 0$.

The proof is in Appendix A.2: (a) is Cramer's rule in the two-dimensional space of polynomials of degree $\leq 2J + 2$ with zero data at $\mathbf{y}$, which is spanned by E_1, E_2 , and (b) is the Schur complement formula.

Positivity of all coefficients of $P_{\mathbf{y} \cup \{Y\}}$ does not imply it for $P_{\mathbf{y}}$: for the true family on the prime powers it holds at $J = 10$ (Fact 2.13) but not at $J = 9$ (Numerical observation 10.1). So (d) cannot be run backwards.

4. THE CHAIN AND INSERTION CALCULUS

Throughout this section φ is a base kernel, and chains, node sets and interpolants are as in Definition 3.1. Every statement is proved for an arbitrary nested family: the proofs use only the linear algebra of the data functionals and the nesting $\mathcal{A}_{K-1} \subset \mathcal{A}_K$ with $\dim \mathcal{A}_K / \mathcal{A}_{K-1} = 1$. So they apply verbatim to the true, Gamma-only and model kernels. This applies to statements about interpolants and their values. Statements about the *coefficients* of polynomial cofactors (Lemma 6.2 and Theorem 6.3) use the polynomial structure of the model and are not claimed for other kernels; nor are far-field statements, which depend on the kernel (Theorems 9.27 and 7.1).

If Z is a regular chain of level K , the elements of $\mathcal{A}_K$ vanishing on Z form the line spanned by $\mathcal{F}_Z$: such an element is $c\sigma_K + G$ with $G \in \mathcal{A}_{K-1}$, and either $c = 0$, so that $G = 0$ by regularity, or it equals $c\mathcal{F}_Z$. For $z \in Z$ of multiplicity μ and $\omega_0(y) := \prod_{w \in Z, w \neq z} (y - w)$ (product over the list) we have

$$(4.1) \quad \delta_z^Z(\mathcal{F}_Z) = \mu! \omega_0(z) f_Z(z),$$

and the same formula with $\mu = 0$ holds for $z \notin Z$.

Lemma 4.1 (Regularity propagation). *Let Z be a regular chain of level $K - 1$, $z \in I$, and δ_z^Z the next data functional at z . If $\delta_z^Z(\mathcal{F}_Z) \neq 0$, then $Z + z$ is regular.*

Proof. If $F \in \mathcal{A}_{K-1}$ vanishes on $Z + z$, then $F = c\mathcal{F}_Z$, and $0 = \delta_z^Z(F) = c\delta_z^Z(\mathcal{F}_Z)$ forces $c = 0$. $\square$

Lemma 4.2 (Christoffel identity). *Let Z be a regular chain of level K with $\ell_0(\mathcal{F}_Z) \neq 0$.*

- (a) *The space $\mathcal{S}(Z) := \{F \in \mathcal{A}_{K+1} : F \text{ vanishes on } Z\}$ has dimension 2, and $\mathcal{S}(Z) \cap \mathcal{A}_K = \text{span}(\mathcal{F}_Z)$.*
- (b) *There is a unique $G_Z \in \mathcal{S}(Z) \cap (\sigma_{K+1} + \mathcal{A}_K)$ with $\ell_0(G_Z) = 0$. For $z \in I$ with $\delta_z^Z(\mathcal{F}_Z) \neq 0$ (points of Z allowed), $Z + z$ is regular and*

$$\mathcal{F}_{Z+z} = G_Z - \chi_Z(z) \mathcal{F}_Z, \quad \chi_Z(z) := \delta_z^Z(G_Z) / \delta_z^Z(\mathcal{F}_Z).$$

For $z \notin Z$, $\chi_Z(z) = G_Z(z) / \mathcal{F}_Z(z)$; the function $\chi_Z := G_Z / \mathcal{F}_Z$ extends real-analytically to every point of I at which $f_Z \neq 0$, with the value $\delta_z^Z(G_Z) / \delta_z^Z(\mathcal{F}_Z)$ at points of Z .

- (c) (Model, cofactor form.) *Let $\varphi = e^{-y}$. Every element of $\mathcal{A}_K$ is e^{-y} times a polynomial of degree $\leq 2K$, so $\mathcal{F}_Z = e^{-y} \omega_Z \mathbf{f}_Z$ with $\mathbf{f}_Z$ a monic polynomial of degree K , and $\ell_0(F) = F(0)$. If $\mathbf{f}_Z(0) \neq 0$, then $G_Z = \mathcal{F}_{Z+0}$ is the interpolant of the chain with one more point at $y = 0$; write $\mathcal{F}_{Z+0} = e^{-y} y \omega_Z \mathbf{g}_Z$. If moreover $\mathbf{f}_Z(z) \neq 0$, then*

$$(y - z) \mathbf{f}_{Z+z}(y) = y \mathbf{g}_Z(y) - \frac{z \mathbf{g}_Z(z)}{\mathbf{f}_Z(z)} \mathbf{f}_Z(y).$$

The proof is in Appendix A.3.

Part (b) is an analogue, for a nested biorthogonal system with the test functional ℓ_0 , of Christoffel's formula for orthogonal polynomials [22, Theorem 2.5]. It also gives the reciprocity $f_Z(x) f_{Z+x}(y) = f_Z(y) f_{Z+y}(x)$ (Proposition 4.5(a)).

Lemma 4.3 (Ladder identity). *Let Z be a regular chain of level K , and let x be a point of Z of multiplicity μ . Let Z^- be Z with the multiplicity of x lowered by one, and assume that Z^- is regular, with interpolant $G := \mathcal{F}_{Z^-}$ and cofactor g , and that $g(x) \neq 0$. Moving all μ copies of x together and keeping the other points fixed,*

$$\partial_x \mathcal{F}_Z = -\frac{\mathcal{F}_Z^{(\mu)}(x)}{G^{(\mu-1)}(x)} G, \quad \partial_x f_Z = \mu \frac{f_Z - (f_Z(x)/g(x))g}{y - x}.$$

The proof is in Appendix A.3.

Lemma 4.4 (Bordered determinant and chain values). *Let Z be a chain of level $K - 1$, not necessarily regular, and $z \in I$. Let $\Phi_Z(y)$ be the determinant of the matrix of $\Delta(Z + z)$ with its last row replaced by $(\sigma_0(y), \dots, \sigma_{K-1}(y))$.*

- (i) $\Phi_Z \in \mathcal{A}_{K-1}$ vanishes on Z , and its σ_{K-1} -coefficient is $\Delta(Z)$; if Z is regular, $\Phi_Z = \Delta(Z)\mathcal{F}_Z$.
- (ii) $\Delta(Z + z) = \delta_z^Z(\Phi_Z)$; if Z is regular, $\Delta(Z + z) = \Delta(Z) \delta_z^Z(\mathcal{F}_Z)$, and $\delta_z^Z(\mathcal{F}_Z) = \mu! \omega_0(z) f_Z(z)$ is the chain value of (4.1).
- (iii) If every prefix $(z_1, \dots, z_k)$, $k < K$, of a chain $(z_1, \dots, z_K)$ is regular, then $\Delta(z_1, \dots, z_K) = \prod_{k=1}^K \delta_{z_k}^{Z_{k-1}}(\mathcal{F}_{Z_{k-1}})$, $Z_{k-1} := (z_1, \dots, z_{k-1})$.

In particular, let $\mathbf{y} = (y_1 < \dots < y_J)$, let Z be the doubled chain of $(y_1, \dots, y_{J-1})$, and assume Z regular. If $\mathcal{F}_Z(y_J) \neq 0$ and the half-step interpolant $\mathcal{F}_{Z+y_J}$ (double at $y_1, \dots, y_{J-1}$, simple at y_J) has a simple zero at y_J , then the doubled chain of $\mathbf{y}$ is regular; if $Z + y_J$ is regular, the converse holds as well.

The proof is in Appendix A.3.

The product formula (iii) needs regular prefixes; for an arbitrary ordering it can fail.

Proposition 4.5 (Pointwise forms: a Chebyshev-pair criterion). *Let Z be a regular chain with $\ell_0(\mathcal{F}_Z) \neq 0$, let G_Z and $\chi_Z = G_Z/\mathcal{F}_Z$ be as in Lemma 4.2, and assume $f_Z > 0$ on I (so χ_Z is real-analytic on I and $Z + z$ is regular for every $z \in I$).*

- (a) $f_{Z+z}(y) = f_Z(y) \frac{\chi_Z(y) - \chi_Z(z)}{y - z}$ for $y \neq z$, and $f_{Z+z}(z) = f_Z(z) \chi'_Z(z)$.
- (b) The following are equivalent: (1) $f_{Z+z} > 0$ on I for every $z \in I$; (2) $\chi'_Z > 0$ on I ; (3) $\text{Wr}(\mathcal{F}_Z, G_Z) > 0$ on $I \setminus Z$, and $f_Z(y) f_{Z+y}(y) > 0$ at the points $y \in Z$; (4) $f_Z(y) f_{Z+y}(y) > 0$ for every $y \in I$. An isolated zero of χ'_Z already violates (1), so “ χ_Z strictly increasing” is not sufficient.
- (c) For one given z : $f_{Z+z} > 0$ on I if and only if $(\chi_Z(y) - \chi_Z(z))/(y - z) > 0$ for $y \in I \setminus \{z\}$ and $\chi'_Z(z) > 0$.
- (d) (Pointwise half-step increment.) Put $\tilde{F} := \mathcal{F}/\ell_0(\mathcal{F})$ and $\tilde{f}_W := \tilde{F}_W / \prod_{w \in W} (1 - y/w)$ for $W = Z, Z + z$. If $\chi_Z(z) \neq 0$, then

$$\frac{\tilde{f}_{Z+z}(y)}{\tilde{f}_Z(y)} = \frac{z(\chi_Z(y) - \chi_Z(z))}{\chi_Z(z)(y - z)} \quad (y \in I \setminus \{z\}).$$

Consequently, if $\tilde{f}_Z > 0$ on I and $\chi_Z(z) > 0$, then $\tilde{f}_{Z+z} \geq \tilde{f}_Z$ on I if and only if $\chi_Z(y)/y - \chi_Z(z)/z$ has the sign of $y - z$ for $y \in I \setminus \{z\}$; if $\chi_Z(z) < 0$, the inequality between $\tilde{f}_{Z+z}$ and $\tilde{f}_Z$ is reversed.

The proof is in Appendix A.3.

So (FR) for every one-point extension of Z says that $(\mathcal{F}_Z, G_Z)$ is a Chebyshev pair on I , and the pointwise half-step increment says the same for $(\mathcal{F}_Z, G_Z/y)$ [16]. For the model on the prime powers χ_Z first decreases, so usable forms restrict to y beyond a point below y_1 (Section 5).

Proposition 4.6 (Insertion calculus). *Assume $(N)_{\mathbf{y}}$, $\mathbf{y} = (y_1 < \dots < y_J)$, and use $E_m, \ell_k, \ell_1^*, \ell_2^*$ and E_* of Definition 3.1. Put $\omega := \prod_j (y - y_j)^2$, $\mathcal{R}_0 := \mathcal{F}_{\mathbf{y}}/\omega$ and $\mathcal{R}_m := \mathcal{F}_{E_m}/\omega$ ($m \geq 1$), analytic on I .*

- (a) (Several new nodes.) *For distinct $z_1, \dots, z_r \in I \setminus \mathbf{y}$, (N) holds for $\mathbf{y} \cup \{z_1, \dots, z_r\}$ if and only if the $2r \times 2r$ matrix $[\mathcal{R}_m(z_i), \mathcal{R}'_m(z_i)]_{i \leq r, 1 \leq m \leq 2r}$ is non-singular. In that case $P_{\mathbf{y} \cup \{z_i\}} = P_{\mathbf{y}} + \sum_{m \leq 2r} a_m E_m$, where (a_m) is the unique vector for which $\mathcal{R}_0 + \sum_m a_m \mathcal{R}_m$ has a double zero at every z_i , and $p_1(\mathbf{y} \cup \{z_i\}) - p_1(\mathbf{y}) = -\sum_m a_m \ell_{2J+m}$.*
- (b) (One new node.) *Let $Y \in I \setminus \mathbf{y}$ and $\mathcal{W} := \text{Wr}(\mathcal{F}_{E_1}, \mathcal{F}_{E_2})$. Then $(N)_{\mathbf{y} \cup \{Y\}}$ holds if and only if $\mathcal{W}(Y) \neq 0$, and in that case $P_{\mathbf{y} \cup \{Y\}} = P_{\mathbf{y}} + aE_1 + bE_2$ with a, b as in Lemma 3.4(a), and*

$$\Delta_{\mathbf{y}}(Y) = \frac{\text{Wr}(\mathcal{F}_{P_{\mathbf{y}}}, \mathcal{F}_{E_*})(Y)}{\mathcal{W}(Y)}, \quad \mathcal{F}''_{\mathbf{y} \cup \{Y\}}(Y) = \frac{\text{Wr}(\mathcal{F}_{P_{\mathbf{y}}}, \mathcal{F}_{E_1}, \mathcal{F}_{E_2})(Y)}{\mathcal{W}(Y)}.$$

The weights of the new node in the $(J+1)$ -node p_1 -rule are $\lambda_Y = -\mathcal{F}'_{E_}(Y)/\mathcal{W}(Y)$ and $\mu_Y = \mathcal{F}_{E_*}(Y)/\mathcal{W}(Y)$, and $\partial_Y p_1(\mathbf{y} \cup \{Y\}) = \Delta'_{\mathbf{y}}(Y) = -\mathcal{F}''_{\mathbf{y} \cup \{Y\}}(Y) \mu_Y$.*

- (c) $\Delta_{\mathbf{y}}$, a and b are unchanged by any change of the variable y and by multiplying every $\mathcal{F}$ by one non-vanishing function; μ_Y and $\mathcal{F}''$ transform as a derivative weight and as a second derivative at a double zero.

The proof is in Appendix A.3: (a) and (b) are Cramer's rule in $\text{span}\{E_1, \dots, E_{2r}\}$, and (c) holds because all these quantities are ratios of Wronskians.

Every node is the inserted node of the configuration without it. So, under (N) for $\mathbf{y}$ and $\mathbf{y} \setminus \{y_j\}$, (b) applied to $\mathbf{y} \setminus \{y_j\}$ gives $\mu_j(\mathbf{y})$ and $\partial p_1/\partial y_j = \Delta'_{\mathbf{y} \setminus \{y_j\}}(y_j)$, consistently with Lemma 3.2(d).

The next proposition describes how the p_1 -problem responds to one or two new points. We use window interpolants normalised in the φ -basis. For a chain B of level K and $r \geq 0$ let

$$D_r(B) := \det[\lambda_i(\varphi_{r+j})]_{1 \leq i \leq K, 0 \leq j < K}$$

be the determinant of the data at B of $(\varphi_r, \dots, \varphi_{r+K-1})$, and when $D_r(B) \neq 0$ let F_r^B be the unique element of $\varphi_{r+K} + \text{span}\{\varphi_r, \dots, \varphi_{r+K-1}\}$ vanishing on B (the *window- r interpolant*). For $z \in I$ write $F[z] := \delta_z^B(F)$; this is $F(z)$ if $z \notin B$, and at a point of B it is the “confluent value”. Put

$$S(B) := \frac{D_0(B)D_2(B)}{D_1(B)^2}, \quad q_r^B(z) := \frac{F_{r+1}^B[z]}{F_r^B[z]}, \quad a_B(z) := \frac{F_0^B[z] F_2^B[z]}{F_1^B[z]^2}.$$

If $D_1(B) \neq 0$ there is a unique polynomial P_B of degree $\leq K$ with $P_B(0) = 1$ and $\mathcal{F}_{P_B}$ vanishing on B ; put $p_1(B) := [u]P_B$. For the doubled chain B of a node set $\mathbf{y}$, $D_1(B) \neq 0$ is $(N)_{\mathbf{y}}$ and $P_B = P_{\mathbf{y}}$. In the σ -normalisation of Definition 3.1 the functions q_r change sign, while $S(B)$ and a_B do not; we use the φ -normalisation throughout.

Proposition 4.7 (Half-step calculus). *Let B be a chain of level K and $z \in I$.*

- (H1) *If $D_r(B)D_{r+1}(B) \neq 0$ and $F_r^B[z] \neq 0$, then $F_r^{B+z} = F_{r+1}^B - q_r^B(z) F_r^B$.*
- (H2) *If $D_r(B) \neq 0$, then $D_r(B+z) = D_r(B) F_r^B[z]$. Hence, if $D_0D_1D_2(B) \neq 0$ and $F_1^B[z] \neq 0$, $S(B+z) = S(B) a_B(z)$.*
- (H3) *If $D_0(B)D_1(B) \neq 0$ and $F_1^B[z] \neq 0$, then $p_1(B+z) - p_1(B) = -S(B) F_0^B[z]/F_1^B[z]$, that is $-S(B)/q_0^B(z)$ when $F_0^B[z] \neq 0$.*
- (H4) *Let $D_0D_1D_2(B) \neq 0$, and let $z \neq x$ be points of I with $F_1^B[z] \neq 0 \neq F_1^B[x]$ and $q_1^B(x) \neq q_1^B(z)$. Then*

$$m_B(z, x) := p_1(B+z+x) - p_1(B+z) - p_1(B+x) + p_1(B) = S(B) \frac{a_B(x) - a_B(z)}{q_1^B(x) - q_1^B(z)}.$$

(H5) (One base.) Let $D_0 D_1 D_2(B) \neq 0$, $y \in I \setminus B$, $F_1^B(y) \neq 0$ and $(q_1^B)'(y) \neq 0$. Put $h_B(y) := p_1(B + y) - p_1(B)$ and $\Delta_B(y) := p_1(B + y + y) - p_1(B)$. Then

$$\Delta_B(y) - 2h_B(y) = S(B) a'_B(y) / (q_1^B)'(y).$$

Hence, if $S(B) > 0$ and $(q_1^B)'(y) < 0$, then $a'_B(y) \geq 0$ if and only if $\Delta_B(y) \leq 2h_B(y)$ (“multiplicity concavity”); if moreover $a_B(y) > 0$, both are equivalent to $(F_0^B)' / F_0^B - 2(F_1^B)' / F_1^B + (F_2^B)' / F_2^B \geq 0$ at y .

(H6) (Doubled bases.) Let B be the doubled chain of a node set $\mathbf{y}$ with $(N)_{\mathbf{y}}$. Then $S(B) = S_{\mathbf{y}}$. If $z \in I \setminus \mathbf{y}$, $D_0 D_1 D_2(B) \neq 0$, $F_r^B(z) \neq 0$ ($r = 0, 1, 2$) and $F_1^{B+z}[z] \neq 0$, then the Turán ratio is

$$\text{Tu}(z) := S_{\mathbf{y} \cup \{z\}} / S_{\mathbf{y}} = a_B(z) a_{B+z}(z),$$

with the confluent value in the second factor. For $z \neq x$ in $I \setminus \mathbf{y}$, whenever p_1 is defined at the nine chains $B + iz + jx$ ($0 \leq i, j \leq 2$),

$$\Delta_{\mathbf{y} \cup \{z\}}(x) - \Delta_{\mathbf{y}}(x) = m_B(z, x) + m_{B+z}(z, x) + m_{B+x}(z, x) + m_{B+z+x}(z, x).$$

The proof is in Appendix A.4. It uses only Cramer’s rule and the fact that, when $D_0(B), \dots, D_m(B)$ are non-zero, the elements of $\text{span}\{\varphi_0, \dots, \varphi_{K+m}\}$ that vanish on B are exactly the combinations of $F_0^B, \dots, F_m^B$.

In words: the two-point interaction of the p_1 -problem is a divided difference of the single function a_B with respect to the single function q_1^B .

Proposition 4.8 (Tail bound from the set-comparison law). Let $y_1 < y_2 < \dots$ be nodes in I , $\mathbf{y}_k := (y_1, \dots, y_k)$, and $J_1 \geq 1$. Assume (N) for $\mathbf{y}_{J_1}$ and $\mathbf{y}_{J_1+1}$, and, for every $k \geq J_1$ and every $n > k + 1$, (N) for $\mathbf{y}_k \cup \{y_n\}$ and for $\mathbf{y}_{k+1} \cup \{y_n\}$, together with

$$(\text{SM})_k : \quad \Delta_{\mathbf{y}_{k+1}}(y_n) \leq \Delta_{\mathbf{y}_k}(y_n).$$

Then for every $K \geq J_1$, $\Delta p_1^{(K)} := p_1(\mathbf{y}_{K+1}) - p_1(\mathbf{y}_K) = \Delta_{\mathbf{y}_K}(y_{K+1}) \leq \Delta_{\mathbf{y}_{J_1}}(y_{K+1})$, and every partial sum satisfies $\sum_{K=J_1}^J \Delta p_1^{(K)} \leq \sum_{n=J_1+1}^{J+1} \Delta_{\mathbf{y}_{J_1}}(y_n)$. If all $\Delta p_1^{(K)} \geq 0$, then $\delta_{J_1} := \sum_{K \geq J_1} \Delta p_1^{(K)} \leq B_{J_1} := \sum_{n > J_1} \Delta_{\mathbf{y}_{J_1}}(y_n)$. (Robust form.) If instead $\Delta_{\mathbf{y}_{k+1}}(y_n) \leq (1 + \varepsilon_k) \Delta_{\mathbf{y}_k}(y_n)$ with $\varepsilon_k \geq 0$, then $\Delta p_1^{(K)} \leq \prod_{J_1 \leq k < K} (1 + \varepsilon_k) \Delta_{\mathbf{y}_{J_1}}(y_{K+1})$; if moreover $\Delta_{\mathbf{y}_{J_1}}(y_n) \geq 0$ for $n > J_1$, the bounds above hold with the factor $\prod_{k \geq J_1} (1 + \varepsilon_k)$.

Proof. The hypotheses give (N) for every $\mathbf{y}_k$, $k \geq J_1$. Apply $(\text{SM})_k$ with $n = K + 1$ for $k = K - 1, \dots, J_1$: $\Delta_{\mathbf{y}_K}(y_{K+1}) \leq \Delta_{\mathbf{y}_{K-1}}(y_{K+1}) \leq \dots \leq \Delta_{\mathbf{y}_{J_1}}(y_{K+1})$; then sum. In the robust form each step multiplies by $1 + \varepsilon_k > 0$. $\square$

Only the diagonal comparisons $\Delta_{\mathbf{y}_K}(y_{K+1}) \leq \Delta_{\mathbf{y}_{J_1}}(y_{K+1})$ are used; (SM) is the natural local sufficient condition. Without a sign hypothesis on the $\Delta p_1^{(K)}$, the proposition bounds $\sum_K \Delta p_1^{(K)}$ from above but not $\sum_K |\Delta p_1^{(K)}|$.

For a node sequence $y_1 < y_2 < \dots$ the *half-step chain* is $B_{2K} :=$ the doubled chain of $\mathbf{y}_K$ and $B_{2K+1} := B_{2K} + y_{K+1}$; the next point is z_{i+1} , so that $B_{i+1} = B_i + z_{i+1}$ and $z_{2K+1} = z_{2K+2} = y_{K+1}$.

Theorem 4.9 (Tail bound from one-base laws). Let $y_1 < y_2 < \dots$ be nodes in I with half-step chain (B_i) , and fix $J_1 \geq 1$. Assume that for every $i \geq 2J_1$:

- (CP) $_i$ $D_1(B_i) \neq 0$ and $S(B_i) > 0$;
- (UM) $_i$ on $[z_{i+1}, \infty)$, $F_1^{B_i} \neq 0$ (with the confluent value at z_{i+1} if $z_{i+1} \in B_i$), a_{B_i} is non-decreasing, and $(q_1^{B_i})' < 0$.

Then for every $K \geq J_1$

$$\Delta p_1^{(K)} \leq 2h_{B_{2J_1}}(y_{K+1}) = -2S(B_{2J_1}) \frac{F_0^{B_{2J_1}}}{F_1^{B_{2J_1}}}(y_{K+1}),$$

$$\delta_{J_1} := \sum_{K \geq J_1} \Delta p_1^{(K)} \leq 2S(B_{2J_1}) \sum_{n > J_1} \left(-\frac{F_0^{B_{2J_1}}}{F_1^{B_{2J_1}}} \right)(y_n).$$

Proof. $(CP)_i$ gives $D_0 D_1 D_2(B_i) \neq 0$, and $(UM)_i$ makes $q_1^{B_i}$ strictly decreasing on $[z_{i+1}, \infty)$. *Chain monotonicity:* for $i \geq 2J_1$ and $z > z_{i+1}$, (H4) at B_i with the points z_{i+1} (a confluent point when i is odd) and z gives

$$h_{B_{i+1}}(z) - h_{B_i}(z) = m_{B_i}(z_{i+1}, z) = S(B_i) \frac{a_{B_i}(z) - a_{B_i}(z_{i+1})}{q_1^{B_i}(z) - q_1^{B_i}(z_{i+1})} \leq 0.$$

Since $z_{i+1} \leq y_K < y_{K+1}$ for $i \leq 2K - 1$, iterating from $i = 2J_1$ to $2K - 1$ at $z = y_{K+1}$ gives $h_{B_{2K}}(y_{K+1}) \leq h_{B_{2J_1}}(y_{K+1})$. *Doubling:* $\Delta p_1^{(K)} = \Delta_{B_{2K}}(y_{K+1})$, and (H5) at B_{2K} , $y = y_{K+1}$ (the left end of the interval in $(UM)_{2K}$, where $a'_{B_{2K}} \geq 0$) gives $\Delta_{B_{2K}}(y_{K+1}) \leq 2h_{B_{2K}}(y_{K+1})$. Combine, use (H3), and sum over K . $\square$

Compared with Proposition 4.8, no two-point inequality is needed: one base and one half-line per level. The strict inequality $(q_1^{B_i})' < 0$ cannot be weakened to monotonicity: by (H2), $D_1(B + y + y) = D_1(B)F_1^B(y)^2(q_1^B)'(y)$, so (N) fails where $(q_1^B)'$ vanishes. For the model, the laws $(UM)_i$ are certified at sampled levels (Proposition 7.8), the sign condition of $(CP)_i$ only at a few levels (see the evidence for Conjecture 11.5), and both were computed at every level $J \leq 40$ (Numerical observation 10.11); for the true kernel nothing has been computed in this form.

5. A POLYNOMIAL MODEL

In this section and the next three the Bessel kernel of the true family is replaced by e^{-y} . The resulting *model* is the instance $\varphi = e^{-y}$ of Definition 3.1(v): a Hermite interpolation problem for polynomials, with the same nodes $y_j = 2\pi n_j$. At leading order the true family *is* such a problem (Proposition 5.5), and the kernel-free calculus of Sections 3 and 4 applies to both. In the model, polynomial structure gives proofs that are not available for the true kernel: the front is proved for every $J \leq 80$ (Theorem 6.5, computer-assisted), and its arithmetic input is visible exactly, through prefix sums of prime powers (Lemma 5.7). Section 7 proves far-node asymptotics and an obstruction to proving the tail by sign regularity alone, and Section 8 shows that a natural sign property of the derivative weights fails at the critical node density.

Nothing in these sections implies a statement about the true family; the relation between the two is the subject of Section 5.2 and Numerical observation 10.13, and the quantitative agreement reported there is numerical.

5.1. Definition, umbral form and a radial Fourier interpretation. Let $T_m(x) = \sum_{i=0}^m S(m, i)x^i$ be the Touchard polynomials [25, 6], where $S(m, i)$ are the Stirling numbers of the second kind. Since $(x \frac{d}{dx})^m e^x = T_m(x)e^x$ and the Euler operator $\theta = y \frac{d}{dy}$ is invariant under $y \mapsto -y$, we have $\theta^m e^{-y} = T_m(-y)e^{-y}$, and so

$$(5.1) \quad (-\theta^2)^k e^{-y} = \tau_k(y) e^{-y}, \quad \tau_k(y) := (-1)^k T_{2k}(-y) = (-1)^k \left[y^{2k} - \binom{2k}{2} y^{2k-1} + \dots \right],$$

because $S(m, m-1) = \binom{m}{2}$. Thus τ_k has degree $2k$, leading coefficient $(-1)^k$ and $\tau_k(0) = 0$ for $k \geq 1$; for instance $\tau_1 = y - y^2$ and $\tau_2 = y^4 - 6y^3 + 7y^2 - y$.

Definition 5.1 (The model). (a) For a polynomial $P(u) = \sum_{k=0}^d p_k u^k$ put $Q_P(y) := e^y P(-\theta^2) e^{-y} = \sum_k p_k \tau_k(y)$, a polynomial of degree $2 \deg P$ with $Q_P(0) = P(0)$. In the notation of Definition 3.1 this is the family with base kernel $\varphi = e^{-y}$ on $I = (0, \infty)$, and $\mathcal{F}_P = Q_P e^{-y}$.

(b) Let $\mathbf{y} = (y_1 < \dots < y_J) \subset (0, \infty)$ satisfy (N) $_{\mathbf{y}}$, i.e. the matrix $[\tau_k(y_j); \tau'_k(y_j)]_{j \leq J, 1 \leq k \leq 2J}$ is non-singular. The *model solution* is the unique $P_{\mathbf{y}} = 1 + \sum_{k=1}^{2J} p_k(\mathbf{y}) u^k$ for which $Q_{\mathbf{y}} := Q_{P_{\mathbf{y}}}$ has a double zero at every y_j . Put $A_{\mathbf{y}}(y) := \prod_j (y - y_j)^2$. If $p_{2J}(\mathbf{y}) \neq 0$, then $\deg Q_{\mathbf{y}} = 4J$, and the *cofactor* $R_{\mathbf{y}} := Q_{\mathbf{y}}/A_{\mathbf{y}}$ is a polynomial of degree $2J$ with leading coefficient p_{2J} and $R_{\mathbf{y}}(0) = \prod_j y_j^{-2}$; we write $\hat{R}_{\mathbf{y}} := R_{\mathbf{y}}/R_{\mathbf{y}}(0)$ and $\tilde{R}_{\mathbf{y}} := R_{\mathbf{y}}/p_{2J}$.

(c) $(R^+)_{\mathbf{y}}$ means $R_{\mathbf{y}} \in \mathbb{R}_{>0}[y]$ (all $2J+1$ coefficients positive, in particular $p_{2J} > 0$); the model front $(\text{FR})_{\mathbf{y}}$ means $Q_{\mathbf{y}} > 0$ on $(0, \infty) \setminus \{y_1, \dots, y_J\}$. Clearly $(R^+)_{\mathbf{y}}$ implies $(\text{FR})_{\mathbf{y}}$.

(d) For an increasing sequence $y_1 < y_2 < \dots$ write $\mathbf{y}_{\leq K} := (y_1, \dots, y_K)$ and $P_K, Q_K, R_K, p_k^{(K)}$ for its objects. TEL_J holds if $\hat{R}_J - \hat{R}_{J-1}$ has non-negative coefficients in degrees $1, \dots, 2J-2$, and *strict* TEL_J if it has positive coefficients in degrees $1, \dots, 2J$.

(e) PT_K is the condition $\sum_{j \leq K} y_j > K(4K-1)$, and PT_{δ} at K is $\sum_{j \leq K} y_j \geq (1+\delta)K(4K-1)$. The progression $y_K^* := \frac{21}{20}(8K-5)$ has $\sum_{j \leq K} y_j^* = 1.05 K(4K-1)$ for every K .

(f) On the prime powers, $y_j = 2\pi n_j$, $\mathbf{y}_J^{\text{PP}} = (2\pi n_1, \dots, 2\pi n_J)$, and P_J, Q_J, R_J denote the objects of $\mathbf{y}_J^{\text{PP}}$.

The degree, leading coefficient and constant term in (b) are read off from (5.1): only $p_{2J}\tau_{2J}$ contributes to degree $4J$, and $Q_{\mathbf{y}}(0) = P_{\mathbf{y}}(0) = 1$.

Proposition 5.2 (Poisson, Mellin and umbral forms). *Let P be a polynomial and $F_P := Q_P e^{-y}$.*

- (a) $F_P(y) = \sum_{j \geq 0} P(-j^2) (-y)^j / j!$.
- (b) For $\text{Re } s > 0$, $\Gamma(s)^{-1} \int_0^\infty y^{s-1} F_P(y) dy = P(-s^2)$. Conversely, a polynomial $Q = \sum_m q_m y^m$ of degree $\leq 4J$ equals Q_P for some P of degree $\leq 2J$ if and only if $M_Q(s) := \sum_m q_m s(s+1) \cdots (s+m-1)$ is an even polynomial.
- (c) Equivalently, Q is of the form Q_P if and only if, for $l = 1, \dots, 2J$, $q_{2l-1} = -\sum_{k=2l}^{4J} |s(k, 2l-1)| q_k$, where $s(k, i)$ are the Stirling numbers of the first kind. For $l = 2J$ this reads $q_{4J-1} = -\binom{4J}{2} q_{4J}$.
- (d) If $P(-j^2) = \sum_m b_m(j)_m$ with falling factorials $(j)_m$, then $Q_P(y) = \sum_m b_m(-y)^m$.

The proof is in Appendix A.7.

Let $\mathcal{H}_0 g(y) := \int_0^\infty J_0(2\sqrt{yw}) g(w) dw$, where J_0 is the Bessel function of order 0, and let $\mathfrak{L}$ be the linear map $y^m \mapsto m! L_m(y)$ on polynomials (L_m the Laguerre polynomials).

Proposition 5.3 (A two-dimensional radial Fourier form). *Let Q be a polynomial with $Q(0) = 1$ and $F := Q e^{-y}$. The following are equivalent: (i) $Q = Q_P$ for some polynomial P ; (ii) $\mathcal{H}_0 F = -F'$; (iii) $\mathfrak{L}Q = Q - Q'$. For $v \in \mathbb{R}^2$ put $f(v) := F(\pi|v|^2)$ and $\hat{f}(\eta) := \int_{\mathbb{R}^2} f(v) e^{-2\pi i \langle v, \eta \rangle} dv$. Then $\hat{f}(\eta) = (\mathcal{H}_0 F)(\pi|\eta|^2)$, so (i) says $\hat{f}(\eta) = -F'(\pi|\eta|^2)$, and a double zero of Q at $y = 2\pi n$ is equivalent to the vanishing of both f and $\hat{f}$ on the circle $|v| = \sqrt{2n}$.*

The proof is in Appendix A.7.

So the model at the prime-power nodes asks for a radial function f on $\mathbb{R}^2$ with $\hat{f} = -F'(\pi|\cdot|^2)$ such that f and $\hat{f}$ vanish on the circles $|v| = \sqrt{2n}$, $n \in \text{PP}_J$. For all radii $\sqrt{n}$, $n \geq 1$, Stoller [21] proved that every Schwartz function on $\mathbb{R}^d$, $d \geq 2$, is determined by its restrictions, and those of its Fourier transform, to the spheres of these radii, extending the one-dimensional interpolation theorem of Radchenko and Viazovska [19, Theorems 1 and 2]; with derivatives at the radii $\sqrt{2n}$ the analogous statement is proved in dimensions 8 and 24 [4, Theorem 1.7], and the general case of derivatives of order $< k$ at the radii $\sqrt{kn}$ is [4, Open Problem 7.1]. Here the radii are $\sqrt{2n}$

for the prime powers n only. Heuristically, all integers n form a node set of critical density for such problems, while the prime powers are sparse; the data of Section 8 are consistent with this picture, but we use it only as motivation. Expansions in Laguerre polynomials of this kind, with prescribed double roots, are also the standard ansatz for sign-uncertainty problems [3].

Lemma 5.4 (The u -coefficient as an integral). *Let P be a polynomial with $P(0) = 1$ and $F = Q_P e^{-y}$. Then $\int_0^\infty (F(y) - e^{-y}) \frac{dy}{y} = 0$ and $p_1 = -\int_0^\infty (F(y) - e^{-y}) \log y \frac{dy}{y}$.*

Proof. By Proposition 5.2(b), $\Gamma(s)(P(-s^2) - 1) = \int_0^\infty (F(y) - e^{-y}) y^{s-1} dy =: I(s)$ for $\operatorname{Re} s > 0$; since $F(y) - e^{-y} = (Q_P(y) - 1)e^{-y} = O(y)$ at 0, I is analytic for $\operatorname{Re} s > -1$. As $P(-s^2) - 1 = -p_1 s^2 + O(s^4)$ and $\Gamma(s) = s^{-1} + O(1)$, $I(s) = -p_1 s + O(s^2)$; so $I(0) = 0$ and $I'(0) = -p_1$. $\square$

Thus p_1 , the scalar that controls the tail in Section 7.1, is a logarithmic moment of $F - e^{-y}$.

5.2. Dictionary to the true kernel. Recall $\tilde{\kappa}(z) = \sqrt{2z/\pi} e^z K_0(z)$ ($z > 0$) from Section 9.5, and from Fact 2.5 that $\Psi(\xi) = 2\pi\sqrt{x} \sum_{N \geq 1} d(N) W_0(2\pi N x)$ with $W_0 = \theta^2(\theta + 1)^2 K_0$, $x = e^{2\pi\xi}$.

Proposition 5.5 (Toy-true dictionary). (a) $\tilde{\kappa}(z) = \pi^{-1/2} \int_0^\infty e^{-v} v^{-1/2} (1 + \frac{v}{2z})^{-1/2} dv$; hence $\tilde{\kappa}$ is increasing, $0 < \tilde{\kappa} < 1$, and $\tilde{\kappa}(z) = 1 - \frac{1}{8z} + \frac{9}{128z^2} + O(z^{-3})$ as $z \rightarrow \infty$.

(b) For a polynomial P , with $P^\Xi(u) = P(u)(u + \frac{1}{4})^2$ as in Fact 2.8, put

$$\mathfrak{F}_P(z) := P^\Xi(-\theta^2)[e^{-z}\tilde{\kappa}(z)].$$

Then, with $y = 2\pi x$,

$$(5.2) \quad \widehat{\Xi^2 P(t^2)}(\xi) = \pi \sum_{N \geq 1} d(N) N^{-1/2} \mathfrak{F}_P(Ny).$$

(c) $\mathfrak{F}_P = \mathfrak{m}[Q_{P^\Xi}(z)e^{-z}]$, where $\mathfrak{m}$ is the Mellin multiplier with symbol

$$m(s) = \frac{\Gamma(\frac{s+1/2}{2})^2}{\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})};$$

for $s > 0$, $0 < m(s) \leq 1$ and $m(s) \rightarrow 1$ as $s \rightarrow \infty$.

In particular, if $\tilde{\kappa}$ is replaced by 1 and only the term $N = 1$ is kept, the true transform becomes $\pi Q_{P^\Xi}(y)e^{-y}$: the true family at leading order is the model for the polynomial P^Ξ , which has a frozen double root at $u = -\frac{1}{4}$.

The proof is in Appendix A.7.

The dictionary does not make the true family a perturbation of the model in any useful norm: the two are different interpolation problems, and a positive smoothing cannot transport positivity between them.

Proposition 5.6 (No transfer by positive smoothing). (a) Let $k > 0$ be measurable on $(1, \infty)$ and $T_k f(z) := \int_1^\infty f(z t) k(t) \frac{dt}{t}$ whenever the integral converges absolutely. If f is continuous, $f \geq 0$ on $[z_0, \infty)$, $f \not\equiv 0$ there, and $T_k f(z_0)$ converges absolutely, then $T_k f(z_0) > 0$.

(b) $e^{-z}\tilde{\kappa}(z) = T_k g(z)$ with $g(w) := w^{1/2}e^{-w}$ and $k(t) := \pi^{-1/2}t^{1/2}(t-1)^{-1/2}(\frac{t+1}{2})^{-1/2} > 0$. Consequently, for every polynomial P ,

$$\mathfrak{F}_P = T_k[w^{1/2}Q_P^{\text{Ab}}(w)e^{-w}],$$

where Q_P^{Ab} is the Abel preimage of Fact 2.8.

(c) Hence, if $P \neq 0$ and $\mathfrak{F}_P(z_0) = 0$ for some $z_0 > 0$, then Q_P^{Ab} is negative somewhere on (z_0, ∞) .

The proof is in Appendix A.7.

By (5.2) with $N = 1$, $\widehat{\Xi_\infty^2 P(t^2)}(\xi) = \pi \mathfrak{F}_P(2\pi x)$. So for the Gamma-only family, whose transform has double zeros at the nodes, the Abel preimage must take negative values beyond every node: the “dips” of the Abel preimage are forced. For the true kernel the transform at a node $y_n = 2\pi n$ equals $-\pi \sum_{N \geq 2} d(N) N^{-1/2} \mathfrak{F}_P(N y_n)$ rather than 0, and these terms are of size e^{-2y_n} up to polynomial factors; we do not formalise the corresponding perturbation statement. In particular, no statement of the form “positivity in the model implies positivity for the true kernel” can follow from positivity of the smoothing alone, and the families are different solutions of different problems (in general the true P_J is not a model solution).

5.3. The trace identity and small cases. The node set enters the top of the cofactor only through the sum of the nodes.

Lemma 5.7 (Trace identities). (a) Let $\mathbf{y} = (y_1, \dots, y_J)$ satisfy $(N)_{\mathbf{y}}$ and $p_{2J}(\mathbf{y}) \neq 0$, and write $R_{\mathbf{y}} = p_{2J}(y^{2J} + \rho_1 y^{2J-1} + \dots)$. Then $\rho_1 = 2 \sum_{j \leq J} y_j - 2J(4J - 1)$; equivalently, the $2J$ roots of $R_{\mathbf{y}}$ sum to $2J(4J - 1) - 2 \sum_j y_j$.

(b) Consequently $(R^+)_{\mathbf{y}}$ implies PT_J , and if (R^+) holds for every prefix $\mathbf{y}_{\leq K}$, $K \leq J$, then PT_K holds for every $K \leq J$.

(c) (chains) For a regular chain Z of level K with monic cofactor $\mathbf{f}_Z$ and 0-augmented cofactor $\mathbf{g}_Z$ (Definition 6.1), $[y^{K-1}] \mathbf{f}_Z = \Sigma Z - K(2K - 1)$ and $[y^K] \mathbf{g}_Z = \Sigma Z - (K + 1)(2K + 1)$, where ΣZ is the sum of the points of Z with multiplicity.

(d) (integer nodes) For $y_j = 2\pi(j + 1)$, PT_J holds if and only if $J \leq 12$. Hence for these nodes (R^+) fails for every $J \geq 13$ at which (N) holds.

Proof. (a) By (5.1), only $p_{2J} \tau_{2J}$ contributes to the coefficients of y^{4J} and y^{4J-1} of $Q_{\mathbf{y}}$, so $Q_{\mathbf{y}} = p_{2J}(y^{4J} - \binom{4J}{2} y^{4J-1} + \dots)$, while $A_{\mathbf{y}} = y^{2J} - 2(\sum_j y_j) y^{2J-1} + \dots$; divide, and note $\binom{4J}{2} = 2J(4J - 1)$. (b) Under (R^+) , $p_{2J} > 0$ and $p_{2J} \rho_1 > 0$. (c) $\omega_Z \mathbf{f}_Z$ is a monic element of $\mathcal{A}_K^{\text{pol}} = \text{span}(\tau_0, \dots, \tau_K)$ of degree $2K$, hence equals $(-1)^K \tau_K$ plus lower τ 's, i.e. $y^{2K} - K(2K - 1)y^{2K-1} + \dots$, and $\omega_Z = y^K - (\Sigma Z)y^{K-1} + \dots$. The second identity is the first for the chain $Z + 0$ of level $K + 1$. (d) $\sum_{j \leq J} 2\pi(j + 1) = \pi J(J + 3)$, and $\pi J(J + 3) > J(4J - 1)$ if and only if $J < (3\pi + 1)/(4 - \pi) = 12.14\dots$; then apply (b). $\square$

For the prime powers, $\sum_{j \leq K} n_j \sim \frac{1}{2} K^2 \log K$, so the trace condition holds with a margin that grows like $K^2 \log K$; this is the only place where the arithmetic of the nodes enters the top coefficient. At $J = 1$ everything is explicit. We use the property (W) of Definition 8.1 below.

Proposition 5.8 (One node). Let $J = 1$, $y_1 = a > 0$ and $D(a) := 2a^3 - 9a^2 + 12a - 6$, which has a single real root $a_D = 2.67765\dots$. Then (N) holds if and only if $D(a) \neq 0$, and in that case

$$p_1 = \frac{4a^3 - 18a^2 + 14a - 1}{a^2 D(a)}, \quad p_2 = \frac{2a - 1}{a^2 D(a)}, \quad R = \frac{(2a - 1)y^2 + 2(2a - 1)(a - 3)y + D(a)}{a^2 D(a)},$$

$$\mu_1 = \frac{a^3 - 6a^2 + 7a - 1}{a D(a)}, \quad Q''(a) = \frac{4(4a^3 - 12a^2 + 9a - 3)}{a^2 D(a)},$$

where μ_1 is the derivative weight of the p_1 -rule (Lemma 3.2). Hence: (a) (R^+) holds if and only if $a > 3$, i.e. if and only if PT_1 ; (b) $Q''(a) > 0$ for every $a > a_D$; (c) for $a > a_D$, (W) holds if and only if $a > 4.49086\dots$, the largest root of $a^3 - 6a^2 + 7a - 1 = \tau_2(a)/a$; (d) the screening factor is $S_{(a)} = (2a - 1)(2a^7 - 33a^6 + 208a^5 - 620a^4 + 908a^3 - 636a^2 + 168a - 24)/(a^2 D(a)^2)$; it equals $1 - 8/a + 8/a^2 + O(a^{-3})$, and for $a > a_D$ it is positive if and only if $a > 3.26513\dots$

The proof (Appendix A.7) is an explicit computation with τ_1, τ_2, τ_3 and Cramer's rule.

So in the model (W) is not a consequence of positivity: for $3 < a < 4.49$ the front holds but $\mu_1 < 0$. Zero slack in the trace conditions is not enough, either.

Remark 5.9 (Zero-slack prefix traces are not sufficient). At $J = 2$ the node sets $(y_1, y_2) = (3.02, 11.3)$, $(3.05, 11.05)$ and $(3.02, 13.0)$ satisfy PT_1 and PT_2 , but the coefficient $[y]R_{\mathbf{y}}$ is negative there (approximately $-5.0 \cdot 10^{-4}$, $-3.6 \cdot 10^{-4}$ and $-1.7 \cdot 10^{-5}$; exact rational computation, `toy/pt/verify_pt0_false.py` with `toy/pt/params/pt0_false.json`, which also certifies (N) at the three points). So PT_K for all $K \leq J$ does not imply (R^+) , and a relative slack is needed. We use $\delta = 0.05$.

The three propositions below are exact certificates. Each one is a finite computation in exact integer arithmetic whose design is described in Appendix B.1; the positivity step is Lemma B.1(b).

Proposition 5.10 (Positive cofactors for two nodes). *Let $J = 2$ and $0 < y_1 < y_2$ with $y_1 \geq \frac{63}{20}$ and $y_1 + y_2 \geq \frac{147}{10}$ (that is, $\text{PT}_{0.05}$ at $K = 1, 2$). Then $(N)_{\mathbf{y}}$ holds and $R_{\mathbf{y}} \in \mathbb{R}_{>0}[y]$; in particular $p_4 > 0$.*

Proposition 5.11 (Positive cofactors for three nodes). *Let $c = \frac{21}{20}$ and $0 < y_1 < y_2 < y_3$ with $y_1 \geq 3c$, $y_1 + y_2 \geq 14c$ and $y_1 + y_2 + y_3 \geq 33c$. Then $(N)_{\mathbf{y}}$ holds, $p_6 > 0$ and $R_{\mathbf{y}} \in \mathbb{R}_{>0}[y]$.*

Proposition 5.12 (Positive cofactors for four nodes). *Let $c = \frac{21}{20}$ and $0 < y_1 < y_2 < y_3 < y_4$ with $\sum_{j \leq K} y_j \geq cK(4K-1)$ for $K = 1, 2, 3, 4$ (i.e. $y_1 \geq 3.15$, $y_1 + y_2 \geq 14.7$, $y_1 + y_2 + y_3 \geq 34.65$, $y_1 + \dots + y_4 \geq 63$). Then $(N)_{\mathbf{y}}$ holds, $p_8 > 0$ and $R_{\mathbf{y}} \in \mathbb{R}_{>0}[y]$.*

Proofs of Propositions 5.10–5.12 (computer-assisted). Fix $J \in \{2, 3, 4\}$ and $V := \prod_{i < j} (y_i - y_j)$. The verifier computes, in exact arithmetic in $\mathbb{Z}[y_1, \dots, y_J]$, polynomials $G_0, \dots, G_{2J}$ (the Cramer numerators of $D \cdot R_{\mathbf{y}}$, divided by V^4) and checks two identities: (i) $A_{\mathbf{y}}(y) \sum_i G_i y^i$ satisfies the $2J$ odd Stirling conditions of Proposition 5.2(c) identically; (ii) the determinant D of the system in $(N)_{\mathbf{y}}$ equals $\epsilon_J (y_1 \cdots y_J)^2 V^4 G_0$ with a sign $\epsilon_J = \pm 1$. It then covers the region by finitely many rational parametrisations of $[0, \infty)^m$ (two pieces for $J = 2$, four for $J = 3$, eleven for $J = 4$; Appendix B.1) and checks that every substituted, homogenised G_i has only positive coefficients and a positive constant term; by Lemma B.1(b), every $G_i > 0$ on the region. Then $D \neq 0$, i.e. (N). By (i), $A_{\mathbf{y}} \sum_i G_i y^i$ is an admissible polynomial of degree $\leq 4J$ with double zeros at the nodes, hence a multiple of $Q_{\mathbf{y}}$ (by (N) the space of such polynomials is one-dimensional); comparing values at 0 gives $R_{\mathbf{y}} = \sum_i G_i y^i / ((y_1 \cdots y_J)^2 G_0) \in \mathbb{R}_{>0}[y]$, and $p_{2J} = G_{2J} / ((y_1 \cdots y_J)^2 G_0) > 0$.

Ancillary data. Script `toy/pt/verify_pt.py` with parameter files `toy/pt/params/pt2.json`, `pt3.json`, `pt4.json` (SHA-256 in Appendix C); exact arithmetic (FLINT multivariate polynomials over $\mathbb{Z}$); the sizes of the polynomials are in Table 4. For $J = 2$ a second, independent route (Cramer's rule with explicit ratios r_i/r_4 , on the two original pieces) is `toy/pt/verify_pt2_cramer.py`; for $J = 4$ the region cover is checked by `toy/pt/check_pt4_region.py`. Re-run commands are listed in Appendix C; the $J = 4$ certificate takes about 40 CPU-minutes. $\square$

Corollary 5.13 (The front for $J \leq 4$). *For every $J \leq 4$ and every increasing $\mathbf{y}$ satisfying $\text{PT}_{0.05}$ at $K = 1, \dots, J$, $(N)_{\mathbf{y}}$, $p_{2J} > 0$ and $(R^+)_{\mathbf{y}}$ hold, hence $(\text{FR})_{\mathbf{y}}$. This applies to $\mathbf{y}_J^{\text{PP}}$, $J \leq 4$, and to every increasing $\mathbf{y}$ with $y_k \geq y_k^*$ for all k .*

Proof. For $J = 1$, $y_1 \geq 3.15 > a_D$ gives (N) and (R^+) by Proposition 5.8(a); for $J = 2, 3, 4$ use Propositions 5.10–5.12. The prime powers satisfy the four inequalities directly (their prefix sums 4π , 10π , 18π , 28π exceed 3.15, 14.7, 34.65, 63), and $y \geq y^*$ coordinatewise gives prefix sums $\geq 1.05K(4K-1)$. $\square$

The prime powers lie above the progression y^* , for every K .

Proposition 5.14 (Prime powers dominate the PT-tight progression). (a) For every $K \geq 1$, $2\pi n_K > y_K^* = \frac{21}{20}(8K - 5)$; the smallest difference, $2\pi n_7 - y_7^* = 18\pi - 53.55 > 2.998$, occurs at $K = 7$. (b) For every $K \geq 1$, $\sum_{j \leq K} 2\pi n_j \geq 1.2366 K(4K - 1)$; the ratio of the two sides is smallest at $K = 9$, where it equals $124\pi/315 = 1.23669\dots$. Consequently $\mathbf{y}^{\text{PP}}$ satisfies $\text{PT}_{0.05}$ (indeed $\text{PT}_{0.2366}$) at every K .

The proof is in Appendix A.7.

Theorem 6.5 below settles the front on the prime powers for $J \leq 80$ by a different route. For general node sets above y^* there is a monotone-path reduction, which we record because it is the only route we know that covers all such node sets.

Remark 5.15 (A monotone-path reduction). Let $\mathcal{U}_K := \{\mathbf{y} \in \mathbb{R}^K : y_1 < \dots < y_K, y_k \geq y_k^* \forall k\}$, regard node sets as doubled chains (Definition 6.1), and let $\mathbf{y} \setminus y_j$ be $\mathbf{y}$ with y_j removed. Suppose that for every $K \leq J$: $(\text{AP}^+)_K$: $y_{\leq K}^*$ is regular and $\tilde{R}(y_{\leq K}^*) \in \mathbb{R}_{>0}[y]$; $(\text{B})_K$: $T_K(y_K^*) > 0$, where T_K is the monic cofactor of the half-step chain $(y_1^*, y_1^*, \dots, y_{K-1}^*, y_{K-1}^*, y_K^*)$; $(\text{LOC})_K$: $\partial \tilde{R} / \partial y_j \in \mathbb{R}_{\geq 0}[y]$ at every regular $\mathbf{y} \in \mathcal{U}_K$ with $\tilde{R}(\mathbf{y}) \in \mathbb{R}_{>0}[y]$ for which $\mathbf{y} \setminus y_j$ is regular with $\tilde{R}(\mathbf{y} \setminus y_j) \in \mathbb{R}_{>0}[y]$. Then every $\mathbf{y} \in \mathcal{U}_J$ is regular with $\tilde{R}(\mathbf{y}) \in \mathbb{R}_{>0}[y]$, so (N), $p_{2J} > 0$ and (FR) hold on $\mathcal{U}_J$.

The proof of this reduction is in Appendix A.7.

The base data $(\text{AP}^+)_K$ and $(\text{B})_K$ are certified for $K \leq 90$ (Proposition 5.16). The local law (LOC) is open; computed evidence is in Numerical observation 10.14(a). For $J \leq 4$ the conclusion holds unconditionally by Corollary 5.13.

Proposition 5.16 (Base data on the progression). For every $K \leq 90$: the node set $y_{\leq K}^*$, $y_k^* = \frac{21}{20}(8k - 5)$, satisfies (N), $p_{2K} > 0$ and $R \in \mathbb{R}_{>0}[y]$ (hence $(\text{AP}^+)_K$); and the half-step cofactor T_K at y^* lies in $\mathbb{R}_{>0}[y]$, in particular $T_K(y_K^*) > 0$ $((\text{B})_K)$.

Proof (computer-assisted). All nodes are rational, so everything is exact. For each $K \leq 90$, `toy/ap/verify_ap.py` solves the $2K \times 2K$ system over $\mathbb{Q}$ (FLINT `fmpq`) and checks the residual exactly, divides Q by A with zero remainder, and checks $p_{2K} > 0$ and the positivity of all $2K + 1$ coefficients of R ; as consistency checks it verifies $r_1/r_0 = 1 - P(-1) + 2 \sum_k 1/y_k^*$ and $\rho_1 = 2\delta K(4K - 1)$ exactly. (N) together with $p_{2K} \neq 0$ gives regularity, and $\tilde{R} = R/p_{2K}$. The script `toy/ap/verify_halfstep.py` computes T_K exactly in the τ -basis and checks its coefficients and $T_K(y_K^*)$. Parameter file `toy/ap/params/ap_delta005.json`; logs and re-run commands in Appendix C. $\square$

6. THE CHRISTOFFEL CHAIN OF THE MODEL AND THE FRONT FOR $J \leq 80$

6.1. The Christoffel chain. In the model the chain calculus of Section 4 can be read on coefficients. Let $\mathcal{A}_K^{\text{pol}} := \text{span}(\tau_0, \dots, \tau_K)$, the admissible polynomials of degree $\leq 2K$, so that $\mathcal{A}_K = \mathcal{A}_K^{\text{pol}} e^{-y}$ for the base kernel e^{-y} .

Definition 6.1 (Chains, cofactors and the chain laws). A *chain of level K* is a multiset $Z \subset (0, \infty)$ of size K ; $\omega_Z(y) := \prod_{z \in Z} (y - z)$. A *cofactor* of Z is a monic polynomial f of degree K with $\omega_Z f \in \mathcal{A}_K^{\text{pol}}$; Z is *regular* if the cofactor is unique, and then it is denoted $\mathbf{f}_Z$ (so that $\omega_Z \mathbf{f}_Z e^{-y}$ is the top-normalised interpolant $\mathcal{F}_Z$ of Definition 3.1, since $e^y \theta^{2K} e^{-y} = (-1)^K \tau_K$ is monic). For regular Z with $\mathbf{f}_Z(0) \neq 0$ put $\mathbf{g}_Z := \mathbf{f}_{Z+0}$ (the *0-augmented cofactor*, of degree $K + 1$; $Z + 0$ is regular by Lemma 4.1) and $\hat{\mathbf{f}}_Z := \mathbf{f}_Z / \mathbf{f}_Z(0)$. For the doubled chain $\mathbf{y} + \mathbf{y}$ of a node set (level $2J$), $\mathbf{f}_{\mathbf{y}+\mathbf{y}} = \tilde{R}_{\mathbf{y}}$ and $\hat{\mathbf{f}}_{\mathbf{y}+\mathbf{y}} = \hat{R}_{\mathbf{y}}$. Writing $\mathbf{f}_Z = \sum_k f_k y^k$ and $\mathbf{g}_Z = \sum_k g_k y^k$:

- (I0) $_Z$: $\mathbf{f}_Z \in \mathbb{R}_{>0}[y]$;
- (I1') $_Z$: $\varrho_k := g_k / f_k$ is non-decreasing for $0 \leq k \leq K$;

- (I1)_Z: $\sigma'_k := g_k/f_{k+1}$ is non-decreasing for $0 \leq k \leq K-1$.

Along a sequence $z_1, z_2, \dots > 0$ with $Z_K := (z_1, \dots, z_K)$, $\mathbf{f}^{(K)} := \mathbf{f}_{Z_K}$ and $\mathbf{g}^{(K)} := \mathbf{g}_{Z_K}$, the condition (Y0)_K is $\mathbf{g}^{(K)}(z_{K+1}) > 0$. The *prime-power chain* is $z_{2j-1} = z_{2j} = 2\pi n_j$; its even levels are the node sets $\mathbf{y}_J^{\text{PP}}$ and its odd levels the half-steps $\mathbf{y}_J^{\text{PP}} + y_{J+1}$.

For the model, the Christoffel identity (Lemma 4.2(c)) reads: if Z is regular, $\mathbf{f}_Z(0) \neq 0$ and $\mathbf{f}_Z(z) \neq 0$ ($z \in Z$ allowed), then $Z + z$ is regular and

$$(6.1) \quad (y - z) \mathbf{f}_{Z+z}(y) = y \mathbf{g}_Z(y) - \lambda(z) \mathbf{f}_Z(y), \quad \lambda(z) := z \mathbf{g}_Z(z) / \mathbf{f}_Z(z).$$

The next two statements use the polynomial structure (the top coefficients $f_{K+1} = 0$, $g_{K+1} = 1$, and normalisation at 0); they have no analogue for the true kernel as stated.

Lemma 6.2 (Half-steps). *Let Z be a regular chain of level K , $f = \mathbf{f}_Z$, $g = \mathbf{g}_Z$, and $z > 0$.*

- (i) (ST) *If (I0)_Z, (I1')_Z and $g(z) > 0$, then $Z + z$ is regular, $\mathbf{f}_{Z+z} \in \mathbb{R}_{>0}[y]$, and $\hat{\mathbf{f}}_{Z+z} - \hat{\mathbf{f}}_Z$ has positive coefficients in every degree $1, \dots, K+1$.*
- (ii) (SC) *If (I0)_Z, (I1)_Z, $g_K > 0$ (equivalently $\Sigma Z > (K+1)(2K+1)$), by Lemma 5.7(c)) and $g(z) > 0$, then $\mathbf{f}_{Z+z} \in \mathbb{R}_{>0}[y]$.*
- (iii) *(I1')_Z, log-concavity of the coefficients of f , and $g_1 > 0$ together imply (I1)_Z.*

The proof (Appendix A.7) reads (6.1) coefficientwise: the coefficients of $\mathbf{f}_{Z+z}$ are partial sums of a sequence with a single sign change.

Theorem 6.3 (The front from one likelihood-ratio law per level). *Let $z_1, z_2, \dots > 0$ and $K_0 \geq 1$. Suppose that (I1')_K and (Y0)_K hold for every $0 \leq K < K_0$. Then for every $K \leq K_0$, Z_K is regular and $\mathbf{f}^{(K)} \in \mathbb{R}_{>0}[y]$, and for $K < K_0$, $\hat{\mathbf{f}}^{(K+1)} - \hat{\mathbf{f}}^{(K)}$ has positive coefficients in degrees $1, \dots, K+1$. For a doubled chain ($z_{2j-1} = z_{2j} = y_j$) and every J with $2J \leq K_0$: (N)_{y_{≤J}}, $p_{2J} > 0$, (R⁺)_{y_{≤J}}, strict TEL_J, and every half-step cofactor $\mathbf{f}^{(2J-1)}$ lies in $\mathbb{R}_{>0}[y]$.*

The proof is in Appendix A.7.

Remark 6.4 (Why the law does not propagate). Let Z be a regular chain of level K with $f := \mathbf{f}_Z$ and $g := \mathbf{g}_Z$, let $z > 0$, and assume $f(0), f(z), g(0), g(z) \neq 0$. Then $Z + 0 + 0$ is regular; put $h := \mathbf{f}_{Z+0+0}$, $\Phi_l := (f_l, g_{l-1}, h_{l-2}) \in \mathbb{R}^3$, $\Phi(z) := \sum_l z^l \Phi_l$ and $S_{\leq m}(z) := \sum_{l \leq m} z^l \Phi_l$. If $\mathbf{f}_{Z+z} \in \mathbb{R}_{>0}[y]$, then

$$(6.2) \quad (\text{I1})_{Z+z} \iff \det[\Phi(z), S_{\leq m}(z), \Phi_{m+1}] / f(z) \geq 0 \quad \text{for } 1 \leq m \leq K.$$

Derivation. By (6.1) applied to Z at z , and to $Z + 0$ at z (whose 0-augmented cofactor is h , and whose cofactor at the next level is $\mathbf{f}_{Z+0+z} = \mathbf{g}_{Z+z}$, since cofactors depend only on the multiset), $f' := \mathbf{f}_{Z+z}$ and $g' := \mathbf{g}_{Z+z}$ satisfy $(y - z)f' = yg - \lambda f$ and $(y - z)g' = yh - \lambda' g$, with $\lambda = zg(z)/f(z)$ and $\lambda' = zh(z)/g(z)$. Expanding $(z - y)^{-1}$ at $y = 0$, as in (A.1), gives $f'_k = z^{-k-1} u \cdot S_{\leq k}(z)$ and $g'_k = z^{-k-2} v \cdot S_{\leq k+1}(z)$ with $u = (\lambda, -1, 0)$ and $v = (0, \lambda', -1)$. Since $S_{\leq m+1} = S_{\leq m} + z^{m+1} \Phi_{m+1}$, the identity $(a \cdot c)(b \cdot d) - (a \cdot d)(b \cdot c) = (a \times b) \cdot (c \times d)$ and $u \times v = (1, \lambda, \lambda \lambda') = \Phi(z)/f(z)$ give

$$f'_m g'_m - f'_{m+1} g'_{m-1} = z^{-m-2} \det[\Phi(z), S_{\leq m}(z), \Phi_{m+1}] / f(z).$$

If $f' \in \mathbb{R}_{>0}[y]$, the left side is non-negative if and only if $g'_{m-1}/f'_m \leq g'_m/f'_{m+1}$, and for $1 \leq m \leq K$ these inequalities are (I1)_{Z+z}. So propagating the law to the next level requires the next augmented cofactor $\mathbf{f}_{Z+0+0}$, which in turn requires $\mathbf{f}_{Z+3,0}$, and so on, and we know no way to close this hierarchy at a fixed depth (for its computed behaviour along the prime powers, and for the failures of (I1') at critical density, see Numerical observation 10.14(b)). This is why Theorem 6.5 below is a finite theorem.

6.2. The front for $J \leq 80$.

Theorem 6.5 (The front of the model for $J \leq 80$). *Let $y_j = 2\pi n_j$, where n_j is the j -th prime power. For every $J \leq 80$:*

- (a) $(N)_{\mathbf{y}^{\text{PP}}}$ holds and $p_{2J} > 0$;
- (b) $R_J \in \mathbb{R}_{>0}[y]$; in particular $Q_J > 0$ on $(0, \infty)$ off the nodes (model (FR));
- (c) strict TEL_J holds: $\hat{R}_J - \hat{R}_{J-1}$ has positive coefficients in every degree $1, \dots, 2J$;
- (d) the half-step cofactor (nodes $y_1, \dots, y_{J-1}$ double, y_J simple) has positive coefficients.

Moreover, (e) along the prime-power chain $(\text{I0})_K$, $(\text{I1}')_K$ and $(\text{Y0})_K$ hold at every level $K \leq 160$; hence, for every $K \leq 160$ and every $z > 0$ with $\mathbf{g}^{(K)}(z) > 0$, $\mathbf{f}_{Z_K+z} \in \mathbb{R}_{>0}[y]$.

The theorem concerns the model only; it says nothing about the true family.

Proof (computer-assisted, two independent certificates). Both certificates use Arb ball arithmetic [14] with the real number π (a ball), so the nodes are exact; every sign decision is a ball that excludes 0, and linear systems are solved by a rigorous ball solver that fails unless it proves the matrix invertible (Appendix B). The prime powers $n_1, \dots, n_{81}$ are read from the parameter file and checked in the script to be the initial segment of the prime powers.

Chain route (`toy/pp80/verify_pp80_chain.py`, parameter file `toy/pp80/params/pp80_chain.json`, 4000 bits). For every $K \leq 160$ the script computes $\mathbf{f}^{(K)}$ and $\mathbf{g}^{(K)}$ by rigorous ball solves of the odd Mellin conditions of Proposition 5.2(c), which proves the regularity of Z_K and $Z_K + 0$, and checks $(\text{I0})_K$, $(\text{I1}')_K$ and $(\text{Y0})_K$. Theorem 6.3 with $K_0 = 160$ then gives (a)–(d) for every $J \leq 80$, and (e) follows from Lemma 6.2(i).

Direct route (`toy/pp80/verify_pp80_tau.py`, parameter file `toy/pp80/params/pp80.json`, 6000 bits, two runs). A different algorithm (Hermite systems in the τ -basis, then division by ω_Z) checks the same chain conditions for every $K \leq 160$, and, without using Theorem 6.3, (N) , $p_{2J} > 0$, strict TEL_J and positivity of the half-step cofactor for every $J \leq 80$.

Both routes end with “ALL CERTIFIED”. The worst relative radius of a cofactor coefficient is below $2.6 \cdot 10^{-649}$ (direct route, $K = 160$), and the smallest relative likelihood-ratio increment $\min_k (\varrho_{k+1} - \varrho_k) / |\varrho_k|$ over all levels is at least 0.0473 (attained at $K = 160$); the margins per level are in Table 5. The design of both certificates is described in Appendix B.2; files, hashes, logs and commands are in Appendix C. Total cost about 19 CPU-minutes. $\square$

Without this theorem, the front of the model on the prime powers is proved here only for $J \leq 4$ (Corollary 5.13). By Theorem 6.3, extending it to all J amounts to one likelihood-ratio law per level (Conjecture 11.4); Remark 6.4 explains why an induction on the level does not close.

7. THE TAIL OF THE MODEL AND AN OBSTRUCTION

7.1. The tail of the model. We use the objects of Definition 3.1(iii) for the model: $\Pi_{\mathbf{y}}$, the p_1 -rule $\mathcal{L}_{\mathbf{y}}$, ℓ_k , ℓ_1^* , ℓ_2^* , the extension elements E_m , $E_* = \ell_1^* E_2 - \ell_2^* E_1$ and the screening factor $S_{\mathbf{y}} = -p_{2J} \ell_1^*$. In the model we take the data of P at a node to be $(Q_P(y_j), Q'_P(y_j))$; this multiplies the derivative weights of the p_1 -rule by the positive factors e^{-y_j} and changes no sign. Put

$$R_0 := R_{\mathbf{y}}, \quad R_m := Q_{E_m} / A_{\mathbf{y}} \quad (m \geq 1), \quad R_* := Q_{E_*} / A_{\mathbf{y}} = \ell_1^* R_2 - \ell_2^* R_1, \quad \mathcal{W} := \text{Wr}(R_1, R_2).$$

The top two coefficients of Q_{E_m} come from τ_{2J+m} alone, and those of $Q_{\mathbf{y}}$ from $p_{2J} \tau_{2J}$. By (5.1), and dividing by $A_{\mathbf{y}} = y^{2J} - \sigma y^{2J-1} + \dots$ with $\sigma := 2 \sum_j y_j$,

$$(7.1) \quad R_0 = p_{2J}(y^{2J} + \kappa_0 y^{2J-1} + \dots), \quad R_m = (-1)^m (y^{2J+2m} + \kappa_m y^{2J+2m-1} + \dots),$$

where $\kappa_m := \sigma - \binom{4J+2m}{2}$. So R_m has degree $2J + 2m$ and leading coefficient $(-1)^m$, and $\mathcal{W}$ has degree $4J + 5$ and leading coefficient -2 . Since Wronskians are multiplied by g^2 when all functions are multiplied by g , Proposition 4.6(b) reads, for $Y \in (0, \infty) \setminus \mathbf{y}$: $(N)_{\mathbf{y} \cup \{Y\}}$ if and only if $\mathcal{W}(Y) \neq 0$, and then $P_{\mathbf{y} \cup \{Y\}} = P_{\mathbf{y}} + aE_1 + bE_2$ with

$$(7.2) \quad \begin{aligned} a &= -\frac{\text{Wr}(R_0, R_2)}{\mathcal{W}}, & b &= \frac{\text{Wr}(R_0, R_1)}{\mathcal{W}}, & \Delta_{\mathbf{y}}(Y) &= \frac{\text{Wr}(R_0, R_*)}{\mathcal{W}}, \\ \mu_Y &= \frac{R_*(Y)}{A_{\mathbf{y}}(Y)\mathcal{W}(Y)}, & Q''_{\mathbf{y} \cup \{Y\}}(Y) &= \frac{A_{\mathbf{y}}(Y) \text{Wr}(R_0, R_1, R_2)(Y)}{\mathcal{W}(Y)}, \end{aligned}$$

where μ_Y is the derivative weight of the new node in the p_1 -rule of $\mathbf{y} \cup \{Y\}$, and $\Delta'_{\mathbf{y}}(Y) = -Q''_{\mathbf{y} \cup \{Y\}}(Y) \mu_Y$.

Theorem 7.1 (Far-node asymptotics in the model). *Assume $(N)_{\mathbf{y}}$ and $p_{2J}(\mathbf{y}) \neq 0$, and let $Y_{\mathcal{W}}(\mathbf{y})$ be the largest real zero of $\mathcal{W}$, or y_J if this is larger. For $Y > Y_{\mathcal{W}}(\mathbf{y})$:*

- (1) $(N)_{\mathbf{y} \cup \{Y\}}$ holds, and $a = 2p_{2J}Y^{-2} + O(Y^{-3})$, $b = p_{2J}Y^{-4} + O(Y^{-5})$; in particular $P_{\mathbf{y} \cup \{Y\}} \rightarrow P_{\mathbf{y}}$ coefficientwise at rate $O(Y^{-2})$;
- (2) $\Delta_{\mathbf{y}}(Y) = 2S_{\mathbf{y}}Y^{-2} + 16JS_{\mathbf{y}}Y^{-3} + O(Y^{-4})$;
- (3) $Q''_{\mathbf{y} \cup \{Y\}}(Y) = 8p_{2J}Y^{4J-2}(1 + O(Y^{-1}))$, $\mu_Y = \frac{S_{\mathbf{y}}}{2p_{2J}}Y^{-4J-1} + O(Y^{-4J-2})$ and $\Delta'_{\mathbf{y}}(Y) = -4S_{\mathbf{y}}Y^{-3} + O(Y^{-4})$.

The implied constants depend on $\mathbf{y}$; $\Delta_{\mathbf{y}}$, a and b are rational functions of Y .

The proof (Appendix A.7) expands the Wronskians in (7.2) to two orders with (7.1).

The relative coefficient $8J$ in (2) does not depend on the node positions. For the true kernel the corresponding constant is $8(J+1)$ (Theorem 9.27): the frozen double root of P^{Ξ} acts as one more node.

Corollary 7.2 (Signs in the far field). *If $p_{2J}(\mathbf{y}) > 0$ and $S_{\mathbf{y}} > 0$, there is $Y_0(\mathbf{y})$ such that for all $Y \geq Y_0$: $(N)_{\mathbf{y} \cup \{Y\}}$, $\mu_Y > 0$, $Q''_{\mathbf{y} \cup \{Y\}}(Y) > 0$ and $\Delta_{\mathbf{y}}(Y) > 0$. If $p_{2J} > 0$ and $S_{\mathbf{y}} < 0$, then $\mu_Y < 0$ and $\Delta_{\mathbf{y}}(Y) < 0$ for all large Y .*

Proof. Immediate from Theorem 7.1. □

Proposition 7.3 (One far node changes S by the factor $1 - 8/z$). *Assume $(N)_{\mathbf{y}}$ and $p_{2J}(\mathbf{y}) \neq 0$. As $z \rightarrow \infty$, $S_{\mathbf{y} \cup \{z\}} = S_{\mathbf{y}}(1 - 8/z) + O(z^{-2})$.*

The proof is in Appendix A.7.

Proposition 7.4 (Two far nodes). *Assume $(N)_{\mathbf{y}}$ and $p_{2J}(\mathbf{y}) \neq 0$. Let $z = L\alpha$ and $s = L\beta$, with (α, β) in a compact subset of $(0, \infty)^2$ (the diagonal allowed), and $L \rightarrow \infty$. Then, uniformly and together with all derivatives,*

$$p_1(\mathbf{y} \cup \{z, s\}) = p_1(\mathbf{y}) + 2S_{\mathbf{y}} \left[\frac{1}{z^2} + \frac{1}{s^2} + 8J \left(\frac{1}{z^3} + \frac{1}{s^3} \right) - \frac{8}{zs(z+s)} \right] + O(L^{-4}),$$

and $\partial_z \partial_s p_1(\mathbf{y} \cup \{z, s\}) = -32S_{\mathbf{y}}(z^2 + 3zs + s^2)/(z^2 s^2 (z+s)^3) + O(L^{-6})$.

The proof is in Appendix A.7.

The case of one node in the same computation reproduces Theorem 7.1(2). In the far field, therefore, $\mu_Y > 0$, $\Delta_{\mathbf{y}} > 0$, the sign of the two-node interaction, and the decrease $S_{\mathbf{y} \cup \{z\}} < S_{\mathbf{y}}$ are all governed by the single sign of $S_{\mathbf{y}}$ (given $p_{2J} > 0$). The screening factor also has an exact multiplicative update.

Proposition 7.5 (Turán-type update of the screening factor). *For $r \geq 0$ let $V_r := u^{r+2J} -$ (Hermite interpolant of u^{r+2J} in $\text{span}(u^r, \dots, u^{r+2J-1})$ at $\mathbf{y}$) and $\text{Wr}_{ab} := \text{Wr}(Q_{V_a}, Q_{V_b})$. Assume (N) for $\mathbf{y}$ and for $\mathbf{y} \cup \{z\}$, $z \notin \mathbf{y}$, and $D_0(\mathbf{y}), D_2(\mathbf{y}), D_3(\mathbf{y}) \neq 0$, where $D_r(\mathbf{y}) := D_r(B)$ for the doubled chain B of $\mathbf{y}$ (Proposition 4.7, $\varphi = e^{-y}$). Then $D_r(\mathbf{y} \cup \{z\}) = e^{-2z} D_r(\mathbf{y}) \text{Wr}_{r,r+1}(z)$ for $r = 0, 1, 2$, and*

$$\text{Tu}(z) := \frac{S_{\mathbf{y} \cup \{z\}}}{S_{\mathbf{y}}} = \frac{\text{Wr}_{01}(z) \text{Wr}_{23}(z)}{\text{Wr}_{12}(z)^2}.$$

Hence $S_{\mathbf{y}_K^{\text{PP}}} = S_{\mathbf{y}_1^{\text{PP}}} \prod_{k < K} \text{Tu}_k$ with $\text{Tu}_k := [\text{Wr}_{01} \text{Wr}_{23} / \text{Wr}_{12}^2]_{\mathbf{y}_k^{\text{PP}}}(y_{k+1})$.

The proof is in Appendix A.7.

Tail bounds in the model. The kernel-free tail statements of Section 4 apply to the model verbatim. By Proposition 4.8, if (N) and the set-comparison law $(\text{SM})_k$: $\Delta_{\mathbf{y}_{k+1}^{\text{PP}}}(y_n) \leq \Delta_{\mathbf{y}_k^{\text{PP}}}(y_n)$ ($n > k+1$) hold for every $k \geq J_1$, then every partial sum of the scalar tail satisfies $\sum_{K=J_1}^J \Delta p_1^{(K)} \leq \sum_{j=J_1+1}^{J+1} \Delta_{\mathbf{y}_{J_1}^{\text{PP}}}(y_j)$, a sum of values of one explicit rational function. By Theorem 4.9, the same kind of bound follows from the one-base laws $(\text{CP})_i$ and $(\text{UM})_i$ at every chain level $i \geq 2J_1$. These laws, and the far-node expansion of Theorem 7.1 with explicit remainders, are certified on finite ranges.

Proposition 7.6 (Certified far-node remainders). *Let $\mathbf{y} = \mathbf{y}_J^{\text{PP}}$ with $J \in \{6, 10, 20\}$. For every $Y \geq y_{J+1}$,*

$$\Delta_{\mathbf{y}}(Y) = \frac{2S_{\mathbf{y}}}{Y^2} \left(1 + \frac{8J}{Y} + \lambda_Y \frac{C_J}{Y^2} \right) \quad \text{for some } \lambda_Y \in [0, 1],$$

with $C_6 = 1300$, $C_{10} = 4000$ and $C_{20} = 11000$; here $S_{\mathbf{y}} = 0.0371185\dots, 0.0140380\dots$ and $0.0056532\dots$

Proposition 7.7 (Certified set-comparison law). *For every $k = 1, \dots, 60$ and every real $x > y_{k+1}$: (N) holds for $\mathbf{y}_k^{\text{PP}} \cup \{x\}$ and $\mathbf{y}_{k+1}^{\text{PP}} \cup \{x\}$, and $\Delta_{\mathbf{y}_{k+1}^{\text{PP}}}(x) < \Delta_{\mathbf{y}_k^{\text{PP}}}(x)$. Consequently $\Delta p_1^{(K)} \leq \Delta_{\mathbf{y}_{J_1}^{\text{PP}}}(y_{K+1})$ for $1 \leq J_1 \leq K \leq 61$.*

Proposition 7.8 (Certified one-base laws at sampled levels). *Let $J \in \{2, 4, 8, 12, 20, 30, 40, 60\}$ and let B be either chain level $\mathbf{y}_J^{\text{PP}}$ (all nodes double) or $\mathbf{y}_J^{\text{PP}} + y_{J+1}$. With $F_r = F_r^B$ as in Proposition 4.7 ($\varphi = e^{-y}$), $q_1 = F_2/F_1$ and $a = a_B = F_0 F_2 / F_1^2$: on $[y_{J+1}, \infty)$ one has $F_1 \neq 0$ (at the level $\mathbf{y}_J^{\text{PP}} + y_{J+1}$, with the confluent value at y_{J+1}), $a'(y) > 4/y^2$ and $q'_1 < 0$, and $0 < a(y_{J+1}) \leq 1 - 4/y_{J+1}$. In particular (UM) holds at both levels, and $S_{\mathbf{y}_{J+1}^{\text{PP}}} / S_{\mathbf{y}_J^{\text{PP}}} \leq (1 - 4/y_{J+1})^2$.*

Proofs of Propositions 7.6–7.8 (computer-assisted). All three are Arb computations with the nodes $2\pi n_j$ as balls (real π), rigorous ball solves (which certify the relevant instances of (N)), and Descartes certificates at $c = y_{J+1}$ (resp. $c = y_{k+1}$) in the sense of Lemma B.1(a); the working precision is raised until every sign is decided. By (7.2), $\Delta_{\mathbf{y}} = N/\mathcal{W}$ off the nodes, with $N := \text{Wr}(R_0, R_*)$ and $\mathcal{W} = \text{Wr}(R_1, R_2)$. Proposition 7.6 (`toy/ft/verify_farbound.py`, `toy/ft/params/farbound.json`): put $\mathcal{E} := Y^3 N / (2S_{\mathbf{y}}) - (Y + 8J)\mathcal{W}$, so that $Y^2 \Delta_{\mathbf{y}} / (2S_{\mathbf{y}}) - 1 - 8J/Y = \mathcal{E} / (Y\mathcal{W})$. The script shows that $-\mathcal{W}$, $-Y\mathcal{E}$ and $Y\mathcal{E} - C_J\mathcal{W}$ have positive Taylor coefficients at c ; the two top coefficients of $Y\mathcal{E}$ vanish identically by Theorem 7.1(2), and the script checks that their balls contain 0 before dropping them. Proposition 7.7 (`toy/sm/verify_smset.py`, `toy/sm/params/smset.json`): with the superscript $+$ for the objects of $\mathbf{y}_{k+1}^{\text{PP}}$, $\Delta_{\mathbf{y}_{k+1}^{\text{PP}}} - \Delta_{\mathbf{y}_k^{\text{PP}}} = (N^+ \mathcal{W} - N \mathcal{W}^+) / (\mathcal{W} \mathcal{W}^+)$, and all Taylor coefficients at c of $\mathcal{W}$, $\mathcal{W}^+$ and $N^+ \mathcal{W} - N \mathcal{W}^+$ are certified negative; $\mathcal{W}(x) \neq 0$ is (N) for the enlarged node set (Proposition 4.6(b)). The consequence is the argument of Proposition 4.8, which uses $(\text{SM})_k$ only for $J_1 \leq k < K$. Proposition 7.8 (`toy/um/verify_um.py`, `toy/um/params/um.json`, 6000 bits): the polynomials

$y^2(F_0 \text{Wr}(F_1, F_2) - F_2 \text{Wr}(F_0, F_1)) - 4F_1^3 = F_1^3(y^2a' - 4)$ and F_1^3 are one-signed with the same sign, and $\text{Wr}(F_1, F_2) = F_1^2q_1'$ has negative Taylor coefficients; at the level $\mathbf{y}_{J+1}^{\text{PP}} + y_{J+1}$ every F_r has the exact factor $y - y_{J+1}$, whose contribution to the lowest Taylor coefficients is checked to vanish and removed. Finally $a(y_{J+1}) > 0$ is certified at both levels, and $a(y_{J+1}) \leq 1 - 4/y_{J+1}$ follows from $a' > 4/y^2$ and $a(y) \rightarrow 1$ as $y \rightarrow \infty$ (by (5.1), F_r has leading term $(-1)^{r+K}y^{2r+2K}e^{-y}$, $K = |B|$). The bound for $S_{\mathbf{y}_{J+1}^{\text{PP}}}/S_{\mathbf{y}_J^{\text{PP}}}$ is Proposition 4.7(H6). $\square$

Further computed tail data of the model (uncertified) are collected in Numerical observation 10.11.

7.2. An obstruction. The laws needed for the tail, the Turán inequality $\text{Tu} < 1$ and the set-comparison law (SM), are statements about five functions at one or two points. We first write them in that form.

For a node set $\mathbf{y}$ let $D_r(\mathbf{y})$ be as in Proposition 7.5 and, when $D_r(\mathbf{y}) \neq 0$, let V_r be the element of $\text{span}(u^r, \dots, u^{r+2J})$, monic in u^{r+2J} , with zero data at $\mathbf{y}$; put $v_r := Q_{V_r}$. Thus $V_0 = P_{\mathbf{y}}/p_{2J}$ and $V_1 = E_1$. For points $z \neq x$ off $\mathbf{y}$ write $[ij]_z := \text{Wr}(v_i, v_j)(z)$ and $[i_1i_2i_3i_4](z, x) := \det[v_{i_b}^{(e)}(w_a)]$, with rows $(w_a, e) = (z, 0), (z, 1), (x, 0), (x, 1)$, and define

$$\text{Tu}(z) := \frac{[01]_z[23]_z}{[12]_z^2}, \quad I(z, x) := -\frac{[0234](z, x)}{[1234](z, x)} + \frac{[02]_z}{[12]_z} + \frac{[02]_x}{[12]_x}.$$

Both are unchanged if all v_r are multiplied by one non-vanishing function.

Lemma 7.9 (Insertion through the companion solutions). *Let $D_r(\mathbf{y}) \neq 0$ for $0 \leq r \leq 4$, let $z \neq x$ be points of $(0, \infty) \setminus \mathbf{y}$, and assume (N) for $\mathbf{y} \cup \{z\}$, $\mathbf{y} \cup \{x\}$ and $\mathbf{y} \cup \{z, x\}$. Then $\Delta_{\mathbf{y}}(x) = -S_{\mathbf{y}}[02]_x/[12]_x$ and*

$$\Delta_{\mathbf{y}}^2(z, x) := \Delta_{\mathbf{y} \cup \{z\}}(x) - \Delta_{\mathbf{y}}(x) = S_{\mathbf{y}} I(z, x).$$

Moreover $\text{Tu}(z) = S_{\mathbf{y} \cup \{z\}}/S_{\mathbf{y}}$ (Proposition 7.5).

The proof is in Appendix A.7.

For the model v_r has degree $4J + 2r$ and, by (5.1), $v_r = (-1)^r y^{4J+2r} (1 + \kappa_r/y + O(y^{-2}))$ with $\kappa_r = -\binom{4J+2r}{2}$, a quadratic function of r with second difference -4 . The next lemma isolates what this quadratic term does.

Lemma 7.10 (First-order law for perturbed monomial systems). *For constants $\kappa_0, \dots, \kappa_4$, an integer $d \geq 2$ and a parameter t , let $v_r(y) := (-1)^r y^{d+2r} (1 + t\kappa_r/y)$, $r = 0, \dots, 4$, and define Tu and I from these v_r as above. At $t = 0$, $\text{Tu} \equiv 1$ and $I \equiv 0$; and*

- (i) $\partial_t \text{Tu}(z)|_{t=0} = (3\kappa_0 - 5\kappa_1 + \kappa_2 + \kappa_3)/(2z)$;
- (ii) if $\kappa_r = \alpha + \beta r + Cr^2$, then $\partial_t I(z, x)|_{t=0} = 8C/(xz(x+z))$.

In particular both derivatives vanish when κ_r is affine in r , and for $\kappa_r = Cr^2$ they equal $4C/z$ and $8C/(xz(x+z))$; neither depends on d .

The proof is in Appendix A.7.

By Lemma 7.10, $x^2 I/(2(\text{Tu} - 1)) \rightarrow x/(x+z)$ at first order, independently of C : the far-field shape of the set-comparison law is universal. For the model ($C = -2$) the first-order terms are $-8/z$ and $-16/(xz(x+z))$, the leading terms of Propositions 7.3 and 7.4.

Proposition 7.11 (Sign regularity cannot prove the tail of the model). *Fix $d \geq 2$ and $c > 0$, and for $\kappa \in \mathbb{R}^5$ let $v_r^\kappa(y) := (-1)^r y^{d+2r} (1 + \kappa_r/y)$ on $[c, \infty)$.*

- (1) *There is $\delta = \delta(c, d) > 0$ such that for $|\kappa| < \delta$ every Wronskian $W_I := \text{Wr}(v_i^\kappa : i \in I)$, $I \subseteq \{0, \dots, 4\}$, has no zero on $[c, \infty)$ and has there the sign of the monomial system $\kappa = 0$, and $\pm W_I(c+t)$ has only positive Taylor coefficients in t .*

- (2) Along $\kappa = s \cdot (r^2)_{r=0}^4$, for fixed $z, x > c$, $z \neq x$: $\text{Tu}(z) - 1 = 4s/z + O(s^2)$ and $I(z, x) = 8s/(xz(x+z)) + O(s^2)$.

Consequently, any family of sign conditions on the five-function system $(v_0, \dots, v_4)$ on $[c, \infty)$ that holds at the monomial system and is stable under small perturbations of κ (for instance: extended Chebyshev properties, non-vanishing of all Wronskians, and positivity of their Taylor coefficients at c) is satisfied both by systems with $\text{Tu} > 1$, $I > 0$ and by systems with $\text{Tu} < 1$, $I < 0$. Such conditions therefore cannot imply the Turán inequality $\text{Tu} < 1$ or, via Lemma 7.9, the set-comparison law $I \leq 0$.

The proof is in Appendix A.7.

The scope of this obstruction is narrow, and deliberately so: it concerns sign conditions on the fixed-window five-function system that are open in the perturbation parameter. Arguments that use more structure (the exact kernel, other windows, the node set as a whole, or a strict second-order input such as concavity of $r \mapsto \kappa_r$, which degenerates to equality at the monomial system) are not excluded. A computed example with $d = 4$, $c = 10$ and $s = \pm \frac{1}{2}$ is in Numerical observation 10.14(c).

8. A SIGN PROPERTY THAT FAILS

The derivative weights μ_j of the p_1 -rule govern how p_1 moves with the nodes (Lemma 3.2(d)). A sign property of these weights would give monotonicity of p_1 in the node set. We show that it fails at and strictly inside the critical density, that it holds for two and three nodes, and that at a new node it is the front property of a companion family.

Definition 8.1 (The sign property W). Let $\mathbf{y}$ satisfy (N) $_{\mathbf{y}}$, and let μ_j ($j \leq J$) be the derivative weights of the p_1 -rule (Lemma 3.2, model data). *Property (W) at $\mathbf{y}$* : $\mu_j > 0$ and $Q''_{\mathbf{y}}(y_j) > 0$ for every $j \leq J$. *Property (W)_{top}* for the insertion of Y : $\mu_Y > 0$ and $Q''_{\mathbf{y} \cup \{Y\}}(Y) > 0$ at the new node. By Lemma 3.2(d), $\partial p_1 / \partial y_j = -Q''_{\mathbf{y}}(y_j) \mu_j$, so (W) makes p_1 strictly decreasing in each node. The class $\mathcal{W}$ consists of the node sets with $y_1 \geq 2\pi$ and $y_{j+1} - y_j \geq 2\pi$ (gaps at least 1 in the units $n = y/2\pi$).

The class $\mathcal{W}$ contains the prime powers and all sets of integers, i.e. it reaches exactly the critical density. One might expect (W) at every node set with (N) in $\mathcal{W}$: together with Corollary 7.2, (W) along a sequence of node sets would give $\Delta p_1^{(K)} > 0$ and a comparison principle in the node set. This expectation is false.

Proposition 8.2 (Counterexamples to W). *In the units $n = y/2\pi$:*

- (a) $\mathbf{y} = 2\pi \cdot (1, 2, \dots, 7)$: (N) holds and $Q''_{\mathbf{y}}(y_j) > 0$ for all j , but $\mu_7 < 0$ ($\mu_7 \approx -2.332 \cdot 10^{-18}$);
- (b) $\mathbf{y} = 2\pi \cdot (1, \dots, 10)$: $\mu_9, \mu_{10} < 0$, and $p_1^{(10)} < p_1^{(9)}$;
- (c) $\mathbf{y} = 2\pi \cdot (1, \dots, 5)$: $S_{\mathbf{y}} < 0$ ($S_{\mathbf{y}} \approx -3.478 \cdot 10^{-6}$); hence, by Theorem 7.1(2), $\Delta_{\mathbf{y}}(Y) < 0$ for all sufficiently large Y ; and $\mu_Y < 0$, $\Delta_{\mathbf{y}}(Y) < 0$ at $Y = 2\pi m$ for $m \in \{10, 15, 20, 50, 100, 1000\}$;
- (d) (strict interior) $y_j = 2\pi(1 + \varepsilon)j$, $j \leq 10$: for $\varepsilon \in \{10^{-4}, 10^{-3}\}$, $\mu_9, \mu_{10} < 0$ and $S_{\mathbf{y}_{\leq 5}} < 0$; for $\varepsilon = 1/200$, $\mu_{10} < 0$;
- (e) (shifted integers) $y_j = 2\pi(\frac{11}{10} + j)$, $0 \leq j < 50$: $\mu_{48}, \mu_{49}, \mu_{50} < 0$;
- (f) (no monotonicity in the node set) $2\pi \cdot (1, \dots, 8)$ satisfies (W), while its subset $2\pi \cdot (1, \dots, 7)$ does not.

All these node sets lie in the class $\mathcal{W}$ of Definition 8.1, and those of (d) in its interior. Hence (W) fails on $\mathcal{W}$, at and strictly inside the critical density.

Proof (computer-assisted). For each node set the verifier builds the model system with the nodes $2\pi n$ as Arb balls (real π), solves the system with $P(0) = 1$ by a rigorous ball solver (which certifies (N)), obtains the weights μ_j from the transposed system (the p_1 -rule of Lemma 3.2), $Q''_{\mathbf{y}}(y_j)$ from the solution, $S_{\mathbf{y}}$ and $\Delta_{\mathbf{y}}(Y)$ from (7.2), and accepts each sign only if the ball excludes 0; it also checks that every node set lies in the class. In (c), $S_{\mathbf{y}} \neq 0$ gives $p_{2J} \neq 0$ (since $S_{\mathbf{y}} = -p_{2J}\ell_1^*$), so Theorem 7.1 applies. Ancillary file: `toy/w/verify_w_cex.py` with `toy/w/params/w_cex.json` (1500–7000 bits; items (a)–(f) as above). $\square$

The failures at critical density do not reach the prime powers (Numerical observation 10.12), and for two and three nodes (W) holds on the whole class.

Proposition 8.3 (W for two nodes). *Let $J = 2$ and $0 < y_1 < y_2$ with either (a) $y_1 \geq \frac{157}{25}$ and $y_2 - y_1 \geq \frac{157}{25}$, or (b) $y_1 \geq 8$. Then (N) holds, $\mu_1, \mu_2 > 0$ and $Q''_{\mathbf{y}}(y_1), Q''_{\mathbf{y}}(y_2) > 0$.*

Proposition 8.4 (W for three nodes). *Let $J = 3$ and $0 < y_1 < y_2 < y_3$ with either (a) $y_1 \geq \frac{157}{25}$ and $y_{j+1} - y_j \geq \frac{157}{25}$ ($j = 1, 2$), or (b) $y_1 \geq 12$. Then (N) holds and $\mu_j > 0$, $Q''_{\mathbf{y}}(y_j) > 0$ for $j = 1, 2, 3$.*

Since $\frac{157}{25} < 2\pi$, both propositions cover the class $\mathcal{W}$; part (b) needs no gap condition.

Proofs of Propositions 8.3 and 8.4 (computer-assisted, exact). Write each region in non-negative slack variables: $y_1 = g + w_1$ and $y_k = y_{k-1} + h + w_k$ ($k \geq 2$), with $(g, h) = (\frac{157}{25}, \frac{157}{25})$ in case (a) and $(g, h) = (8, 0)$, resp. $(12, 0)$, in case (b). Two independent exact certificates are shipped. *Cramer route* (`toy/w/verify_w_small.py`): with M the matrix of $(N)_{\mathbf{y}}$, Cramer's rule gives $\mu_j = -N_j / \det M$ with explicit $N_j, \det M \in \mathbb{Z}[y_1, \dots, y_J]$, and $\det M = (y_1 \cdots y_J)^2 V^4 D_c$ with $V = \prod_{i < j} (y_i - y_j)$. After the substitution, D_c and every N_j have coefficients of one fixed sign, strictly (a non-zero constant term in case (a); in case (b) a non-zero monomial free of w_1 , positive because the gaps are positive), and the signs give $\det M \neq 0$ and $\mu_j > 0$ (Lemma B.1). Each region lies in $\text{PT}_{0.05}$, so Propositions 5.10 and 5.11 give $R_{\mathbf{y}} \in \mathbb{R}_{>0}[y]$ and hence $Q''_{\mathbf{y}}(y_j) = 2R_{\mathbf{y}}(y_j) \prod_{i \neq j} (y_j - y_i)^2 > 0$. *Insertion route* (`toy/w/verify_w_insertion.py`): every node is the inserted node of the configuration without it (Proposition 4.6 allows insertion anywhere off the nodes), and by (7.2), after clearing the base determinant, $\mu_Y = A/(\omega W_d)$ and $Q''_{\mathbf{y} \cup \{Y\}}(Y) = \omega W_{3d} / (\det M \cdot W_d)$ with explicit polynomials $A, W_d, W_{3d}, \det M$ and $\omega = \prod_i (Y - y_i)^2$; for every insertion position these four polynomials are one-signed on the region, with the signs that give (N), $\mu_Y > 0$ and $Q'' > 0$, without using Propositions 5.10 and 5.11. Both formulas are checked against a direct rational solve at one point. Parameter file `toy/w/params/w_small.json`. $\square$

The μ -half of (W) is a front property of a companion family.

Proposition 8.5 (W at a new node is companion positivity). *Let $\mathbf{y}$ satisfy $(N)_{\mathbf{y}}$, $J \geq 1$. Then $E_* = \ell_1^* E_2 - \ell_2^* E_1$ lies in $u^2 \mathbb{R}[u]$ and has zero data at $\mathbf{y}$; $R_* = Q_{E_*}/A_{\mathbf{y}}$ has degree $2J + 4$ and leading coefficient ℓ_1^* ; and for $Y \in (0, \infty) \setminus \mathbf{y}$, $(N)_{\mathbf{y} \cup \{Y\}}$ holds if and only if $\mathcal{W}(Y) \neq 0$, and then $\mu_Y = R_*(Y)/(A_{\mathbf{y}}(Y)\mathcal{W}(Y))$.*

- (a) *If $\ell_1^* \neq 0$ and $\mathcal{I} \subset (0, \infty)$ is an interval, unbounded above, on which R_* and $\mathcal{W}$ have no zeros, then for every $Y \in \mathcal{I} \setminus \mathbf{y}$, $(N)_{\mathbf{y} \cup \{Y\}}$ holds and $\text{sign } \mu_Y = -\text{sign } \ell_1^*$, which is $\text{sign}(S_{\mathbf{y}}/p_{2J})$ when $p_{2J} \neq 0$.*
- (b) *If $[u^2]E_* \neq 0$, then $E_*/[u^2]E_* = u^2 \tilde{h}$ with $\tilde{h}(0) = 1$ and $\deg \tilde{h} \leq 2J$, and $\tilde{h}(-\theta^2)[\theta^4 e^{-y}]$ has double zeros at $\mathbf{y}$: $\tilde{h}$ solves the Hermite problem at the same nodes for the companion kernel $\theta^4 e^{-y} = \tau_2(y)e^{-y}$. So “ R_* has no zeros on $\mathcal{I}$ ” is the front property on $\mathcal{I}$ of the companion family.*

- (c) For a node y_j , with all objects built from $\mathbf{y} \setminus y_j$ (assuming (N) for $\mathbf{y} \setminus y_j$ and for $\mathbf{y}$): $\mu_j = R_*^{(j)}(y_j)/(A^{(j)}(y_j)\mathcal{W}^{(j)}(y_j))$. Hence $\mu_j > 0$ for every j follows if, for each j , $S_{\mathbf{y} \setminus y_j}/p_{2J-2}(\mathbf{y} \setminus y_j) > 0$ and $R_*^{(j)}, \mathcal{W}^{(j)}$ have no zeros on $[y_j, \infty)$ (old nodes included).

The proof is in Appendix A.7.

The companion kernel $\theta^4 e^{-y}$ changes sign at the zeros $0.16575\dots, 1.34337\dots, 4.49086\dots$ of $\tau_2(y)/y$, so its front property can hold only beyond about 4.4; at $J = 1$, (W) is exactly $\tau_2(a) > 0$, i.e. $a > 4.49086\dots$ (Proposition 5.8(c)). Thus (W) splits into the companion front, (N) along the insertions, and one sign at infinity, $S_{\mathbf{y}} > 0$; the Q'' -half of (W) is the model front itself.

9. THE TRUE FAMILY: ZERO SIDE, FRONT, FAR FIELD AND TAIL

We return to the exact family $(P_J)_{J \in \mathcal{J}}$ of the true kernel (Fact 2.7) and to the four hypotheses (N), (G_a) , (Z_∞) , (Mg) of the criterion recalled in Fact 2.10. Section 9.2 derives (G_a) and (Z_∞) from root counts and from a coefficientwise step law; Section 9.3 derives (Mg) near the front $x \approx n_J$ from a step law (TEL) and a summability condition (ES); Sections 9.4–9.6 treat (Mg) away from the front, where it reduces to a finite certificate plus the summability of the scalar increments $\Delta p_1^{(K)}$, and prove the far-node asymptotics of the true kernel; Section 9.7 gives exact identities for the coefficient step. Every statement of this section is either proved for all J under explicitly named hypotheses, or a finite certificate; none of the named hypotheses is known for all J . Functions of ξ are also viewed as functions of $x = e^{2\pi\xi}$, and $I_X := [\xi_2, \xi_X]$.

9.1. Voronoi decoupling and a Gamma-only sufficient condition. The transform of $\Xi^2 H$ can be written through the Gamma-only kernel alone, at shifted points. This gives a sufficient condition for an exact magic function in which neither the zeros nor the primes appear.

Theorem 9.1 (Voronoi decoupling). *Let $H \in \mathcal{H}$, analytic on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, and put $G_H := \Xi_\infty^2 H$. Then $\widehat{G}_H$ is real-valued, and for every real ξ*

$$\widehat{\Xi^2 H}(\xi) = \sum_{m \geq 1} d(m) m^{-1/2} \widehat{G}_H(\xi + \xi_m),$$

where for every $\tau \in (\frac{1}{2}, \frac{1}{2} + \delta]$ the m -th term is bounded by $C_\tau e^{-2\pi\tau\xi} m^{-1/2-\tau}$. Moreover

$$\mathcal{A}(\Xi^2 H) = \frac{1}{2\pi} \sum_{k \geq 2} d(k) (\log k) k^{-1/2} \widehat{G}_H(\xi_k), \quad \int_{\mathbb{R}} \Xi^2 H = \sum_{k \geq 1} d(k) k^{-1/2} \widehat{G}_H(\xi_k).$$

The proof (Appendix A.1) moves the contour to $\operatorname{Im} w = -\tau$, where the series $\Xi(w)^2 = \Xi_\infty(w)^2 \sum_m d(m) m^{-1/2-iw}$ converges absolutely, integrates termwise, and moves each term back to $\mathbb{R}$.

Remark 9.2. By Theorem 9.1, $\mathcal{A}(\Xi^2 H)/\int \Xi^2 H$ is the average of $(\log k)/2\pi$ over $k \geq 1$ with weights $d(k)k^{-1/2}\widehat{G}_H(\xi_k)$. It is a mean in the strict sense when these weights are non-negative, for instance under hypothesis (a) of Proposition 9.3. The primes have disappeared: they are absorbed into the Dirichlet series $\zeta^2 = \sum d(m)m^{-s}$, and only the Gamma factor and the integers remain. The mechanism (shift to $\operatorname{Re} s > 1$, expand, shift back) is the classical one behind Koshliakov-type transformations; Theorem 9.1 packages it for the class $\Xi^2 \mathcal{H}$. For a related identity, the map $\mathcal{E}$ of Connes and Consani, see [7].

Proposition 9.3 (A Gamma-only sufficient condition). *Let $H \in \mathcal{H}$ with $H \geq 0$ on $\mathbb{R}$, and suppose that (a) $\widehat{G}_H \geq 0$ on $[\xi_2, \infty)$ (resp. on $[0, \infty)$); (b) $\widehat{G}_H(\xi_k) = 0$ for every integer $k \geq 2$; and (c) $\widehat{G}_H(0) > 0$. Then $\Xi^2 H \in \mathcal{C}$ (resp. $\mathcal{C}_{\text{OPS}}$), $\mathcal{A}(\Xi^2 H) = 0$, $\int \Xi^2 H = \widehat{G}_H(0) > 0$, and $\kappa^* \leq 0$. If moreover (d) H has finitely many real zeros and (e) $\widehat{G}_H$ has finitely many zeros in*

$[\xi_2, \infty) \setminus \{\xi_k : k \geq 2\}$, then $\widehat{\Xi^2 H}$ vanishes on $[\xi_2, \infty)$ only at the points ξ_k , $k \geq 2$, which lie in $\mathcal{M}$, and $\mathcal{K} \subseteq \{p_\zeta\}$; if in addition (E) holds, then RH holds.

Proof. For $\xi \geq \xi_2$ (resp. $\xi \geq 0$) every argument $\xi + \xi_m$ in Theorem 9.1 is $\geq \xi_2$ (resp. ≥ 0), so $\widehat{\Xi^2 H}(\xi) \geq 0$ by (a); evenness gives the same for $\xi \leq -\xi_2$ (resp. $\xi \leq 0$). Since $\Xi^2 H \geq 0$, we get $\Xi^2 H \in \mathcal{C}$ (resp. $\mathcal{C}_{\text{OPS}}$). By (b) every term of the formula for $\mathcal{A}$ in Theorem 9.1 vanishes, and only $k = 1$ survives in $\int \Xi^2 H$. Hence $\kappa^* \leq \mathcal{A}(\Xi^2 H) / \int \Xi^2 H = 0$. For the second part, let $\xi \geq \xi_2$ with $\widehat{\Xi^2 H}(\xi) = 0$. All terms of the Voronoi series are non-negative, so $\widehat{G}_H(\xi + \xi_m) = 0$ for every $m \geq 1$. By (e), for all large m the point $\xi + \xi_m$ lies beyond the finitely many exceptional zeros, so $xm \in \mathbb{Z}$ for all large m , where $x = e^{2\pi\xi}$. Hence $x = x(m+1) - xm \in \mathbb{Z}$, and $\xi = \xi_k$ with $k = x \geq 2$. These points lie in $\mathcal{M}$, since $\xi_k = \log(k^2)/4\pi$. The real zeros of $\Xi^2 H$ are Z_ζ together with the finitely many real zeros of H . Fact 2.4 applies to $F = \Xi^2 H$ and gives $\mathcal{K} \subseteq \{p_\zeta\}$, and RH under (E). $\square$

Remark 9.4. Proposition 9.3 asks for zeros of $\widehat{G}_H$ at all integers $k \geq 2$, including 6, 10, 12, $\dots$. The near-extremal functions that certify the bound on κ^* in [15] do not vanish there, so the condition is sufficient but not necessary. More generally, by Theorem 9.1 the infimum of $\mathcal{A}(\Xi^2 H) / \int \Xi^2 H$ over $H \geq 0$ with $\widehat{G}_H \geq 0$ on $[\xi_2, \infty)$ is an upper bound for κ^* that involves only the Gamma factor, the functions K_0, K_1 and the integers.

9.2. The zero side. Hypotheses (G_a) and (Z_∞) concern the polynomials P_J alone. They follow from a coefficient bound, or from information on the roots u_j of P_J (so $P_J(u) = \prod_j (1 - u/u_j)$, since $P_J(0) = 1$).

Lemma 9.5 (Left-half-plane roots are harmless). *If $\operatorname{Re} u \leq 0$, then $|1 - t^2/u| \geq 1$ for every real t ; if $\operatorname{Re} u > 0$, then $|1 - t^2/u| \geq |\sin \arg u|$. Hence, if P_J has real coefficients, $|H_J(t)| \geq \prod |\sin^2(\arg u_j)|$ on $\mathbb{R}$, the product running over the pairs $\{u_j, \bar{u}_j\}$ of roots with $\operatorname{Re} u_j > 0$ and $\operatorname{Im} u_j \neq 0$, each pair counted once, provided P_J has no root in $(0, \infty)$. In that case H_J is real and has no zero on $\mathbb{R}$, and $H_J(0) = 1$, so the bound holds for $H_J = |H_J|$ itself. (If P_J has a simple root $u_0 > 0$, then H_J changes sign at $\sqrt{u_0}$, and no positive lower bound holds.) In particular, if for $J \geq J_0$ the polynomial P_J has no root in $(0, \infty)$ and at most K root pairs in $\operatorname{Re} u > 0$, all with $|\arg u| \geq \theta_0 > 0$, then $H_J \geq (\sin \theta_0)^{2K}$ on $\mathbb{R}$, which is (Z^+) with a constant h .*

Proof. If $\operatorname{Re} u \leq 0$ then $\operatorname{Re}(1/u) \leq 0$ and $\operatorname{Re}(1 - t^2/u) \geq 1$. If $\operatorname{Re} u > 0$ then $|u - t^2| \geq |\operatorname{Im} u| = |u| |\sin \arg u|$. For a conjugate pair, $|1 - t^2/u| |1 - t^2/\bar{u}| = |1 - t^2/u|^2$. Multiplying over all roots bounds $|H_J(t)|$ from below. If P_J has no root in $(0, \infty)$, then $H_J(t) = P_J(t^2) \neq 0$ for real t , so H_J has constant sign on $\mathbb{R}$, and $H_J(0) = P_J(0) = 1 > 0$. $\square$

Lemma 9.6 (Root counting gives growth). *Let $P(u) = \prod_{j \leq d} (1 - u/u_j)$, $H(t) = P(t^2)$ and $N(r) := \#\{j : |u_j| \leq r^2\}$. If $N(r) \leq Ar + n_0$ for all $r > 0$ and $N(r) = 0$ for $r < r_0$, then $|H(t)| \leq (1 + |t|^2/r_0^2)^{n_0} e^{\pi A|t|}$ for all $t \in \mathbb{C}$. Hence (G_a) holds on every strip, for every $a > \pi A$, if the counting bound holds uniformly in J with $A < \frac{1}{2}$, and either $n_0 = 0$ or there is in addition a uniform root-free disc $|u| < r_0^2$.*

Proof. $|1 - t^2/u_j| \leq 1 + \rho^2/|u_j|$ with $\rho = |t|$, and integration by parts gives $\sum_j \log(1 + \rho^2/|u_j|) = \int_0^\infty N(r) \frac{2\rho^2}{r(r^2 + \rho^2)} dr \leq \pi A \rho + n_0 \log(1 + \rho^2/r_0^2)$. $\square$

Lemma 9.7 (Strip bounds from a sector). *Let P, H be as in Lemma 9.6, with real coefficients, every root satisfying $|\arg u_j| \geq \theta_0 \in (0, \pi/2]$, and put $\beta := \sum_j 1/|u_j|$. (a) For real T, η : $\log|H(T + i\eta)| \leq \log|H(T)| + (2|T||\eta| + \eta^2)\beta/\sin \theta_0$. So a real-line bound with exponent a , together with a uniform sector and a uniform bound on β , gives $(G_{a'})$ on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$ with $a' = a + (1 +$*

$2\delta)/\sin \theta_0$. (b) If moreover $|u_j| \geq r_0^2$ for all j , every zero t of H has $|\operatorname{Im} t| \geq r_0 \sin(\theta_0/2)$; so a uniform sector and disc give (R_η) .

Proof. (a) $|u_j - (T + i\eta)^2| \leq |u_j - T^2| + 2|T||\eta| + \eta^2$ and $|u_j - T^2| \geq \operatorname{dist}(u_j, [0, \infty)) \geq |u_j| \sin \theta_0$; use $\log(1 + s) \leq s$. (b) The zeros are $t = \pm u_j^{1/2}$, with $|\arg t| \in [\theta_0/2, \pi - \theta_0/2]$. $\square$

Proposition 9.8 (Limits under linear root counting). *Assume (N) and, uniformly in $J \in \mathcal{J}$, the counting bound $N_J(r) \leq Ar + n_0$ with $A < \frac{1}{2}$ (with a uniform root-free disc $|u| < r_0^2$ if $n_0 > 0$). Then $\{P_J\}$ is locally bounded on $\mathbb{C}$, and every subsequence has a further subsequence with $P_J \rightarrow P_\infty$ locally uniformly, where P_∞ is entire of order $\leq \frac{1}{2}$, $P_\infty(0) = 1$, and $P_\infty(u) = \prod_k (1 - u/u_k^\infty)$; the u_k^∞ are exactly the limits of bounded sequences of roots of P_J . Hence (Z_∞) holds along this subsequence if and only if P_∞ has no zero in $(0, \infty)$, and in that case $H_\infty(t) = P_\infty(t^2) \geq \prod_{\operatorname{Re} u_k^\infty > 0} \sin^2(\arg u_k^\infty)$ on $\mathbb{R}$.*

Proof. By Lemma 9.6 with $t^2 = u$, $|P_J(u)| \leq (1 + |u|/r_0^2)^{n_0} e^{\pi A|u|^{1/2}}$, so Montel applies on $\mathbb{C}$. The limit has order $\leq \frac{1}{2}$, hence genus 0, and Hadamard's theorem gives the product; by Hurwitz's theorem its zeros are the limits of roots of P_J [8, VII.2.5]. The last statement is Lemma 9.5 for P_∞ . $\square$

The roots of the exact members are certified at three values of J .

Proposition 9.9 (Certified root geometry). *Let $J \in \{60, 110, 111\}$, and write $P_J(u) = \prod_j (1 - u/u_j)$. Then P_J has exactly three pairs of conjugate roots in $\operatorname{Re} u > 0$; for these six roots, $|\arg u|$ lies in $[1.0351074, 1.4522567]$ for $J = 60$, in $[1.0351421, 1.4522859]$ for $J = 110$ and in $[1.0351423, 1.4522861]$ for $J = 111$, so in $[1.0351, 1.4523]$ in all three cases, and P_J has no root in $(0, \infty)$. Every root satisfies $|\arg u| \geq 1.0351$ and $|u| \geq 108.25$; $\sum_j 1/|u_j| \leq 0.0581684$; every zero of $H_J(t) = P_J(t^2)$ satisfies $|\operatorname{Im} t| \geq 9.6386$; and $N_J(r) := \#\{j : |u_j| \leq r^2\} \leq 0.41658r$ for all $r > 0$.*

Proof (computer-assisted). The script `exact/verify_roots.py` reads the certified enclosure of P_J (Fact 2.13) and isolates all $2J$ roots by validated disjoint balls, one root in each, valid for every polynomial in the coefficient balls, in particular for the exact P_J ; it certifies the sign of $\operatorname{Re} u$ for every root and prints outward-rounded bounds for $|\arg u|$ over the roots with $\operatorname{Re} u > 0$, for $\min|\arg u|$, $\min|u|$, $\sum 1/|u|$, $\min|\operatorname{Im} t|$ over the zeros of H_J , and $\sup_r N_J(r)/r$. The printed values are $\min|u| \geq 108.25343, 108.25821, 108.25825$; $\sum 1/|u| \leq 0.05816834, 0.05815321, 0.05815310$; $\min|\operatorname{Im} t| \geq 9.6386206, 9.6388900, 9.6388919$; and $\sup_r N_J(r)/r \leq 0.41658, 0.41623, 0.41623$ for $J = 60, 110, 111$. Command, for instance: `python exact/verify_roots.py exact/out/P60_ball.txt --prec 2000`. $\square$

These root data alone give finite- J instances of the hypotheses (Z^+) , (G_a) and (R_η) . At these J , $H_J \geq (\sin 1.0351)^6$ on $\mathbb{R}$ by Lemma 9.5, and $|H_J(t)| \leq e^{0.41658\pi|t|}$ on $\mathbb{C}$ by Lemma 9.6 with $n_0 = 0$ (note $0.41658\pi < \pi/2$). For the zeros of H_J , Lemma 9.7(b) gives only $|\operatorname{Im} t| \geq 108.25^{1/2} \sin(1.0351/2) > 5.14$; the distance 9.6386 is certified directly. The coefficient bounds of Fact 2.13 give more at these J , namely $H_J \geq 1$ and $|H_J(t)| \leq \cosh(0.71636|t|)$; the point of the bounds above is that they come from the roots alone. Uniform versions in J would give (Z_∞) and (G_a) ; none is known.

The next theorem reduces (POS_a) to a local step law. Inequalities between polynomials are coefficientwise.

Theorem 9.10 (Coefficient positivity from a sandwich step). (a) (Growth.) *Let $J_0 \geq 1$ and suppose, for every $J \geq J_0$: (G1) P_J has non-negative coefficients; (G2) $P_{J+1} \leq P_J \cdot (1 + c_{1,J}u + c_{2,J}u^2)$ with $c_{1,J}, c_{2,J} \geq 0$; (G3) with $d_J := \max(c_{1,J}/2, c_{2,J}^{1/2}) \leq D$, the count $\#\{J \geq$*

$J_0 : d_J \geq 1/R\}$ is at most $A\sqrt{R} + B$ ($A, B \geq 0$). Then $H_J(t) \leq H_{J_0}(t)(1 + Dt^2)^{2B}e^{2\pi A|t|}$ on $\mathbb{R}$, and (POS_a) holds for every $a > 2\pi A$ (for every $a > 0$ if the count is $o(\sqrt{R})$).

- (b) (Positivity.) Let $J_0 \geq 1$, assume $(N)_J$ for every $J \geq J_0$, and assume that P_{J_0} has positive coefficients. Suppose that for every $J \geq J_0$ there is a root pair $u_J, \bar{u}_J$ of P_{J+1} with $\text{Re } u_J \leq 0$ (a non-real conjugate pair, or a negative real double root) such that $P_J^\natural := P_{J+1}/|1 - u/u_J|^2$ satisfies (LR-step): $p_k^\natural/p_k^{(J)}$ is positive and non-increasing in k ($0 \leq k \leq 2J$). Then $P_J > 0$ for all $J \geq J_0$, $0 < P_J^\natural \leq P_J$, and $P_{J+1} \leq P_J|1 - u/u_J|^2$. The same conclusions hold, without the base case, if (LR-step) is replaced by (SW-step): $0 < p_k^\natural \leq p_k^{(J)}$ ($0 \leq k \leq 2J$). If moreover $(\text{CNT}_1) \#\{J \geq J_0 : |u_J| \leq R\} \leq A\sqrt{R} + B$, then (POS_a) holds for every $a > 2\pi A$; under $o(\sqrt{R})$ it holds for every $a > 0$, and every limit H_∞ has exponential type 0.

Here $|1 - u/u_J|^2$ denotes the polynomial $(1 - u/u_J)(1 - u/\bar{u}_J) = 1 - 2\text{Re}(1/u_J)u + |u_J|^{-2}u^2$.

The proof is in Appendix A.5.

Corollary 9.11 (The zero side from one step law). Assume $(N)_J$ for every $J \geq 10$, that P_{10} has positive coefficients, that (LR-step) or (SW-step) holds for every $J \geq 10$, and that (CNT_1) holds with $A < \frac{1}{4}$. Then (POS_a) holds with some $a < \pi/2$; hence, by Fact 2.12, (G_a) holds on every strip, $H_J \geq 1$ on $\mathbb{R}$, and (Z_∞) holds. Positivity of the coefficients of P_{10} and $(N)_{10}$ are computer-assisted facts (Fact 2.13).

Proof. Theorem 9.10(b) with $J_0 = 10$, and $(2\pi A, \pi/2) \neq \emptyset$ since $A < \frac{1}{4}$. $\square$

9.3. The front: a step law. Hypothesis (Mg) of Fact 2.10 asks for positivity of $\liminf_J \hat{F}_J$ at every point. On a fixed compact range of x it reduces to a summable tail of increments (Section 9.4). Near the front $x \approx n_J$, however, the relative decrements $(\hat{F}_J - \hat{F}_{J+1})/\hat{F}_J$ are close to 1, and a statement uniform in J is needed there. We formulate it as a step law. In this subsection $\hat{F}_J$ is viewed as a function of $x = e^{2\pi\xi}$, derivatives are taken in x , and $\hat{F}_0 := \Psi$.

Hypothesis 9.12 (Step law). (TEL_J) : $\hat{F}_J(x) \geq (1 - x/n_J)^2 \hat{F}_{J-1}(x)$ for every $x \geq 1$. Equivalently, with the bottom-normalised cofactor $\mathcal{G}_J := \hat{F}_J / \prod_{j \leq J} (1 - x/n_j)^2$: $\mathcal{G}_J \geq \mathcal{G}_{J-1}$ on $[1, \infty)$. (EVEN-TEL_J) : $\hat{F}_J(x) \geq (1 - x^2/n_J^2)^2 \hat{F}_{J-1}(x)$ for every $x \geq 1$.

Hypothesis 9.13 (Erosion summability). (ES) : for every $x \in [1, \infty) \setminus \text{PP}$, $\hat{F}_K(x) > 0$ for all sufficiently large K , and $\sum_K \max(\varepsilon_K(x), 0) < \infty$, where $\varepsilon_K(x) := \log(\hat{F}_{K-1}(x)/\hat{F}_K(x))$ and the sum runs over those K .

Hypothesis 9.14 (Base). $(B^+)_{J_0}$: $(N)_{J_0}$ holds; $\hat{F}_{J_0}(x) > 0$ for every $x \in [1, \infty) \setminus \{n_1, \dots, n_{J_0}\}$, in particular at the later prime powers n_K , $K > J_0$; and $\hat{F}_{J_0}'' > 0$ at each of its nodes.

Parts of $(B^+)_{60}$ and $(B^+)_{10}$ are certified (Proposition 9.19), but not on $[1, 2)$ nor on the short intervals $(199, 199.39)$ and $(16, 16.45)$; so $(B^+)_{J_0}$ remains a hypothesis below.

Theorem 9.15 (The front from the step law). Let $J_0 \geq 1$. Assume $(N)_J$ for every $J \geq J_0$, $(B^+)_{J_0}$, and (TEL_J) for every $J > J_0$. Then for every $J \geq J_0$ and $x \geq 1$,

$$\hat{F}_J(x) \geq \hat{F}_{J_0}(x) \prod_{J_0 < K \leq J} (1 - x/n_K)^2.$$

Hence: (i) $\hat{F}_J > 0$ on $[1, \infty) \setminus \{n_1, \dots, n_J\}$ (the front statement (FR), globally); (ii) $\hat{F}_J''(n_K) \geq (1 - n_K/n_J)^2 \hat{F}_{J-1}''(n_K) > 0$ for $K < J$, and $\frac{1}{2} \hat{F}_J''(n_J) \geq \hat{F}_{J-1}(n_J)/n_J^2 > 0$; (iii) $p_{2J}^{(J)} \geq 0$, and $p_{2J}^{(J)} > 0$ if and only if the doubled chain of $\mathbf{y}_J^{\text{PP}}$ is regular; (iv) that chain is regular

if the doubled chain of $\mathbf{y}_{J-1}^{\text{PP}}$ is regular and the half-step interpolant (double at $n_1, \dots, n_{J-1}$, simple at n_J) has a simple zero at n_J . If (ES) holds as well, then $\liminf_J \hat{F}_J(x) > 0$ for every $x \in [1, \infty) \setminus \text{PP}$ (pointwise (Mg)). If (EVEN-TEL_J) holds for every $J > J_0$ in place of (TEL_J), then $\hat{F}_J(x) \geq \hat{F}_{J_0}(x) \prod_{J_0 < K \leq J} (1 - x^2/n_K^2)^2$, and pointwise (Mg) holds without (ES).

Proof. The product bound follows by induction, multiplying (TEL_J) by the inductive bound for $J - 1$. (i) follows from $(B^+)_{J_0}$. (ii) Both $\hat{F}_J$ and $\hat{F}_{J-1}$ vanish doubly at n_K ($K < J$); divide (TEL_J) by $(x - n_K)^2$ and let $x \rightarrow n_K$. Divided by $(x - n_J)^2$, (TEL_J) gives the second inequality, and $\hat{F}_{J-1}(n_J) > 0$ by (i). The positivity of $\hat{F}_{J-1}''(n_K)$ follows by induction from $(B^+)_{J_0}$ and the second inequality. (iii) $\hat{F}_J = \sum_k p_k^{(J)} \varphi_k$ is positive near $+\infty$ by (i), and if $p_{2J}^{(J)} \neq 0$ the term $p_{2J}^{(J)} \varphi_{2J}$ dominates there with $\varphi_{2J} > 0$ (fact (e) after Definition 3.1); so $p_{2J}^{(J)} \geq 0$. The equivalence is fact (d) there. (iv) is Lemma 4.4, since $\hat{F}_{J-1}(n_J) > 0$ by (i). Under (ES): for $x \notin \text{PP}$ the values $\hat{F}_K(x)$ are positive for $K \geq J_0$ by (i), and $\log \hat{F}_J(x) \geq \log \hat{F}_{K_x}(x) - \sum_{K > K_x} \max(\varepsilon_K(x), 0)$ for $J > K_x := \max(J_0, J(x))$, where $J(x)$ is the least J with $n_J > x$. Under (EVEN-TEL) the same induction gives the second product, which converges, as $J \rightarrow \infty$, to a positive number for $x \notin \text{PP}$, because $\sum_K n_K^{-2} < \infty$. $\square$

Lemma 9.16 (Positivity implies non-singularity). *Let $\mathbf{y}$ be a node set for the true family whose doubled chain Z is regular. If the top-normalised interpolant $\mathcal{F}_Z$, viewed as a function of ξ , is ≥ 0 for $\xi \geq 0$ (that is $x \geq 1$) and not identically zero there, then its φ_0 -coefficient $c_0 := \ell_0(\mathcal{F}_Z)$ is positive. Hence (N) _{$\mathbf{y}$} holds and $p_{2J}(\mathbf{y}) = 1/c_0 > 0$.*

Proof. $\mathcal{F}_Z = \hat{F}$ with $F = \Xi^2 P(t^2)$, where P has constant term c_0 . By Fourier inversion and evenness, $\Xi(0)^2 c_0 = F(0) = \int_{\mathbb{R}} \hat{F} = 2 \int_0^\infty \hat{F} > 0$. Then fact (d) after Definition 3.1 gives (N) _{$\mathbf{y}$} and $P_{\mathbf{y}} \leftrightarrow \mathcal{F}_Z/c_0$, whose top coefficient is $1/c_0$. $\square$

This is the true-kernel analogue of a fact about the model, where positivity of the cofactor's coefficients gives $p_{2J} > 0$ (Section 5); there it is a point value, here an integral.

By Lemma 9.16, hypothesis (N)_J can be traded for regularity of the doubled chains (item (iv)) together with (TEL) in the top normalisation.

Lemma 9.17 (Positivity beyond the front from the Abel preimage). *Let P be a polynomial and Q_P^{Ab} its Abel preimage (Fact 2.8). If the leading coefficient of Q_P^{Ab} is positive and Q_P^{Ab} has no root in (y_0, ∞) , then $\widehat{\Xi^2 P(t^2)} > 0$ at every $x \geq y_0/2\pi$.*

Proof. $Q_P^{\text{Ab}} > 0$ on (y_0, ∞) , so $g_P(z) > 0$ for $z \geq y_0$ by the integral formula of Fact 2.8. In $\widehat{\Xi^2 P(t^2)}(\xi) = 2\pi\sqrt{x} \sum_N d(N) g_P(2\pi Nx)$ every argument satisfies $2\pi Nx \geq 2\pi x \geq y_0$. $\square$

A sufficient check is that all Taylor coefficients of $Q_P^{\text{Ab}}(y_0 + v)$, as a polynomial in v , are positive (Lemma B.1); certified instances are in Proposition 9.19.

We also record an exact identity for the Abel preimage of Fact 2.8, the true-kernel analogue of the trace identity of the model (Lemma 5.7).

Lemma 9.18 (Trace of the Abel preimage). *For every polynomial P of exact degree $2J$, the $4J + 4$ roots of Q_P^{Ab} , counted with multiplicity, sum to $8(J + 1)^2$.*

Proof. Write $\Omega(\theta) = P(-(\theta + \frac{1}{2})^2) \theta^2 (\theta + 1)^2 = \sum_m \omega_m \theta^m$, of degree $M = 4J + 4$. The term $p_{2J}(-(\theta + \frac{1}{2})^2)^{2J} = p_{2J}(\theta^{4J} + 2J\theta^{4J-1} + \dots)$ is the only one contributing to the top two coefficients of $P(-(\theta + \frac{1}{2})^2)$, and $\theta^2(\theta + 1)^2 = \theta^4 + 2\theta^3 + \theta^2$, so $\omega_{M-1}/\omega_M = 2J + 2$. Since $e^y \theta^m e^{-y} = T_m(-y) = (-y)^m + \binom{m}{2}(-y)^{m-1} + \dots$, the sum of the roots of Q_P^{Ab} is $\binom{M}{2} + \omega_{M-1}/\omega_M = 2(J + 1)(4J + 3) + 2(J + 1) = 8(J + 1)^2$. $\square$

In the model the corresponding sum is $2J(4J-1)$. The computed root structure of the Abel preimages along the prime powers is in Numerical observation 10.7.

Proposition 9.19 (Certified base data). *Let $d(x) := \text{dist}(x, \text{PP})$.*

- (a) $\hat{F}_{60}(x) > 0$ for all $x \geq 199.39$, and $\hat{F}_{10}(x) > 0$ for all $x \geq 16.45$.
- (b) For $(J_1, X, \theta) = (10, 16, 0.786684)$, $(60, 199, 0.787979)$ and $(110, 59, 0.787979)$: $\hat{F}_{J_1}(\xi) \geq \theta \Psi_1(\xi) \min(1, d(x)^2)$ on $[\xi_2, \xi_X]$, and $\hat{F}_{J_1}'' > 0$ at every node in $[2, X]$.
- (c) The ratio $|\hat{F}_{J_1+1} - \hat{F}_{J_1}| / (|\Delta p_1^{(J_1)}| \hat{F}_{J_1})$ is bounded on $[2, X] \setminus \text{PP}$ by $R_1(X)$, where $R_1(31) = 1.021 \cdot 10^9$ for $J_1 = 60$, and $R_1(31) = 1.037 \cdot 10^9$, $R_1(43) = 2.701 \cdot 10^9$, $R_1(47) = 3.476 \cdot 10^9$, $R_1(53) = 4.844 \cdot 10^9$, $R_1(59) = 6.474 \cdot 10^9$ for $J_1 = 110$.

Proof (computer-assisted). All three parts use the certified enclosures of P_J of Fact 2.13 (computed in [15]), and hold for every polynomial in those enclosures, in particular for the exact P_J . (a) By Lemma 9.17, it suffices that all Taylor coefficients of $Q_P^{\text{Ab}}(2\pi x_0 + v)$ are positive balls (Lemma B.1); `exact/verify_bf.py` checks this at 4600 bits for $x_0 = 199.39$ (245 coefficients) and at 1000 bits for $x_0 = 16.45$ (45 coefficients), and prints negative controls at $x_0 = 199.35$ and 16.44 , where the check fails. (b) `exact/verify_base.py` certifies $\hat{F}'' > 0$ at each node by a Taylor model centred at the node, with the exactly known value and slope 0 there, and the lower bound on each gap by Taylor models on subintervals (6313, 484 and 142 pieces for $J_1 = 60, 110, 10$). (c) `exact/verify_r1.py` bounds the ratio on each gap by Taylor models of numerator and denominator, and on punctured neighbourhoods of the nodes through the limit value of Lemma 3.3(c); the certified suprema are $1.02023 \cdot 10^9$ ($J_1 = 60$, $X = 31$) and $1.03543, 2.68925, 3.46048, 4.79036, 6.45773$ ($\times 10^9$) for $J_1 = 110$. Commands, for instance: `python exact/verify_base.py exact/out/P60_ball.txt --X 199 --theta 0.787979 --prec 1800` and `python exact/verify_r1.py exact/out/P60_ball.txt exact/out/P61_ball.txt --target 31:1.021e9 --prec 1800`. $\square$

By (a) and (b), hypothesis $(B^+)_{60}$ of Theorem 9.15 holds except possibly on $[1, 2)$ and on $(199, 199.39)$, and $(B^+)_{10}$ except possibly on $[1, 2)$ and $(16, 16.45)$. These short intervals are not certified, so $(B^+)_{J_0}$ remains a hypothesis in Corollary 9.21.

The same certified enclosures of P_{60} , P_{61} , P_{110} and P_{111} determine the first increments of the family.

Proposition 9.20 (Certified increments at $J = 60$ and $J = 110$). *Write $\Delta p_1^{(J)} := p_1^{(J+1)} - p_1^{(J)}$. Then $\Delta p_1^{(60)} = 2.08079540774023 \cdot 10^{-11} \pm 3.6 \cdot 10^{-26}$ and $\Delta p_1^{(110)} = 2.91831836483267 \cdot 10^{-12} \pm 2.2 \cdot 10^{-27}$. For $J \in \{60, 110\}$ the increment $p_k^{(J+1)} - p_k^{(J)}$ is negative exactly for $k \in \{2, 4, 6, 10, 11\}$ and positive for every other $k \leq 2J$.*

Proof (computer-assisted). The enclosures of Fact 2.13 contain the exact coefficients of P_J and P_{J+1} . The increments $\Delta p_1^{(J)}$ are printed by `exact/verify_r1.py` (the same runs as in Proposition 9.19(c)), and the signs of all increments are certified by `exact/verify_increments.py` from the enclosures at J and $J+1$ (a sign is accepted only if the ball of the difference excludes 0). Command, for instance: `python exact/verify_increments.py exact/out/P60_ball.txt exact/out/P61_ball.txt --expect-negative 2,4,6,10,11`. $\square$

Corollary 9.21 (What remains for (S) and (U)). *Assume $(N)_J$ for every $J \geq J_0$, (POS_a) for all large J with some $a < \pi/2$, $(B^+)_{J_0}$, (TEL_J) for every $J > J_0$, and (ES). Then (S) and (U) hold, and $\text{RH} \Leftrightarrow (\text{E})$.*

Proof. By Theorem 9.15, pointwise (Mg) holds on $[1, \infty) \setminus \text{PP} \supseteq [2, \infty) \setminus \text{PP}$. By Fact 2.12, (POS_a) gives (G_a) and (Z^+) with $h \equiv 1$; and $\mathcal{J} \supseteq [J_0, \infty)$ is infinite. Corollary 2.11(M⁺) applies. $\square$

None of the five hypotheses of Corollary 9.21 is known for all J , and they are not all localised at the front: (POS_a) constrains every coefficient of P_J , (ES) concerns every $x \notin \text{PP}$, and $(\text{N})_J$ is a global non-singularity statement. Only the step law (TEL_J) is a statement about the front $x \approx n_J$, where the relative decrements are close to 1; it is there that the problem is hardest. The certified finite- J data are in Fact 2.13 and Propositions 9.19 and 9.20. The base J_0 is free (for instance $J_0 = 60$), and with (EVEN-TEL) in place of (TEL) , (ES) can be dropped. Heuristically, (ES) should follow from the summability of the scalar tail $\sum_K |\Delta p_1^{(K)}|$ via the rank-one structure of the increments (Numerical observation 10.4); deriving it that way, however, requires sign control of the increments of the kind discussed in Section 9.6.

Remark 9.22 (EVEN-TEL and the geometric form of TEL). (a) (EVEN-TEL_J) is stronger than (TEL_J) and, with $(\text{B}^+)_{J_0}$, gives pointwise (Mg) without (ES) (Theorem 9.15); on the computed data its margin shrinks (Numerical observation 10.14(e)), so we do not conjecture it. (b) The following is proved, for any base kernel. Assume (N) at $J-1$ and J and $\hat{F}_{J-1}(n_J) \neq 0$, let $\mathcal{F}_{E_1}, \mathcal{F}_{E_2}$ be the extension elements of $\mathbf{y}_{J-1}^{\text{PP}}$, and put $\gamma(x) := (\mathcal{F}_{E_1}/\hat{F}_{J-1}, \mathcal{F}_{E_2}/\hat{F}_{J-1})(x) \in \mathbb{R}^2$ where $\hat{F}_{J-1}(x) \neq 0$. Then, with $Y = n_J$,

$$\frac{\hat{F}_J(x)}{\hat{F}_{J-1}(x)} = \frac{\det(\gamma(x) - \gamma(Y), \gamma'(Y))}{\det(-\gamma(Y), \gamma'(Y))}.$$

Indeed $\hat{F}_J = \hat{F}_{J-1} + a\mathcal{F}_{E_1} + b\mathcal{F}_{E_2}$ (Lemma 3.4), so $\hat{F}_J/\hat{F}_{J-1} = 1 + (a, b) \cdot \gamma$; the two conditions at Y say $(a, b) \cdot \gamma'(Y) = 0$ and $(a, b) \cdot \gamma(Y) = -1$, and the determinant of this system is $\det(\gamma(Y), \gamma'(Y)) = \mathcal{W}(Y)/\hat{F}_{J-1}(Y)^2 \neq 0$. Under $(\text{FR})_{J-1}$ and $\hat{F}_{J-1}'' \neq 0$ at the nodes, γ is a smooth curve on $[1, \infty)$ (Lemma 3.3(c)), and (FR) at J says that γ stays on the same side of its tangent line at $\gamma(Y)$ as the origin, while (TEL_J) says that the ratio of the two distances to that line is at least $(1 - x/Y)^2$.

9.4. The increment identity and a reduction on compacts. Write $D_K := \hat{F}_{K+1} - \hat{F}_K$ for the increments of the exact family (Lemma 3.3).

Theorem 9.23 (Tail-inward reduction). *Let $X > 2$, let $w \geq 0$ be a function on I_X , and let J_1 be such that every prime power $\leq X$ is among $n_1, \dots, n_{J_1}$. Assume $(\text{N})_J$ for all $J \geq J_1$ and (a) $\hat{F}_{J_1} \geq \mu w$ on I_X for some $\mu > 0$; (b) for every $K \geq J_1$ there is $\varepsilon_K \in [0, 1)$ with $D_K \geq -\varepsilon_K \hat{F}_K$ on I_X ; (c) $\sum_K \varepsilon_K < \infty$. Then $\hat{F}_J \geq \mu \prod_{J_1 \leq K < J} (1 - \varepsilon_K) w \geq \mu \prod_{K \geq J_1} (1 - \varepsilon_K) w$ on I_X for every $J \geq J_1$. With $w = \Psi_1 \varrho$, where $\varrho(x) := ((x - n_i)(n_{i+1} - x)/(n_{i+1} - n_i))^2$ on each gap (n_i, n_{i+1}) , one has $d^2/4 \leq \varrho \leq d^2$ with $d = \text{dist}(x, \text{PP})$, and the conclusion is a quantitative form of (Mg) on $[2, X]$.*

Proof. By induction, $\hat{F}_{K+1} = \hat{F}_K + D_K \geq (1 - \varepsilon_K) \hat{F}_K$, and $\hat{F}_K \geq 0$ on I_X by the inductive bound. On a gap of length L , $(x - n_i)(n_{i+1} - x)/L = d(L - d)/L$ with $(L - d)/L \in [\frac{1}{2}, 1]$. $\square$

Theorem 9.23 is a statement on one compact interval. Since (Mg) is pointwise, it follows on all of $[\xi_2, \infty)$ by exhaustion once the hypotheses of Theorem 9.23 hold for every finite X , with μ, w and the starting index J_1 allowed to depend on X ; one fixed X , such as a computed one, does not suffice. Near the front the relative decrements are close to 1, which makes (b) and (c) hard to verify unless $J_1(X)$ is taken well beyond the prime powers up to X , with the base (a) at that index. A theorem uniform in the front, such as Theorem 9.15, is one way to supply these bases for every X at once; it is not logically necessary. Hypothesis (b) is a pointwise inequality without division, so each single instance is certifiable.

Theorem 9.24 (Far-field reduction on compacts). *Let X, I_X and J_1 be as in Theorem 9.23, and assume $(\text{N})_K$ for $K \geq J_1$ and:*

- (B) $\widehat{F}_{J_1} > 0$ on I_X off the nodes, and $\widehat{F}_{J_1}'' > 0$ at the nodes in I_X ;
- (Φ) there is $R < \infty$ with $|D_K| \leq R|\Delta p_1^{(K)}|\widehat{F}_{J_1}$ on I_X for all $K \geq J_1$;
- (p) $\delta := \sum_{K \geq J_1} |\Delta p_1^{(K)}| < 1/R$.

Then $\widehat{F}_J \geq (1 - R\delta)\widehat{F}_{J_1}$ on I_X for all $J \geq J_1$. Hence (Mg) holds on $[2, X]$, quantitatively with $\theta(X) := (1 - R\delta) \min_{I_X} \widehat{F}_{J_1} / (\Psi_1 \min(1, d^2)) > 0$; the minimum exists, because at a node the quotient tends to $\widehat{F}_{J_1}'' / (2\Psi_1(dx/d\xi)^2) > 0$. If (N) fails for some K , the same holds along $\mathcal{J}$, with differences taken between consecutive members.

Proof. $|\widehat{F}_J - \widehat{F}_{J_1}| \leq \sum_{J_1 \leq K < J} |D_K| \leq R\delta \widehat{F}_{J_1}$ pointwise on I_X . The statement about $\theta(X)$ follows from (B) and Lemma 3.3(c). $\square$

Writing $D_K = \Delta p_1^{(K)} \Phi_K$ with $\Phi_K := \widehat{\Xi^2 q_K}$, $q_K(t) := (P_{K+1} - P_K)(t^2) / \Delta p_1^{(K)}$ (so $q_K(0) = 0$ and $\frac{1}{2}q_K''(0) = 1$), hypothesis (Φ) is a uniform bound on the normalised profiles Φ_K , and (p) is a single scalar series.

Corollary 9.25 (Finite part plus two infinite inputs). *Let X , I_X and J_1 be as in Theorem 9.23, assume (N) $_K$ for $K \geq J_1$, and let $R_1 \geq \sup_{I_X \setminus \text{PP}} |D_{J_1}| / (|\Delta p_1^{(J_1)}|\widehat{F}_{J_1})$. Assume (B) as in Theorem 9.24 and the two infinite hypotheses*

- (Φ -mono) $|D_K| / |\Delta p_1^{(K)}| \leq (1 + \eta)|D_{J_1}| / |\Delta p_1^{(J_1)}|$ on I_X for all $K \geq J_1$, for some $\eta \geq 0$;
- (p) $(1 + \eta)R_1\delta < 1$, where $\delta := \sum_{K \geq J_1} |\Delta p_1^{(K)}|$.

Then $\widehat{F}_J \geq (1 - (1 + \eta)R_1\delta)\widehat{F}_{J_1}$ on I_X for all $J \geq J_1$; hence (Mg) holds on $[2, X]$. By Proposition 9.19, (B) and R_1 are certified for the exact members with $(J_1, X) = (60, 31)$, $R_1 = 1.021 \cdot 10^9$, and $(J_1, X) = (110, 53)$, $R_1 = 4.844 \cdot 10^9$.

Proof. (Φ -mono) and the definition of R_1 give (Φ) with $R = (1 + \eta)R_1$. $\square$

The hypotheses (Φ -mono) and (p) concern all $K \geq J_1$ and are open; computed values are in Numerical observation 10.14(d). Theorem 9.28 below replaces the extrapolation used there by a conditional bound.

9.5. The far field. Let $\tilde{\kappa}(y) := \sqrt{2y/\pi} e^y K_0(y)$; it is increasing, with values in $(0, 1)$, and $\tilde{\kappa}(y) = 1 - 1/(8y) + O(y^{-2})$ (Proposition 5.5(a)). For a polynomial P let $P^\Xi(u) = P(u)(u + \frac{1}{4})^2$ be as in Fact 2.8, and let $Q_{P^\Xi}(y) := e^y P^\Xi(-\theta^2) e^{-y}$ be its transform in the model (Definition 5.1).

Lemma 9.26 (Moment asymptotics). *For each fixed polynomial P , as $y \rightarrow \infty$,*

$$\frac{e^y}{\pi} \widehat{\Xi^2 P(t^2)} = \tilde{\kappa}(y) Q_{P^\Xi}(y) + O(y^{2 \deg P^\Xi - 2}),$$

where the left side is evaluated at $x = y/2\pi$; the expansion may be differentiated termwise. In particular, for $P = u^k$,

$$\frac{e^y}{\pi} \widehat{t^{2k} \Xi^2} = (-1)^k y^{2k+4} [1 - \gamma_k/y + O_k(y^{-2})], \quad \gamma_k := \binom{2k+4}{2} + \frac{1}{8}.$$

The divisor terms $N \geq 2$ contribute $O(e^{-y} \text{poly}(y))$ relative to the term $N = 1$, with all derivatives, and the same holds for the Gamma-only kernel (which is the term $N = 1$). The expansion is not uniform in k (nor in J or in the nodes): the $O_k(y^{-2})$ term is of size about $(2k+4)^4/(8y^2)$, so the expansion is in powers of $(2k+4)^2/y$ and is informative only for $y \gg k^2$.

The proof is in Appendix A.6.

At the front of the computed family ($J = 110$, $k \leq 222$, $y \approx 3010$) one has $\gamma_k/y \approx 33$, so Lemma 9.26 says nothing about individual moments there.

Theorem 9.27 (Far-node asymptotics for the true kernel). *Let φ be the true kernel Ψ or the Gamma-only kernel Ψ_1 , let $\mathbf{y}$ be a fixed node set of size J with $(N)_{\mathbf{y}}$ and $p_{2J}(\mathbf{y}) \neq 0$, and let $Y \rightarrow \infty$ (in y -units). Then:*

- (a) $(N)_{\mathbf{y} \cup \{Y\}}$ holds for all large Y , and $P_{\mathbf{y} \cup \{Y\}} = P_{\mathbf{y}} + aE_1 + bE_2$ with $a = 2p_{2J}Y^{-2}(1 + 8(J+1)/Y + O(Y^{-2}))$ and $b = p_{2J}Y^{-4}(1 + (16J+20)/Y + O(Y^{-2}))$;
- (b) $\Delta_{\mathbf{y}}(Y) = 2S_{\mathbf{y}}/Y^2 + 16(J+1)S_{\mathbf{y}}/Y^3 + O(Y^{-4})$;
- (c) $\mathcal{F}_{\mathbf{y} \cup \{Y\}}''(Y) = 8\pi p_{2J}e^{-Y}Y^{4J+2}(1 + O(1/Y))$ and $\mu_Y = S_{\mathbf{y}}e^Y Y^{-4J-5}/(2\pi p_{2J}) + O(e^Y Y^{-4J-6})$ (y -derivatives; in ξ -derivatives these read $32\pi^3 p_{2J}e^{-Y}Y^{4J+4}(1 + O(1/Y))$ and $S_{\mathbf{y}}e^Y Y^{-4J-6}/(4\pi^2 p_{2J}) + O(e^Y Y^{-4J-7})$);
- (d) $S_{\mathbf{y} \cup \{Y\}} = S_{\mathbf{y}}(1 - 8/Y) + O(Y^{-2})$;
- (e) if $p_{2J} > 0$ and $S_{\mathbf{y}} > 0$, then for all large Y : $\mu_Y > 0$, $\mathcal{F}_{\mathbf{y} \cup \{Y\}}''(Y) > 0$ and $\Delta_{\mathbf{y}}(Y) > 0$. If $p_{2J} > 0$ and $S_{\mathbf{y}} < 0$, then $\mu_Y < 0$ and $\Delta_{\mathbf{y}}(Y) < 0$ for all large Y .

The O -constants depend on $\mathbf{y}$: these are asymptotics for a fixed node set, not uniform in $\mathbf{y}$ or J . In the model the same statements hold with $16(J+1)$ replaced by $16J$ (Theorem 7.1).

Proof. Put $f_0 := e^y \mathcal{F}_{P_{\mathbf{y}}} / \pi$ and $f_m := e^y \mathcal{F}_{E_m} / \pi$. Since $\Pi_{\mathbf{y}} u^{2J+m}$ has degree $\leq 2J$, Lemma 9.26 gives $f_0 = p_{2J}y^M(1 - \gamma_{2J}/y + \dots)$, $f_1 = -y^{M+2}(1 - \gamma_{2J+1}/y + \dots)$ and $f_2 = y^{M+4}(1 - \gamma_{2J+2}/y + \dots)$, $M = 4J+4$, with differentiable expansions. The quantities $a, b, \Delta, \mathcal{F}'', \mu_Y$ and S are given by Proposition 4.6 and Proposition 4.7 as ratios of Wronskians, in which the common factor πe^{-y} cancels (Proposition 4.6(c)). Expanding the Wronskians to two orders gives the theorem; the computation is in Appendix A.6. The constant $8(J+1)$ comes from $5a_1 + c_1 - 6b_1 = 32(J+1)$ with a_1, b_1, c_1 the first-order coefficients of f_0, f_1, f_2 . $\square$

Heuristically, the frozen factor $(u + \frac{1}{4})^2$ in P^{Ξ} raises every u -degree by two and acts like one extra node; this is why $8J$ in the model becomes $8(J+1)$. The sign of $S_{\mathbf{y}}$ is the single far-field sign: by (e), positive increments, positive derivative weights and the Turán defect of (d) all follow from $S_{\mathbf{y}} > 0$, and all fail together if $S_{\mathbf{y}} < 0$.

9.6. The tail. Let $\mathbf{y}_k := \mathbf{y}_k^{\text{PP}}$ and $y_n := 2\pi n$. Proposition 4.8 bounds each $\Delta p_1^{(K)}$ from above. Hypothesis (p) of Corollary 9.25 needs $\sum_K |\Delta p_1^{(K)}|$, so this route needs a sign law as well.

Theorem 9.28 (Conditional tail bound). *Let $J_1 \in \{60, 110\}$, and assume for the true family:*

- (N) for $\mathbf{y}_k$ ($k \geq J_1$) and for $\mathbf{y}_k \cup \{2\pi n\}$, $n \in \text{PP}$, $n > n_k$;
- (SM)_{true} for every $k \geq J_1$: $\Delta_{\mathbf{y}_{k+1}}(2\pi n) \leq \Delta_{\mathbf{y}_k}(2\pi n)$ for $n \in \text{PP}$, $n > n_{k+1}$;
- (W)_{top} $\Delta p_1^{(K)} = \Delta_{\mathbf{y}_K}(2\pi n_{K+1}) \geq 0$ for every $K \geq J_1$.

Then $\delta_{J_1} := \sum_{K \geq J_1} |\Delta p_1^{(K)}| = \sum_{K \geq J_1} \Delta p_1^{(K)} \leq B_{J_1} := \sum_{n \in \text{PP}, n > n_{J_1}} \Delta_{\mathbf{y}_{J_1}}(2\pi n)$. Consequently, if $B_{J_1} < 1/((1+\eta)R_1)$ with R_1 from Proposition 9.19, hypothesis (p) of Corollary 9.25 holds on $[2, X]$, for $(J_1, X) = (110, 53)$ or $(60, 31)$.

Proof. Proposition 4.8 with nodes $y_j = 2\pi n_j$ gives $\sum_{K \geq J_1} \Delta p_1^{(K)} \leq B_{J_1}$ (its hypotheses are (N) and (SM)_{true}), and (W)_{top} makes every term non-negative, so the sum is δ_{J_1} . $\square$

(W)_{top} cannot be dropped: without it, Proposition 4.8 bounds the increments only from above. The three hypotheses are open for all k ; Numerical observation 10.9 records them along the prime powers in a finite range. By the robust form of Proposition 4.8, the conclusion survives if (SM)_{true} fails by a total factor $\prod_k (1 + \varepsilon_k)$, provided the constant still fits.

The constants B_{60} and B_{110} have been computed but not certified (Numerical observation 10.10); every use of them is conditional. Certifying B_{110} would be a finite computation once explicit remainder bounds for Theorem 9.27 are available, as they are for the model in

Proposition 7.6: enclose P_{110} , Πu^{221} and Πu^{222} , evaluate the insertion formula in interval arithmetic at the 2218 prime powers up to 20000, and bound the remaining tail with those remainders (Section 11.3).

Remark 9.29 (The solution pencil). Assume (Q): there are C , $a_1 < \pi/2$ and $\delta_0 > 0$ with $|q_K(t)| \leq Ce^{a_1|\operatorname{Re} t|}$ on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta_0$ for all large K , where $q_K := (P_{K+1} - P_K)(t^2)/\Delta p_1^{(K)}$. Then every subsequential limit q (which exists by Montel's theorem) satisfies $q(0) = 0$, $q''(0) = 2$, and $\widehat{\Xi^2}q$ has a double zero at every ξ_n , $n \in \text{PP}$: the zero data pass to the limit by the domination argument of Fact 2.9. If H_∞ is a subsequential limit of the exact family under (N) and (G_a) (such limits exist and solve the infinite interpolation problem [15]), then $H_\infty + \lambda q$ solves the same normalised problem ($H(0) = 1$, double zeros of $\widehat{\Xi^2}H$ at every ξ_n) for every $\lambda \in \mathbb{R}$. So the normalised solutions in $\mathcal{H}_{\max(a, a_1)}$ contain an affine line, the solutions of the homogeneous problem form a linear space of dimension at least 2 (it contains H_∞ and q , which are independent because $q(0) = 0 \neq H_\infty(0)$), and, since $[t^2]q = 1$, the coefficient p_1 is not determined by the limiting interpolation problem alone. Heuristically, positivity at infinity should pin $\lambda = 0$; we do not use this. For computed evidence on (Q) see Numerical observation 10.14(g).

9.7. Coefficient positivity: exact identities. The zero side (Section 9.2) reduces to a coefficientwise sandwich $P_{J+1} \leq P_J \cdot N_J$ with a quadratic N_J . Here we give exact identities for the natural choice of N_J , determined by the insertion. Let φ be a base kernel and $\varphi^{(1)} := -\theta^2\varphi$, $\varphi^{(2)} := \theta^4\varphi$ its first two companions, so that $\mathcal{F}_{uR}^\varphi = \mathcal{F}_R^{\varphi^{(1)}}$ and $\mathcal{F}_{u^2R}^\varphi = \mathcal{F}_R^{\varphi^{(2)}}$.

Proposition 9.30 (Companion identities). *Let $\mathbf{y}$ be a node set of size J , assume (N) $_{\mathbf{y}}$ for φ , $\varphi^{(1)}$ and $\varphi^{(2)}$, and $S_{\mathbf{y}} \neq 0$. Let $\bar{P} := P_{\mathbf{y}}/p_{2J}(\mathbf{y})$, and let $\bar{W}_i$ be the monic Hermite solution at $\mathbf{y}$ for the kernel $\varphi^{(i)}$ (the solution of Definition 3.1 divided by its top coefficient). Then*

$$E_1 = u\bar{W}_1, \quad E_2 - (\ell_2^*/\ell_1^*)E_1 = u^2\bar{W}_2, \quad \Pi_{\mathbf{y}}(uP_{\mathbf{y}}) = u p_{2J}(\mathbf{y})(\bar{P} - \bar{W}_1),$$

and $S_{\mathbf{y}} = p_{2J}(\mathbf{y}; \varphi)/p_{2J}(\mathbf{y}; \varphi^{(1)})$. In particular $[u]\Pi_{\mathbf{y}}(uP_{\mathbf{y}}) = 1 - S_{\mathbf{y}}$, and if $p_{2J}(\mathbf{y}) > 0$, then $\Pi_{\mathbf{y}}(uP_{\mathbf{y}})$ has non-negative coefficients if and only if $\bar{P} \geq \bar{W}_1$ coefficientwise.

Proof. $E_1 = u^{2J+1} - \Pi_{\mathbf{y}}u^{2J+1}$ vanishes at $u = 0$, so $E_1 = uV$ with V monic of degree $2J$, and $\mathcal{F}_{uV}^\varphi = \mathcal{F}_V^{\varphi^{(1)}}$ has zero data at $\mathbf{y}$. By (N) $_{\mathbf{y}}$ for $\varphi^{(1)}$ the polynomials of degree $\leq 2J$ with zero $\varphi^{(1)}$ -data form a line; it contains the monic V , so its top coefficient is non-zero and $V = \bar{W}_1$. Since $S_{\mathbf{y}} = -p_{2J}\ell_1^* \neq 0$, $\ell_1^* \neq 0$, and $E_2 - (\ell_2^*/\ell_1^*)E_1$ has zero constant and linear terms; the same argument with $\varphi^{(2)}$ gives the second identity. Next, $uP_{\mathbf{y}} - p_{2J}E_1$ has degree $\leq 2J$, no constant term, and the data of $uP_{\mathbf{y}}$, so it equals $\Pi_{\mathbf{y}}(uP_{\mathbf{y}})$. Its u -coefficient is $p_{2J}(1/p_{2J} - 1/w_{2J})$, where $w_{2J} = p_{2J}(\mathbf{y}; \varphi^{(1)})$, and it equals $\mathcal{L}_{\mathbf{y}}(uP_{\mathbf{y}}) = 1 - S_{\mathbf{y}}$. $\square$

Proposition 9.31 (The sandwich deficit). *Let $\mathbf{y}$ have size J , $Y \in I \setminus \mathbf{y}$, assume (N) for $\mathbf{y}$ and $\mathbf{y} \cup \{Y\}$, and let $b := p_{2J+2}(\mathbf{y} \cup \{Y\})$ with $p_{2J}(\mathbf{y})/b > 0$. Put $\hat{z} := (p_{2J}(\mathbf{y})/b)^{1/4}$, $c_1 := 2/\hat{z}^2$, $c_2 := \hat{z}^{-4}$ and $\tau := c_1p_{2J} + c_2p_{2J-1} - p_{2J+1}(\mathbf{y} \cup \{Y\})$, the coefficients p_k being those of $P_{\mathbf{y}}$. Then*

$$P_{\mathbf{y}}(1 + u/\hat{z}^2)^2 - P_{\mathbf{y} \cup \{Y\}} = \frac{2}{\hat{z}^2} \Pi_{\mathbf{y}}(uP_{\mathbf{y}}) + \hat{z}^{-4} \Pi_{\mathbf{y}}(u^2P_{\mathbf{y}}) + \tau E_1.$$

If moreover (N) $_{\mathbf{y}}$ holds for $\varphi^{(1)}$, $\varphi^{(2)}$ and $S_{\mathbf{y}} \neq 0$, then, with $\lambda := \Delta_{\mathbf{y}}(Y)/S_{\mathbf{y}}$,

$$P_{\mathbf{y}}(1 + u/\hat{z}^2)^2 - P_{\mathbf{y} \cup \{Y\}} = p_{2J}(\mathbf{y}) u [c_1\bar{P} + c_2u(\bar{P} - \bar{W}_2) - \lambda\bar{W}_1].$$

Proof. By Lemma 3.4(a) the left side is $(c_1u + c_2u^2)P_{\mathbf{y}} - aE_1 - bE_2$, with $a = p_{2J+1}(\mathbf{y} \cup \{Y\})$. As in the proof of Proposition 9.30, $uP_{\mathbf{y}} = p_{2J}E_1 + \Pi_{\mathbf{y}}(uP_{\mathbf{y}})$ and $u^2P_{\mathbf{y}} = p_{2J}E_2 + p_{2J-1}E_1 + \Pi_{\mathbf{y}}(u^2P_{\mathbf{y}})$. Since $c_2p_{2J} = b$, the E_2 terms cancel, and the coefficient of E_1 is τ . For the second form, insert

Proposition 9.30: the coefficient of $u\bar{W}_1$ becomes $-b\ell_2^*/\ell_1^* - a = \Delta_{\mathbf{y}}(Y)/\ell_1^* = -p_{2J}\lambda$, using $\Delta_{\mathbf{y}}(Y) = -a\ell_1^* - b\ell_2^*$ and $S_{\mathbf{y}} = -p_{2J}\ell_1^*$. $\square$

For orientation, consider the idealised problem in which P itself is interpolated: at each node Y_j the data are the value and the derivative of $s \mapsto P(-s^2)$ at $s = Y_j$, so that the solution is $P_J(u) = \prod_{j \leq J} (1 + u/Y_j^2)^2$. There $\Pi(uP_J) = \Pi(u^2P_J) = 0$, $S = 1$, $\tau = 0$ and $\hat{z} = Y_{J+1}$, and the identity reduces to $P_{J+1} = P_J(1 + u/\hat{z}^2)^2$. For the true kernel and for the model the extra terms measure the departure from this idealised problem.

Corollary 9.32 (The sandwich step from companion comparisons). *(a) In the setting of the second identity of Proposition 9.31, assume $p_{2J}(\mathbf{y}) > 0$, $b > 0$, all coefficients of $P_{\mathbf{y}}$ positive, $S_{\mathbf{y}} > 0$, $\Delta_{\mathbf{y}}(Y) \geq 0$, and*

$$(C01) \ \bar{P} \geq \bar{W}_1, \quad (C02) \ \bar{P} \geq \bar{W}_2 \text{ (coefficientwise)}, \quad (Z) \ \lambda \leq 2/\hat{z}^2.$$

Then (SW $_{\hat{z}}$) holds: $P_{\mathbf{y} \cup \{Y\}} \leq P_{\mathbf{y}}(1 + u/\hat{z}^2)^2$ coefficientwise. For $\lambda \geq 0$, (Z) holds if and only if $1 + \lambda u + c_2 u^2$ has non-positive discriminant. (b) For the true family with $Y = y_{J+1}$: if for all $J \geq J_0$ the coefficients of P_J are positive, (SW $_{\hat{z}}$) $_J$ holds, and $\hat{z}_J \geq c y_{J+1}$ for a fixed $c > 0$, then (POS $_a$) holds for every $a > 0$, and every limit H_{∞} has exponential type 0.

Proof. (a) The bracket in Proposition 9.31 is $(c_1 - \lambda)\bar{P} + \lambda(\bar{P} - \bar{W}_1) + c_2 u(\bar{P} - \bar{W}_2)$, a sum of polynomials with non-negative coefficients; the discriminant of $1 + \lambda u + c_2 u^2$ is $\lambda^2 - 4/\hat{z}^4$. (b) Theorem 9.10(a) with $c_{1,J} = 2/\hat{z}_J^2$ and $c_{2,J} = \hat{z}_J^{-4}$, for which $d_J = \hat{z}_J^{-2} \leq (4\pi c)^{-2}$. The count $\#\{J : d_J \geq 1/R\}$ is at most the number of prime powers up to $\sqrt{R}/(2\pi c)$, which is $o(\sqrt{R})$ because the prime powers have density zero. $\square$

The hypotheses (C01), (C02) and (Z) are open for all J ; the data are in Numerical observation 10.8. Condition (Z) is thin at the front: its first-order far-field form $c_1/\lambda = 1 + 2/Y + O(Y^{-2})$ follows for fixed $\mathbf{y}$ from Theorem 9.27(a),(b), but it is not a first-order statement at the actual front, where its computed margin is small (Numerical observation 10.8).

9.8. What transfers between the kernels.

Remark 9.33 (What transfers). Kernel-free, hence valid for the true kernel: the Christoffel identity and its pointwise Chebyshev-pair forms (Lemma 4.2, Proposition 4.5), the chain calculus (Lemmas 4.3, 4.1 and 4.4), the insertion calculus (Proposition 4.6), the half-step calculus (Proposition 4.7) with Proposition 7.5 and Lemma 7.9, the tail bounds (Proposition 4.8, Theorem 4.9), and the identities of Section 9.7. Polynomial (model) only: Lemma 6.2, Theorems 6.3 and 6.5, the Descartes and Taylor certificates, and Propositions 5.10–5.12, 8.3 and 8.4. The far-node constants differ: $8J$ in the model (Theorem 7.1) and $8(J+1)$ for the true kernel (Theorem 9.27).

10. NUMERICAL OBSERVATIONS

This section collects everything in this paper that was computed but not certified. Nothing here is used in a proof; each item says what was computed and, where it matters, in which arithmetic and whether the computation is reproduced by the ancillary files. The certified statements are in Sections 5–9.

10.1. The exact family of the true kernel.

Numerical observation 10.1 (The exact family for $J \leq 110$). The following were computed in high-precision arithmetic (at most 4600 bits), not certified except where Fact 2.13 and Proposition 9.20 say so. For every $10 \leq J \leq 110$ all coefficients $p_k^{(J)}$ are positive, and $a_J := \max_k ((2k)! p_k^{(J)})^{1/2k}$ increases with J , up to 0.7164 at $J = 110$ (attained near $k \approx 20$ –24);

so (POS_a) holds numerically with $a = 0.7164$ and $C = 1$ in this range. For $3 \leq J \leq 9$ the coefficients $p_{2J-1}^{(J)}$ (and, for some of these J , $p_{2J-3}^{(J)}$) are negative, while the top coefficient $p_{2J}^{(J)}$ is positive for every computed J . For $J \geq 10$, P_J has exactly three root pairs in $\text{Re } u > 0$, with $|\arg u|$ close to 1.035, 1.245 and 1.452 (certified at $J = 60, 110, 111$, Proposition 9.9); $H_J \geq 1 = H_J(0)$ on $\mathbb{R}$; and $\sup_t \log |H_J(t)|/|t|$ is about 0.7027 at $J = 110$, and the same on $\text{Im } t = 0.6$. The increments $p_k^{(J+1)} - p_k^{(J)}$ are ≥ 0 except for $k \in \{2, 4, 6, 10, 11\}$ (for $10 \leq J \leq 12$ also $k = 12$), where their relative size decays roughly like J^{-3} , with cumulative relative effect at most 0.0114 over $10 \leq J \leq 109$; this sign pattern is certified at $J = 60$ and $J = 110$ (Proposition 9.20). The (LR-step) data are in Numerical observation 10.8.

Numerical observation 10.2 (Predicted constants of the exact family). The following were computed in high-precision arithmetic; they are not part of Conjecture 2.14, and the ranges and certified values are in Table 1. $a^* \approx 0.7164$; for every $J \geq 10$, P_J has exactly three pairs of conjugate roots in $\text{Re } u > 0$, with $|\arg u|$ tending to 1.035, 1.245, 1.452, and the zeros of H_J satisfy $\min |\text{Im } t| \rightarrow 9.639$; $\sup_r N_J(r)/r \leq 0.4166 < \frac{1}{2}$, where $N_J(r)$ counts the roots of P_J with $|u| \leq r^2$; $H_J \geq 1 = H_J(0)$ on $\mathbb{R}$; $\Delta p_1^{(J)} > 0$ for every J , with a local exponent α_J ($\Delta p_1^{(J)} \approx n_{J+1}^{-\alpha_J}$) decreasing towards 2; and $\Delta p_1^{(J)}/\Delta p_1^{(J),\Gamma} \rightarrow 0.50950$, where $\Delta p_1^{(J),\Gamma}$ is the same increment for the Gamma-only kernel Ψ_1 at the same nodes. For $3 \leq J \leq 9$ some coefficients just below the top, p_{2J-1} and p_{2J-3} , are negative, while $p_{2J} > 0$ for every computed J ; this is why (b) starts at $J = 10$. Evidence for Conjecture 2.14 in the same computations: (a), (b) and the root geometry were computed for every $J \leq 110$ (floating-point table `computed/family_J1_110.tsv` in the ancillary files); (a) and the finite- J instances of (b) are certified at $J \in \{10, 60, 61, 110, 111\}$ (Fact 2.13, with $\max_k ((2k)!p_k)^{1/2k} \leq 0.71636$), and the root geometry at $J \in \{60, 110, 111\}$ (Proposition 9.9). For the front statement (FR) (Section 2.2), dense scans at $J = 20, 30, 40, 50, 60$ ($x \in [2, 1.15n_J + 5]$, step 0.05, and up to $x = 425$ at $J = 60$) found no sign change off the nodes; certified base data are in Proposition 9.19. $\Delta p_1^{(J)} > 0$ was computed for $J \leq 109$ and is certified at $J = 60, 110$ (Proposition 9.20); α_J decreases from 3.82 to 2.37 up to $n = 443$, and the decrease is slowing down. The ratio 0.50950 is constant to all printed digits for $J = 30, \dots, 99$.

Numerical observation 10.3 (TEL along the prime powers). In midpoint arithmetic at 4600 bits (not certified), (TEL_J) holds for every $2 \leq J \leq 110$ on a grid of 657 points in $[1.5, 1.35n_J]$ and at all nodes. Write $\tau_J(x) := \log(\hat{F}_J/\hat{F}_{J-1})(x) - 2\log|1 - x/n_J|$, so that (TEL_J) says $\tau_J \geq 0$. For $x < n_J$ the ratio of the erosion $\log(\hat{F}_{J-1}/\hat{F}_J)(x)$ to the allowance $-2\log|1 - x/n_J|$ is at most 0.855, and on the newest gap $[n_{J-1}, n_J]$ one has $\tau_J \geq 1.556$ for $10 \leq J \leq 110$. An independent re-computation for $J = 2, \dots, 12, 20, 30, \dots, 110$, in some cases up to $x = 1000n_J$, agrees. Also $\hat{F}_J > 0$ at large x for $3 \leq J \leq 9$.

Numerical observation 10.4 (Rank-one rigidity). For $J \leq 109$, computed with three independent codes: on every compact range of x , $\hat{F}_{J+1} - \hat{F}_J$ equals $\Delta p_1^{(J)}$ times a profile that is independent of J to 5–6 digits, on the prime side and, for $P_{J+1} - P_J$, on the zero side. For example $(\hat{F}_{J+1} - \hat{F}_J)/(\Psi_1 \Delta p_1^{(J)})$ at $x = 2.505$ equals -31.5241 for every $J \geq 40$. The newest cardinal functions A_{J+1}, B_{J+1} of Lemma 3.3, restricted to a fixed compact, are numerically of rank one (singular-value ratio $2.4 \cdot 10^{-8}$); the scaled matrix M_J has condition number about $10^{0.81n_J}$; and the influence of nodes to the right of x is super-exponentially small. The cardinal basis is not localised (it is not diagonally dominant) in this scaling.

This is the heuristic basis for expecting (ES) to follow from the summability of the scalar tail.

Numerical observation 10.5 (Universality of the scalar tail). With the Gamma-only kernel Ψ_1 at the same nodes, $\Delta p_1^{(J)}(\Psi)/\Delta p_1^{(J)}(\Psi_1) = 0.50950$ for every $J = 30, \dots, 99$; and the kernel e^{-y} at

TABLE 1. Predicted constants of Conjecture 2.14 for the exact family: computed behaviour, and certified values where available (Fact 2.13, Propositions 9.9 and 9.20).

Quantity	Predicted	Computed	Certified
$a_J := \max_k ((2k)! p_k^{(J)})^{1/2k}$	$\rightarrow 0.7164 < \pi/2$	$J \leq 110$	≤ 0.71636 at $J = 10, 60, 61, 110, 111$
pairs of roots of P_J in $\operatorname{Re} u > 0$	3 for $J \geq 10$	$J \leq 110$	3 at $J = 60, 110, 111$
their $ \arg u $	$\rightarrow$ 1.035, 1.245, 1.452	$J \leq 110$	in $[1.0351, 1.4523]$
$\min \operatorname{Im} t $ over zeros of H_J	$\rightarrow 9.639$	$J \leq 110$	≥ 9.6386
$\sup_r N_J(r)/r$	$\leq 0.4166 < \frac{1}{2}$	$J \leq 110$	≤ 0.41658 at $J = 60, 110, 111$
$\min_{\mathbb{R}} H_J$	$= 1 = H_J(0)$	$J \leq 110$	where (POS) is certified
$\Delta p_1^{(J)}$	> 0 for all J	$J \leq 109$	$\approx 2.0807954 \cdot 10^{-11}$ ($J = 60$), $\approx 2.9183184 \cdot 10^{-12}$ ($J = 110$)
local exponent α_J	decreases to 2	$3.82 \rightarrow 2.37$, $n \leq 443$	—
$\Delta p_1^{(J)}/\Delta p_1^{(J),\Gamma}$	$\rightarrow 0.50950$	constant, $30 \leq J \leq 99$	—

the same nodes gives the same sequence of local decay exponents. The decay is set by the node density: for integer nodes it is faster than any power, for squares the exponent approaches 2 in the computed range, and the prime powers lie in between. For the true family $\Delta p_1^{(J)} > 0$ for all $J \leq 109$, and the local exponent of its decay falls from 3.82 to 2.37 up to $n = 443$, decelerating, consistently with a limit 2. For the analogous families built from the data of $\mathbb{Q}(\sqrt{5})$, of $\mathbb{Q}(\sqrt{-3})$, and of ζ^2 with Ξ^4 , $p_1^{(J)}$ shows no sign of convergence in the computed range.

So the one scalar input of Corollary 9.25 appears to be a statement about interpolation by a Gamma-type kernel at the prime powers (heuristic).

Numerical observation 10.6 (Monotone decrease and the first gap). $\hat{F}_{J+1} - \hat{F}_J < 0$ at every tested point of $[2, \infty)$ off the nodes, so $\hat{F}_J$ decreases in J there; and $\hat{F}_J'' > 0$ at every node, for $J \leq 60$. On the first gap, with ϱ as in Theorem 9.23, the normalised margin $\hat{F}_J/(\Psi_1 \varrho)$ is 0.96292 at $x = 2.5$ and tends to 2.4249 as $x \rightarrow 2^+$, stable in J (relative increments about 10^{-7}). The binding region is the front $x \approx n_J$, not the first gap.

Numerical observation 10.7 (The front). Near $x \approx n_J$ the erosion $\log(\hat{F}_{J-1}/\hat{F}_J)$, which is stable in J in the computed range, is about $0.4x$, against a fresh amplification of about $(9.7/\log x)x$. For $15 \leq J \leq 110$ the Abel preimage $Q_{P_J}^{\text{Ab}}$ has exactly $2J+3$ positive roots: three small ones, and one pair close to each node (offsets ≤ 0.46). Its cofactor $Q_{P_J}^{\text{Ab}}/(y \prod_i (y-r_i))$, r_i the positive roots, has positive coefficients for $J = 1, 2$ and at every tested $J \geq 9$ ($J = 9, \dots, 15, 20, 30, 45, 60$), and not for $3 \leq J \leq 8$. By Lemma 9.18 the roots of this cofactor sum to $8(J+1)^2 - \sum_i r_i$, so positivity of its coefficients requires $\sum_i r_i > 8(J+1)^2$; in the data this fails exactly for $3 \leq J \leq 8$. A heuristic extrapolation suggests that $\hat{F}_\infty/\Psi_1$ may approach 0 at mid-gaps near $x \sim 10^7$ – 10^8 , so that a fixed margin in Ψ_1 -normalisation may fail even if (Mg) holds; Fact 2.10 needs only pointwise (Mg).

Numerical observation 10.8 (Coefficient steps along the prime powers). For the exact P_J , $10 \leq J \leq 109$ (with independent root-finding): (LR-step) of Theorem 9.10 holds with u_J the newest root pair for every J , and in fact for all but one of the left-half-plane root pairs; for the passing pairs $|t_J| \geq 2.57 n_{J+1}$, where $u_J = t_J^2$. P_J is log-concave for $10 \leq J \leq 110$ (minimum ratio 1.029, migrating to the top like $1 + 4.2/k$). $(\text{SW}_{\hat{z}})$ holds for every $8 \leq J \leq 109$ (and fails for $3 \leq J \leq 7$), with $\hat{z}_J \geq 0.6 y_{J+1}$. (C01) and (C02) hold for $10 \leq J \leq 109$, with the smallest relative gaps at $k = 2J - 1$ (from 0.733 to 0.00316 and from 0.453 to 0.00632 as J goes from 10 to 109); (Z) holds with $c_1/\lambda - 1 = 8.3 \cdot 10^{-4}$ at $J = 109$; and the comparison $\bar{W}_1 \geq \bar{W}_2$ fails at every J .

Numerical observation 10.9 (Insertion sign laws along the prime powers). $(\text{SM})_{\text{true}}$ holds along the prime powers for the sampled $K \leq 121$ ($K = 1-40, 50, 60, \dots, 100, 105, 108-121$, on grids of $x > n_{K+1}$), with relative margin about $-0.8/n_{K+1}$ at the confluent end; $(\text{W})_{\text{top}}$ and (N) hold for insertions at $(n_K, 10^4 n_K]$, $K = 5, 8, 10, 20, 40, 60, 110$. On these grids, with $Y = y_{K+1}$ and $X > Y$, the two-node interaction satisfies $\Delta_{\mathbf{y}_K}^2(Y, X) \leq -16S_K/(YX(Y+X))$ for $K = 2, \dots, 121$ and the Turán ratio satisfies $\text{Tu}_K \leq 1 - 8/Y$. The ratio $r_K := Y^2 \Delta_{p_1}^{(K)}/(2S_K)$ is at least 1.279 at every tested $K \leq 109$ (its minimum over that range, attained at $K = 109$; the maximum is 3.02, at $K = 6$), and $r_{110} = 1.27459$ (Numerical observation 10.10); the values decrease towards 1 roughly like $1 + 0.93 \cdot 8(K+1)/Y$. In this computed range, therefore, each of the three quantities is bounded by its first-order far-field value (“the far field is the worst case”); since $8(K+1)/y_{K+1} \rightarrow 0$ along the prime powers, this is a statement about the computed range only. For a fixed node set, the interaction is $-16S/(Ys(Y+s))$ to first order on cones, a constant computed symbolically for $J \leq 6$.

Numerical observation 10.10 (The tail constants B_{60} and B_{110}). The insertion formula of Proposition 4.6 was evaluated at every prime power $n \in (n_{J_1}, 20000]$ in 3000-bit arithmetic, and reproduced independently to $5 \cdot 10^{-15}$ per point. The tail $n > 20000$ was bounded under the assumption that $Y^2 \Delta_{\mathbf{y}_{J_1}}(Y)/(2S_{\mathbf{y}_{J_1}})$ does not exceed its value at $n = 20000$; this step is heuristic and covers 1.3% of the total. The computed values are $B_{110} \approx 1.8672 \cdot 10^{-10}$ and $B_{60} \approx 6.7195 \cdot 10^{-10}$ (with the cruder tail assumption $r \leq 2$, $B_{110} \leq 1.891 \cdot 10^{-10}$). With the certified $R_1 = 4.844 \cdot 10^9$ ($X = 53$) this gives $R_1 B_{110} \approx 0.905$, so (p) would hold for $\eta < 0.105$; with $R_1 = 1.021 \cdot 10^9$ ($X = 31$), $R_1 B_{60} \approx 0.686$. Also $S_{\mathbf{y}_{110}} \approx 1.036964 \cdot 10^{-5}$, $S_{\mathbf{y}_{60}} \approx 1.332823 \cdot 10^{-5}$, and the ratio $Y^2 \Delta/(2S)$ at the first tail node is 1.27459 for $J_1 = 110$, against $1 + 8 \cdot 111/Y \approx 1.295$ from Theorem 9.27.

10.2. The model.

Numerical observation 10.11 (Further tail data of the model). Along the prime-power chain of the model:

- (a) at every level $J \leq 40$, on 120 log-spaced points of $[y_{J+1}, 1000y_{J+1}]$, the hypotheses of Theorem 4.9 hold;
- (b) $S_K > 0$ at $K \in \{20, 40, 60, 80, 100, 120\}$, with $S_{20} \approx 5.65 \cdot 10^{-3}$ and $S_{120} \approx 1.99 \cdot 10^{-3}$;
- (c) in Proposition 7.6 the second coefficient grows roughly like $30J^2$, so the expansion parameter is J/Y ;
- (d) the bound of Theorem 4.9 exceeds that of Proposition 4.8 by 3.1% ($J_1 = 5$, $K \leq 40$) and 0.42% ($J_1 = 20$, $K \leq 50$).

Items (a) and (b) were obtained with interval arithmetic and (c), (d) in high-precision floating point; they are not reproduced by the ancillary files.

Numerical observation 10.12 (W on sparse sets). (W) holds at every node for the prime powers ($J = 20, 40$), the squares ($J = 15$), the primes ($J = 30$) and $n_j \approx j \log j$ ($J = 30$). Along

$n_j = s + (j - 1)$ the first J at which sign failures appear grows with the deficiency $s - 1$: about 5, 10, 21, 45 for $s - 1 = 0, 0.02, 0.05, 0.1$. The integers from 2 pass for every sampled $J \leq 90$, while S_J decays quickly ($S_{90} \approx 3.9 \cdot 10^{-11}$).

Heuristically, (W) needs a quantitative deficiency below the critical density, which the prime powers have in abundance; at critical density the truncated systems become nearly degenerate and the signs of S_J and of the weights are decided by exponentially small quantities. We do not use this reading.

10.3. The model versus the true kernel. For the TEL comparison put $\tau_J(x) := \log(\hat{F}_J/\hat{F}_{J-1})(x) - 2 \log|1 - x/n_J|$ for the true family (so that $\text{TEL}_J^{\text{true}}$ of Conjecture 11.1 is $\tau_J \geq 0$), the same quantity with $Q_J e^{-y}/(Q_{J-1} e^{-y})$, $y = 2\pi x$, for the model, and $\tau_J(n_J)$ for its limit at the new node.

Numerical observation 10.13 (Quantitative agreement). On the same prime-power nodes:

- (a) (front profile of the step law) $\tau_J(n_J) = 2.009, 1.860, 1.705, 1.646$ in the model and $2.208, 1.917, 1.722, 1.655$ for the true family at $J = 10, 20, 40, 60$; at $J = 60$ and $x = qn_J$, $q = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$, $\tau_J = 0.508, 0.953, 1.327$ (model) and $0.474, 0.931, 1.315$ (true). The two agree to within 0.03 for $J \geq 40$, and the difference decreases with J .
- (b) (set-comparison profile) In both models the far field is the worst case in the computed range: for the true kernel, with $z = y_{K+1}$, $\Delta_{\mathbf{y}_K}^2(z, x) \leq -16S_K/(zx(z+x))$ at every grid point $x > z$ and $S_{K+1}/S_K \leq 1 - 8/z$, for $K = 2, \dots, 121$.
- (c) (companion comparisons) The minimal relative gaps in the comparisons (C01), (C02) of Corollary 9.32 for the true kernel coincide to 4–5 digits with those of the model's companion runs $2 \rightarrow 3$ and $2 \rightarrow 4$: $0.7325/0.7328$ and $0.4527/0.4550$ at $J = 10$; $0.08764/0.08763$ and $0.1750/0.1750$ at $J = 20$; $0.01745/0.01745$ and $0.03489/0.03489$ at $J = 40$; $5.1692 \cdot 10^{-3}/5.1692 \cdot 10^{-3}$ and $1.034 \cdot 10^{-2}/1.034 \cdot 10^{-2}$ at $J = 80$ (true/model).

A heuristic explanation of (c): at the front $u + \frac{1}{4} \approx u$, so $P^\Xi \approx u^2 P$ and the true family behaves like the model two companion rungs up (Proposition 5.5). Making this precise, with error terms, is an open problem (Section 11.3).

10.4. Further computed data quoted in the text.

Numerical observation 10.14 (Computed data quoted in the text). (a) (Remark 5.15) In random tests at $J = 2, 3, 4$ the local law (LOC) held in all 6331 trials.

- (b) (Remark 6.4) Along the prime powers the hierarchy of augmented cofactors $\mathbf{f}_{Z+0+0}, \mathbf{f}_{Z+3+0}, \dots$ is not uniformly sign-regular. The law (I1') fails at critical density: for the integer nodes $2\pi(2, 3, \dots)$ it first fails, at the top index, at $K = 13$, while (I0) first fails at $K = 25$; and it fails on the progression y^* at every level checked.
- (c) (Proposition 7.11) An explicit pair of systems with $d = 4$, $c = 10$ and $s = \pm \frac{1}{2}$ exhibits both signs of $\text{Tu} - 1$ and of I , while every Wronskian stays one-signed with positive Taylor coefficients at c (an exact computation, not reproduced in the ancillary files).
- (d) (Corollary 9.25) (Φ -mono) holds for $K = 59, \dots, 109$ with $\eta \leq 10^{-4}$ (computed values, not bounds). With the extrapolated tail $\delta(60) \approx 5.7\text{--}6.1 \cdot 10^{-10}$ one gets $R_1 \delta \approx 0.58\text{--}0.62$ on $[2, 31]$, and with $\delta(110) \approx 1.5\text{--}1.9 \cdot 10^{-10}$, $R_1 \delta \approx 0.73\text{--}0.92$ on $[2, 53]$.
- (e) (Remark 9.22) (EVEN-TEL $_J$) holds on the data of Numerical observation 10.3, but its margin shrinks roughly like $n_J^{-0.38}$, so it may be asymptotically tight (a heuristic reading of the data).
- (f) (Lemma 7.10) The expansion in the proof of Lemma 7.10, carried out symbolically for general $\kappa \in \mathbb{R}^5$, gives $\partial_t I|_{t=0} = \sum_i \lambda_i \kappa_i / (2xz(x+z)^3)$ with $\lambda_0 = 6x^2 + 13xz + 6z^2$,

$\lambda_1 = -10x^2 - 24xz - 10z^2$, $\lambda_2 = 2x^2 + 10xz + 2z^2$, $\lambda_3 = 2(x^2 + z^2)$, $\lambda_4 = xz$; on quadratic κ it agrees with Lemma 7.10(ii). This computation is not reproduced in the ancillary files.

- (g) (Remark 9.29) The bound (Q) holds for the computed normalised increments q_K , $K \leq 109$, with exponent $a_1 \approx 1.0265$.

11. CONJECTURES AND OPEN PROBLEMS

We collect the open statements of this paper. For each conjecture we give the claim, what it would imply and through which result, the evidence with its range, and what a finite prefix of the computed family can and cannot decide. All numerical values in this section are computed, not certified, unless a certified statement is cited. None of these conjectures implies the Riemann Hypothesis: the strongest consequence of the conjectures on the true family is Conjecture U of [15] (Fact 2.3), and Conjecture U gives only $\text{RH} \Leftrightarrow (\text{E})$.

The named conditions of the paper, with their status, are indexed in Table 2.

11.1. Step laws for the true family. The front hypothesis of Theorem 9.15 and Corollary 9.21 is the step law TEL of Hypothesis 9.12.

Conjecture 11.1 (The true step law). *For every $J \geq 2$ such that $(\text{N})_{J-1}$ and $(\text{N})_J$ hold (so that F_{J-1} and F_J are defined; by Conjecture 2.14(a) this should be every J), and every $x \geq 1$: $\hat{F}_J(x) \geq (1 - x/n_J)^2 \hat{F}_{J-1}(x)$, where $x = e^{2\pi\xi}$.*

Implication. With $(\text{N})_J$ for every $J \geq J_0$ and the base hypothesis $(\text{B}^+)_{J_0}$ (Hypothesis 9.14), which is certified only in part for $J_0 = 10, 60$ (Proposition 9.19), Theorem 9.15 gives the front (FR) for every $J \geq J_0$; adding (ES) (Conjecture 11.3) and (POS_a) , Corollary 9.21 gives (S) and (U). *Evidence.* Computed (not certified) for every $J = 2, \dots, 110$ on a grid of 657 points in $[1.5, 1.35n_J]$ and at all nodes, so the range is short for small J and $[1, 1.5)$ is not covered; repeated with independent code for $J = 2, \dots, 12, 20, 30, \dots, 110$, in some cases up to $x = 1000n_J$ (Numerical observation 10.3). The log-margin $\tau_J(n_J)$ at the new node lies between 1.58 and 2.21 (Numerical observation 10.13). The stronger law with $(1 - x^2/n_J^2)^2$ also holds on these data, but its margin shrinks like a small power of n_J (Numerical observation 10.14(e)), so we do not conjecture it. *Refutation.* One J and one $x \geq 1$ violating the inequality.

Conjecture 11.2 (The true insertion laws beyond $J = 110$). *Let $\mathbf{y}_k := \mathbf{y}_k^{\text{PP}}$ for the true kernel. (a) For every $k \geq 110$, (N) holds for $\mathbf{y}_k$, and for every prime power $n > n_{k+1}$ it holds for $\mathbf{y}_k \cup \{2\pi n\}$ and $\mathbf{y}_{k+1} \cup \{2\pi n\}$, and $\Delta_{\mathbf{y}_{k+1}}(2\pi n) \leq \Delta_{\mathbf{y}_k}(2\pi n)$. (b) $\Delta p_1^{(K)} \geq 0$ for every $K \geq 110$.*

Implication. Part (a) gives the hypotheses (N) and $(\text{SM})_{\text{true}}$ of Theorem 9.28 with $J_1 = 110$, and part (b) its hypothesis $(\text{W})_{\text{top}}$; so $\delta := \sum_{K \geq 110} |\Delta p_1^{(K)}| \leq B_{110}$. With the computed value $B_{110} \approx 1.8672 \cdot 10^{-10}$ (Numerical observation 10.10) and the certified ratio bound $R_1 = 4.844 \cdot 10^9$ on $[2, 53]$ at $J_1 = 110$ (Proposition 9.19), one gets $R_1 \delta \leq R_1 B_{110} \approx 0.905$ (computed), so hypothesis (p) of Corollary 9.25, $(1 + \eta)R_1 \delta < 1$, would hold on $[2, 53]$ provided $(\Phi\text{-mono})$ holds with $\eta < 0.105$. *Evidence.* (a) computed at $K = 1, \dots, 121$ on sampled grids, with relative margin about $0.8/n$ (Numerical observation 10.9); (b) computed for $J \leq 109$ and certified at $J = 60, 110$ (Proposition 9.20). *Refutation.* A single k and prime power n violating (a), or $\Delta p_1^{(K)} < 0$ for some $K \geq 110$.

Conjecture 11.3 (Erosion summability). *Hypothesis 9.13 holds: for every $x \in [1, \infty) \setminus \text{PP}$, $\hat{F}_K(x) > 0$ for all sufficiently large K , and $\sum_K \max(\varepsilon_K(x), 0) < \infty$, where $\varepsilon_K(x) := \log(\hat{F}_{K-1}(x)/\hat{F}_K(x))$ and the sum runs over those K .*

TABLE 2. The named conditions of this paper: where each is defined, where it is used, and its status. “Open” means not known for all J (or all levels); finite certificates and computations are as indicated.

Condition	Defined in	Used in	Status
$(N), (G_a), (Z_\infty), (Mg)$	Section 2.2	Fact 2.10	open; finite- J instances of the first three certified at $J \in \{10, 60, 61, 110, 111\}$ (Fact 2.13, Proposition 9.9)
$(Z^+), (R_\eta), (R_k), (R_E), (Mg\mathcal{M})$	Corollary 2.11	Corollary 2.11	open
(POS_a)	Fact 2.12	Corollary 9.21	open; certified at $J \in \{10, 60, 61, 110, 111\}$ with $a = 0.71636$
$(G1)-(G3), (LR\text{-}step), (SW\text{-}step), (CNT_1)$	Theorem 9.10	Corollary 9.11	open; computed for $10 \leq J \leq 109$ (Numerical observation 10.8)
$(SW_z), (C01), (C02), (Z)$	Corollary 9.32	Corollary 9.32	open; computed (Numerical observation 10.8)
$(TEL_J), (EVEN\text{-}TEL_J)$	Hypothesis 9.12	Theorem 9.15	open (Conjecture 11.1); computed for $J \leq 110$
(ES)	Hypothesis 9.13	Theorem 9.15, Corollary 9.21	open (Conjecture 11.3)
$(B^+)_{J_0}$	Hypothesis 9.14	Theorem 9.15	certified for $J_0 = 10, 60$ except on two short intervals (Proposition 9.19)
$(B), (\Phi), (\Phi\text{-}mono), (p)$	Theorem 9.24, Corollary 9.25	the same	(B) certified for $(J_1, X) = (60, 31), (110, 53)$; the others open
$(SM)_k, (SM)_{true}, (W)_{top}$	Proposition 4.8, Theorem 9.28	Theorem 9.28	open (Conjecture 11.2); computed for $K \leq 121$; for the model $(SM)_k$ is certified for $k \leq 60$ (Proposition 7.7)
$(CP)_i, (UM)_i$	Theorem 4.9	Theorem 4.9	model: open (Conjecture 11.5); certified at sampled levels (Proposition 7.8)
(Q)	Remark 9.29	Remark 9.29	open; computed for $K \leq 109$
$(N)_y, (R^+), (FR), TEL_J, PT_K, PT_\delta$	Definitions 3.1 and 5.1	Sections 5–8	model: (R^+) and (FR) proved for $J \leq 80$ on the prime powers (Theorem 6.5)
$(I0), (I1'), (I1), (Y0)$	Definition 6.1	Theorem 6.3	open (Conjecture 11.4); certified for $K \leq 160$
$(AP^+)_K, (B)_K, (LOC)$	Remark 5.15	Remark 5.15	the first two certified for $K \leq 90$ (Proposition 5.16); (LOC) open
$(W), (W_{top})$	Definition 8.1	Section 8	(W) fails on the class $\mathcal{W}$ (Proposition 8.2); holds for $J = 2, 3$

Implication. With Conjecture 11.1, (N) , (POS_a) and $(B^+)_{J_0}$: (S) and (U) by Corollary 9.21. *Evidence.* At fixed x the erosion is, to 5–6 digits, $\Delta p_1^{(K)}$ times a profile independent of K (Numerical observation 10.4), so it decays like $n_K^{-2.4}$ in the computed range; at $x = 1.25$ and $x = 1.5$ it is negative for $J \geq 10$. Heuristically, (ES) would follow from $\sum_K |\Delta p_1^{(K)}| < \infty$ and a bounded relative profile at fixed x , but this derivation needs sign control of $\Delta p_1^{(K)}$ of the type in Conjecture 11.2(b). *Refutation.* No finite prefix of the family decides it alone (a rigorous tail bound could); a sustained decay of the erosion slower than $1/K$ at some fixed x would be evidence against it.

11.2. Laws for the model.

Conjecture 11.4 (Front law of the model). *Along the prime-power chain of the model (Definition 6.1), $(I1')_K$ and $(Y0)_K$ hold for every $K \geq 0$.*

Implication. By Theorem 6.3: for every J , (N) , $p_{2J} > 0$, (R^+) and strict TEL_J hold on $\mathbf{y}_J^{\text{PP}}$.
Evidence. Certified for every $K \leq 160$ (Theorem 6.5). The relative margins decrease from 0.39 at $K = 20$ to 0.047 at $K = 160$, roughly like $1/K$, and relative to that scale they are still decreasing at $K = 160$ (Table 5 and the comment after it). The law fails at critical density and on the progression y^* (computed; Numerical observation 10.14(b)), so a proof must use a property of the prime powers that these node sets lack, presumably their growth well above the PT-tight scale ($y_K = 2\pi n_K \approx 2\pi K \log K$, against $y_K^* \approx 8.4K$), rather than sparsity as such.
Refutation. A level K with $\varrho_{k+1} < \varrho_k$ for some k , or with $\mathbf{g}^{(K)}(z_{K+1}) \leq 0$.

Conjecture 11.5 (Tail law of the model). *Let B_i be the prime-power half-step chain of the model (B_{2K} the doubled chain of $\mathbf{y}_K^{\text{PP}}$, $B_{2K+1} = B_{2K} + y_{K+1}$) and z_{i+1} its next point. For every $i \geq 0$: $(\text{CP})_i$ $D_1(B_i) \neq 0$ and $S(B_i) > 0$; and $(\text{UM})_i$ on $[z_{i+1}, \infty)$: $F_1^{B_i} \neq 0$, a_{B_i} is non-decreasing, and $(q_1^{B_i})' < 0$.*

Implication. By Theorem 4.9, for every J_1 , in the model,

$$\sum_{K \geq J_1} \Delta p_1^{(K)} \leq 2S_{J_1} \sum_{n \in \text{PP}, n > n_{J_1}} (-F_0/F_1)^{B_{2J_1}} (2\pi n).$$

Evidence. $(\text{UM})_i$ is certified at both chain levels $i = 2J$ and $i = 2J + 1$ for $J \in \{2, 4, 8, 12, 20, 30, 40, 60\}$ (Proposition 7.8), with the non-vanishing $D_1(B_i) \neq 0$ certified by the ball solves. The sign condition $S(B_i) > 0$ of $(\text{CP})_i$ is certified only at $i = 2J$ for $J \in \{6, 10, 20\}$ (the values of $S_{\mathbf{y}_J^{\text{PP}}}$ in Proposition 7.6) and, for $J = 20$, also at $i = 41$, by Proposition 4.7(H2), since $a_{B_{40}}(y_{21}) > 0$ is certified in Proposition 7.8. So both laws are certified together only at $i = 40, 41$. The hypotheses of Theorem 4.9 were computed (not certified) on sampled points at every level $J \leq 40$, and $S_K > 0$ at six further levels (Numerical observation 10.11). By Proposition 7.11, no proof can rest on open sign conditions on the five-function system alone. *Refutation.* A level with $S(B_i) \leq 0$, or a point of $[z_{i+1}, \infty)$ where $a'_{B_i} < 0$ or $(q_1^{B_i})' \geq 0$.

11.3. Open problems.

- (1) *A global description of the chain pair $\mathcal{F}_{Z+0}/\mathcal{F}_Z$* that proves the per-level laws (Conjectures 11.4 and 11.5 in the model, Conjectures 11.1 and 11.2 for the true kernel) at all levels at once; inductive arguments meet the window obstruction of Remark 6.4.
- (2) *Model and true kernel.* Make “the true kernel behaves like the model two companion rungs up” (Numerical observation 10.13(c)) precise, with error terms.
- (3) *Finite certifications.* Certify B_{110} , which would make (p) of Corollary 9.25 on $[2, 53]$ conditional only on Conjecture 11.2; close the base $(B^+)_{60}$ on $[1, 2)$ and on the node sliver $(199, 199.39)$; and certify uniqueness windows at a non-arithmetic critical point q_c itself.
- (4) *Coefficient positivity.* Prove the comparisons (C01), (C02) and the front inequality (Z) of Corollary 9.32 for all J , or the sandwich step directly (Section 9.7).

Table 3 summarises what each conjecture would give and how a finite computation could refute it; the tests listed are sufficient, not a complete account of how a conjecture could fail.

AI-USE STATEMENT

This manuscript was produced with substantial use of AI. The mathematical arguments, the numerical computations and the text were generated by instances of Claude Opus 5.5 (Anthropic) working as coordinated agents: one set of agents developed the arguments, separate

TABLE 3. The conjectures of this paper: consequence, evidence, and finite tests that would refute them.

Statement	Gives (via)	Evidence	Refuted by
Conj. 11.1	front for $J \geq 60$ (Thm. 9.15)	computed $J \leq 110$	one (J, x) violating it
Conj. 11.2	tail at $J_1 = 110$ (Thm. 9.28)	computed $K \leq 121$	one (k, n) ; $\Delta p_1^{(K)} < 0$
Conj. 11.3	(Mg); with the above, (S), (U) (Cor. 9.21)	rank-one profile, $J \leq 109$	no finite prefix of the family alone
Conj. 11.4	model front, all J (Thm. 6.3)	certified $K \leq 160$	a decreasing ratio, or $\mathbf{g}^{(K)}(z_{K+1}) \leq 0$
Conj. 11.5	model tail bound (Thm. 4.9)	(UM) certified at 8 levels $J \leq 60$; (CP) at $i = 40, 41$	$S(B_i) \leq 0$; $a' < 0$; $q'_1 \geq 0$

agents acted as independent hostile referees of each component, and the referees' requested repairs were incorporated. The numerical certificates were computed by AI-written scripts that are included as supplementary material and can be rerun. The author initiated and directed this project, designed its verification process, and reviewed the results and the verification record, but did not independently check every step of the written proofs. The author takes full responsibility for the content of this manuscript, including any errors, and welcomes independent verification and corrections.

APPENDIX A. PROOFS

A.1. Voronoi decoupling.

Proof of Theorem 9.1. By (2.1), $\Xi^2 H$ is analytic on $-\tau \leq \operatorname{Im} w \leq 0$ and decays like $e^{-(\pi/2-a)|\operatorname{Re} w|}$ there, so $\widehat{\Xi^2 H}(\xi) = \int_{\operatorname{Im} w = -\tau} \Xi(w)^2 H(w) e^{-2\pi i \xi w} dw$. On this line $s = \frac{1}{2} + iw$ has $\operatorname{Re} s = \frac{1}{2} + \tau > 1$, so $\Xi(w)^2 = \Xi_\infty(w)^2 \sum_m d(m) m^{-1/2-iw}$ with absolute convergence, and Fubini gives $\sum_m d(m) m^{-1/2} \int_{\operatorname{Im} w = -\tau} G_H(w) e^{-2\pi i (\xi + \xi_m) w} dw$. The poles of $\Xi_\infty(w) = \frac{1}{2} s(s-1) \Gamma_{\mathbb{R}}(s)$ lie at $s = -2k$, $k \geq 1$ (the pole of $\Gamma(s/2)$ at $s = 0$ is cancelled by s), that is, at $w = i(2k + \frac{1}{2})$ in the upper half-plane. So each integral may be moved back to $\mathbb{R}$, where it equals $\widehat{G}_H(\xi + \xi_m)$. On $\operatorname{Im} w = -\tau$ we have $|e^{-2\pi i \eta w}| = e^{-2\pi \tau \eta}$, which gives the bound with $C_\tau = \|G_H\|_{L^1(\operatorname{Im} w = -\tau)}$. For real t , $\overline{\Xi_\infty(t)} = \Xi_\infty(-t)$, so $\overline{G_H(t)} = G_H(-t)$ and $\widehat{G}_H$ is real. For the last two identities, insert the expansion into Fact 2.6 and into $\int \Xi^2 H = \widehat{\Xi^2 H}(0)$, and regroup by $k = nm$; the double series converge absolutely. The regrouping uses $\sum_{nm=k} \Lambda(n) d(m) = \sum_{e|k} \log e = \frac{1}{2} d(k) \log k$. $\square$

A.2. Nested Hermite families.

Proof of Lemma 3.2. (a) By $(N)_y$ the data map $\mathcal{D}_y$ is bijective on $\operatorname{span}\{u, \dots, u^{2J}\}$, so $G \mapsto [u] \mathcal{D}_y^{-1} \mathcal{D}_y G$ is a linear functional of the data of G ; on $\operatorname{span}\{u, \dots, u^{2J}\}$ it is $[u]$, and a functional of the data is determined by its values on a space on which the data map is bijective. Since P_y has zero data, $P_y - 1 = -\Pi_y 1$, so $p_1(y) = -\mathcal{L}_y(1)$. (b) $D := P_{y'} - P_y \in \operatorname{span}\{u, \dots, u^{2J+2}\}$ has the data of $-P_y$ at y' (both polynomials have zero data at y), so $D = -\Pi_{y'} P_y$ and $[u] D = -\mathcal{L}_{y'}(P_y)$. Only the weights at y_{J+1} contribute. (c) $P_{y'}$ has zero data at y , so $0 = \mathcal{L}_y(P_{y'}) = -p_1(y) + p_1(y') + \sum_{k>2J} p_k(y') \ell_k$. (d) The coefficients are rational functions of the data (Cramer), with non-vanishing denominator. Move y_j with velocity 1 and differentiate the identities $\mathcal{F}_P(y_j) =$

$\mathcal{F}'_P(y_j) = 0$: the derivative $\dot{P} \in \text{span}\{u, \dots, u^{2J}\}$ has zero data except $\mathcal{F}'_{\dot{P}}(y_j) = -\mathcal{F}''_{\mathbf{y}}(y_j)$ (the value condition gives $\mathcal{F}_{\dot{P}}(y_j) = -\mathcal{F}'_{\mathbf{y}}(y_j) = 0$). Hence $\dot{p}_1 = \mathcal{L}_{\mathbf{y}}(\dot{P}) = -\mu_j \mathcal{F}''_{\mathbf{y}}(y_j)$. $\square$

Proof of Lemma 3.4. (a) $D := P_{\mathbf{y} \cup \{Y\}} - P_{\mathbf{y}}$ lies in $\text{span}\{u, \dots, u^{2J+2}\}$ and has zero data at $\mathbf{y}$. By (N) $_{\mathbf{y}}$ the data map at $\mathbf{y}$ is onto from $\text{span}\{u, \dots, u^{2J}\}$, so the elements of $\text{span}\{u, \dots, u^{2J+2}\}$ with zero data at $\mathbf{y}$ form a two-dimensional space, which contains E_1, E_2 (independent, of exact degrees $2J+1, 2J+2$). So $D = aE_1 + bE_2$, and a, b are the top two coefficients of $P_{\mathbf{y} \cup \{Y\}}$. The two conditions at Y read $a\mathcal{F}_{E_1} + b\mathcal{F}_{E_2} = -\mathcal{F}_{P_{\mathbf{y}}}$ in value and derivative, and Cramer's rule gives the formulas. (b) Delete column k from the $(2J+2) \times (2J+3)$ data matrix at $\mathbf{y} \cup \{Y\}$. The rows of $\mathbf{y}$ and the columns $\{0, \dots, 2J\} \setminus \{k\}$ form a square block with determinant $\mathfrak{D}_k(\mathbf{y}) \neq 0$; the rows δ_Y, δ'_Y and the columns $2J+1, 2J+2$ come last. By Schur's determinant formula for a block matrix [12, Ch. II, §5, formula (I), p. 46] the determinant is $\mathfrak{D}_k(\mathbf{y})$ times the 2×2 determinant of the data at Y of the two reduced columns, and the reduced column of u^{2J+m} is the data of $G_{k,m}$; this gives $\mathfrak{D}_k(\mathbf{y})\mathfrak{W}_k(Y)$. The ratio formula follows from $p_k = (-1)^k \mathfrak{D}_k / \mathfrak{D}_0$ at $\mathbf{y}$ and at $\mathbf{y} \cup \{Y\}$. The difference $\Pi_{\mathbf{y}} u^{2J+m} - I_k(u^{2J+m})$ has zero data and lies in $\text{span}\{1, \dots, u^{2J}\}$, so it is a multiple $cP_{\mathbf{y}}$; comparing u^k -coefficients gives $c = \pi_k^{(m)} / p_k$. For $k = 0$, $G_{0,m} = E_m$. (c) Divide $\mathcal{F}_{\mathbf{y}} = \mathcal{F}_{\mathbf{y}^-} + (\mathcal{F}_{\mathbf{y}} - \mathcal{F}_{\mathbf{y}^-})$ by $\prod_{j \leq J} (1 - y/y_j)^2$. (d) By (a) and (b), $p_{2J+1}(\mathbf{y} \cup \{Y\}) = a$, $p_{2J+2}(\mathbf{y} \cup \{Y\}) = b$, and $p_k(\mathbf{y} \cup \{Y\}) = p_k(\mathbf{y})\mathfrak{W}_k(Y)/\mathcal{W}(Y)$ for $1 \leq k \leq 2J$. $\square$

A.3. The chain and insertion calculus.

Proof of Lemma 4.2. (a) Vanishing on Z imposes K conditions on the $(K+2)$ -dimensional space $\mathcal{A}_{K+1}$, so $\dim \mathcal{S}(Z) \geq 2$; the intersection with $\mathcal{A}_K$ is $\text{span}(\mathcal{F}_Z)$ by the remark above, and has codimension at most 1 in $\mathcal{S}(Z)$. (b) By (a) some $G \in \mathcal{S}(Z)$ has σ_{K+1} -coefficient 1; then $G_Z := G - (\ell_0(G)/\ell_0(\mathcal{F}_Z))\mathcal{F}_Z$ has the required properties, and the difference of two such elements lies in $\text{span}(\mathcal{F}_Z)$ and has $\ell_0 = 0$, hence vanishes. $Z + z$ is regular by Lemma 4.1. The right-hand side of the identity lies in $\sigma_{K+1} + \mathcal{A}_K$, vanishes on Z , and is annihilated by δ_z^Z , so it is $\mathcal{F}_{Z+z}$. The extension of χ_Z follows from $\chi_Z = (G_Z/\omega_Z)/f_Z$ and (4.1). (c) For $\varphi = e^{-y}$, $\ell_0(\mathcal{F}_P) = P(0) = (e^y \mathcal{F}_P)(0)$, and $0 \notin I$ is harmless because all functions involved are entire. $Z + 0$ is regular by Lemma 4.1, since $\mathcal{F}_Z(0) = \omega_Z(0)f_Z(0) \neq 0$, and $\mathcal{F}_{Z+0} \in \mathcal{S}(Z) \cap (\sigma_{K+1} + \mathcal{A}_K)$ has $\ell_0(\mathcal{F}_{Z+0}) = 0$, so $\mathcal{F}_{Z+0} = G_Z$. Divide the identity of (b) by $e^{-y}\omega_Z$. $\square$

Proof of Lemma 4.3. $\mathcal{F}_Z$ depends analytically on x (Cramer's rule, $\Delta(Z) \neq 0$), and its σ_K -coefficient is fixed, so $\partial_x \mathcal{F}_Z \in \mathcal{A}_{K-1}$. Differentiating $\mathcal{F}_Z^{(i)}(x) = 0$ ($i < \mu$) in x gives $(\partial_x \mathcal{F}_Z)^{(i)}(x) = -\mathcal{F}_Z^{(i+1)}(x)$, which is 0 for $i < \mu - 1$ and $-\mathcal{F}_Z^{(\mu)}(x)$ for $i = \mu - 1$; the data at the other points stay zero. So $\partial_x \mathcal{F}_Z$ is an element of $\mathcal{A}_{K-1}$ vanishing on Z^- , hence equal to cG , and evaluating the $(\mu - 1)$ -st derivative at x determines c . For the cofactor form write $\omega_Z = (y - x)^\mu \omega_0$; then $\mathcal{F}_Z^{(\mu)}(x) = \mu! \omega_0(x)f_Z(x)$, $G^{(\mu-1)}(x) = (\mu - 1)! \omega_0(x)g(x)$, $\partial_x \omega_Z = -\mu \omega_Z/(y - x)$ and $G = \omega_Z g/(y - x)$; divide by ω_Z . $\square$

Proof of Lemma 4.4. (i) Applying a data functional of Z to Φ_Z produces a determinant with two equal rows. The cofactor of $\sigma_{K-1}(y)$ in the last row is the minor on the first $K - 1$ rows and columns, which is $\Delta(Z)$; if Z is regular, $\Phi_Z/\Delta(Z) \in \sigma_{K-1} + \mathcal{A}_{K-2}$ vanishes on Z . (ii) The determinant is linear in the last row. (iii) Induction on K with (ii). For the last statement apply (ii) twice: $\Delta(Z + y_J + y_J) = \Delta(Z) \mathcal{F}_Z(y_J) \mathcal{F}'_{Z+y_J}(y_J)$ when Z and $Z + y_J$ are regular, and $Z + y_J$ is regular when $\mathcal{F}_Z(y_J) \neq 0$. $\square$

Proof of Proposition 4.5. (a) By Lemma 4.2, $\mathcal{F}_{Z+z} = \mathcal{F}_Z(\chi_Z - \chi_Z(z))$; divide by $\omega_{Z+z} = \omega_Z \cdot (y - z)$. (b) (1) $\Rightarrow$ (2): $\chi'_Z(z) = f_{Z+z}(z)/f_Z(z) > 0$. (2) $\Rightarrow$ (1): χ_Z is strictly increasing, so the difference quotient in (a) is positive, and at $y = z$ it equals $\chi'_Z(z) > 0$. (2) $\Leftrightarrow$ (3) $\Leftrightarrow$ (4):

$\text{Wr}(\mathcal{F}_Z, G_Z) = \mathcal{F}_Z^2 \chi'_Z = \omega_Z^2 f_Z^2 \chi'_Z$ and $f_Z(y) f_{Z+y}(y) = f_Z(y)^2 \chi'_Z(y)$. (c) is (a) read at one z . (d) $\ell_0(\mathcal{F}_{Z+z}) = \ell_0(G_Z) - \chi_Z(z) \ell_0(\mathcal{F}_Z) = -\chi_Z(z) \ell_0(\mathcal{F}_Z)$, so $\tilde{F}_{Z+z} = \tilde{F}_Z (1 - \chi_Z/\chi_Z(z))$; divide by $(1 - y/z) \prod_{w \in Z} (1 - y/w)$. Then $\tilde{f}_{Z+z}/\tilde{f}_Z - 1 = yz (\chi_Z(y)/y - \chi_Z(z)/z) / (\chi_Z(z)(y - z))$, and $y, z > 0$. $\square$

Proof of Proposition 4.6. (a) $D := P_{\mathbf{y} \cup \{z_i\}} - P_{\mathbf{y}}$ lies in $V := \text{span}\{u, \dots, u^{2J+2r}\}$ and has zero data at $\mathbf{y}$. As in the proof of Lemma 3.4(a), the elements of V with zero data at $\mathbf{y}$ form the $2r$ -dimensional space $\text{span}\{E_1, \dots, E_{2r}\}$. Since $\omega(z_i) \neq 0$, $\omega\rho$ has a double zero at z_i if and only if ρ has. So (N) for the enlarged set, which says that no non-zero element of V has zero data at all nodes, is the non-singularity of the matrix, and $D = \sum a_m E_m$ with $\mathcal{R}_0 + \sum a_m \mathcal{R}_m$ vanishing doubly at the z_i . Finally $[u]E_m = -\ell_{2J+m}$. (b) The case $r = 1$ of (a), with $\text{Wr}(\mathcal{R}_1, \mathcal{R}_2) = \mathcal{W}/\omega^2$. From a and b , $\Delta_{\mathbf{y}}(Y) = -a\ell_1^* - b\ell_2^* = \text{Wr}(\mathcal{F}_{P_{\mathbf{y}}}, \ell_1^* \mathcal{F}_{E_2} - \ell_2^* \mathcal{F}_{E_1})(Y)/\mathcal{W}(Y)$. In the 3×3 Wronskian of $(\mathcal{F}_{P_{\mathbf{y}}}, \mathcal{F}_{E_1}, \mathcal{F}_{E_2})$, add a times the second and b times the third column to the first: the first column becomes the data of $\mathcal{F}_{\mathbf{y} \cup \{Y\}}$, which is $(0, 0, \mathcal{F}_{\mathbf{y} \cup \{Y\}}''(Y))$ at Y . For the weights: the old-node part of $\mathcal{L}_{\mathbf{y} \cup \{Y\}}$ vanishes on E_1, E_2 , and $\mathcal{L}_{\mathbf{y} \cup \{Y\}}(E_m) = [u]E_m = -\ell_{2J+m}$; solve the 2×2 system $\lambda_Y \mathcal{F}_{E_m}(Y) + \mu_Y \mathcal{F}_{E_m}'(Y) = -\ell_{2J+m}$. The last formula is Lemma 3.2(d) for the node Y of $\mathbf{y} \cup \{Y\}$. (c) Under $y = y(s)$ a 2×2 Wronskian is multiplied by dy/ds , and under $\mathcal{F} \mapsto h\mathcal{F}$ by h^2 ; a, b and $\Delta_{\mathbf{y}}$ are ratios of such Wronskians. $\square$

A.4. The half-step calculus.

Proof of Proposition 4.7. Zero spaces. If $D_0(B) \neq 0$, the data map at B is injective on $\text{span}\{\varphi_0, \dots, \varphi_{K-1}\}$, so the elements of $\text{span}\{\varphi_0, \dots, \varphi_{K+m}\}$ vanishing on B form a space of dimension exactly $m+1$; if $D_0, \dots, D_m$ are non-zero it is $\text{span}\{F_0^B, \dots, F_m^B\}$, since these functions are independent (their top terms differ). By Cramer's rule, $[F_r^B]_r = (-1)^K D_{r+1}(B)/D_r(B)$, so $S(B) = [F_1^B]_1/[F_0^B]_0$; and $[F_r^B]_j = 0$ for $j < r$.

(H2) In the matrix of $D_r(B+z)$, add to the last column (the data of φ_{r+K}) the combination of the other columns that turns it into the data of F_r^B ; it becomes $(0, \dots, 0, F_r^B[z])$, and the expansion along it gives the formula. (H1) The right side lies in $\varphi_{r+K+1} + \text{span}\{\varphi_r, \dots, \varphi_{r+K}\}$, vanishes on B and is annihilated by δ_z^B ; by (H2), $D_r(B+z) \neq 0$, so it is the window- r interpolant of $B+z$.

(H3) By (H2), $D_1(B+z) \neq 0$, so P_{B+z} exists. $\mathcal{F}_{P_{B+z}}$ lies in the zero space $\text{span}\{F_0^B, F_1^B\}$: $\mathcal{F}_{P_{B+z}} = \alpha F_0^B + \beta F_1^B$ with $\alpha[F_0^B]_0 = 1$ and $\alpha F_0^B[z] + \beta F_1^B[z] = 0$. Hence $p_1(B+z) = \alpha[F_0^B]_1 + \beta[F_1^B]_1 = p_1(B) - S(B)F_0^B[z]/F_1^B[z]$.

(H4) Write $F_r = F_r^B$. As in (H3), $\mathcal{F}_{P_{B+z+x}} = \alpha F_0 + \beta F_1 + \gamma F_2$ with $\alpha = 1/[F_0]_0$ and $\beta F_1[w] + \gamma F_2[w] = -\alpha F_0[w]$ for $w = z, x$ (the functional at x is the same for B and $B+z$, because $x \neq z$). The determinant of this system is $F_1[z]F_1[x](q_1(x) - q_1(z)) \neq 0$, so $p_1(B+z+x)$ is defined and equals $p_1(B) + \beta[F_1]_1$, with $\beta = -\alpha(F_0[z]F_2[x] - F_0[x]F_2[z])/(F_1[z]F_1[x](q_1(x) - q_1(z)))$. Subtracting the two instances of (H3) and using $q_1 = F_2/F_1$, the numerator of $m_B(z, x)/S(B)$ over the common denominator $F_1[z]F_1[x](q_1(x) - q_1(z))$ is

$$-F_0[z]F_2[x] + F_0[x]F_2[z] + (F_0[z]F_1[x] + F_0[x]F_1[z])(q_1(x) - q_1(z)) = F_1[z]F_1[x](a(x) - a(z)).$$

(H5) The same computation with the conditions “value and derivative at y ” gives the determinant $F_1(y)^2(q_1^B)'(y) \neq 0$ and $\Delta_B(y)/S(B) = -(F_0F_2' - F_0'F_2)/(F_1^2q_1')$, while $h_B(y)/S(B) = -F_0/F_1$. With $q_1' = (F_1F_2' - F_1'F_2)/F_1^2$ one checks $[-(F_0F_2' - F_0'F_2)/F_1^2 + 2(F_0/F_1)q_1'] = a'$, which is the identity. The equivalences follow from the signs, and from $a'/a = F_0'/F_0 - 2F_1'/F_1 + F_2'/F_2$.

(H6) For B doubled ($K = 2J$), Cramer's rule gives $p_{2J}(\mathbf{y}) = D_0/D_1$, and $\mathcal{F}_{E_1} = F_1^B$ (both lie in $\varphi_{2J+1} + \text{span}\{\varphi_1, \dots, \varphi_{2J}\}$ and vanish on B), so $\ell_1^* = -[F_1^B]_1 = -D_2/D_1$ and $S_{\mathbf{y}} = -p_{2J}\ell_1^* = D_0D_2/D_1^2$ by fact (c) after Definition 3.1. The doubled chain of $\mathbf{y} \cup \{z\}$ is $B+z+z$ up to the order of rows, which changes every D_r by the same sign; apply (H2) twice. The

four-term formula is the telescoping of the second difference $p(2, 2) - p(2, 0) - p(0, 2) + p(0, 0)$, $p(i, j) := p_1(B + iz + jx)$, over the four unit squares. $\square$

A.5. The zero side.

Proof of Theorem 9.10. (a) $1 + c_1 u + c_2 u^2 \leq (1 + du)^2$ coefficientwise, and multiplying a coefficientwise inequality by a polynomial with non-negative coefficients preserves it; so by induction and (G1), $P_J \leq P_{J_0} \prod_{J_0 \leq j < J} (1 + d_j u)^2$. At $u = t^2$, with $N(R) := \#\{j : d_j \geq 1/R\}$ (which vanishes for $R < 1/D$),

$$2 \sum_j \log(1 + d_j t^2) = 2 \int_{1/D}^{\infty} N(R) \frac{t^2}{R(R + t^2)} dR \leq 2\pi A|t| + 2B \log(1 + Dt^2),$$

using $\int_0^{\infty} R^{-1/2} t^2 (R + t^2)^{-1} dR = \pi|t|$. Hence $H_J(t) \leq C_\varepsilon e^{a't}$ for $t > 0$, with $a' = 2\pi A + \varepsilon$, and by (G1) $p_k^{(J)} \leq C_\varepsilon \inf_{t>0} e^{a't} t^{-2k} = C_\varepsilon (a'e/2k)^{2k} \leq C_\varepsilon e\sqrt{2k} a'^{2k}/(2k)!$, by $n! \leq e\sqrt{n}(n/e)^n$. As $\sqrt{2k}(a'/a)^{2k}$ is bounded for $a > a'$, (POS_a) follows. (b) Induction on J . If $P_J > 0$, the ratios $r_k := p_k^\natural/p_k^{(J)}$ are positive, non-increasing, and $r_0 = 1$, so $0 < P_J^\natural \leq P_J$. The quadratic $N_J := |1 - u/u_J|^2$ has non-negative coefficients because $\operatorname{Re}(1/u_J) \leq 0$, and positive constant and top coefficients; hence $P_{J+1} = P_J^\natural N_J$ has positive coefficients and $P_{J+1} \leq P_J N_J$. Under (SW-step) positivity of P_J is immediate. The last part is (a) with $c_{1,J} = -2\operatorname{Re}(1/u_J)$, $c_{2,J} = |u_J|^{-2}$, for which $d_J = 1/|u_J|$; under (CNT₁) only finitely many $|u_J|$ are ≤ 1 , so D exists. $\square$

A.6. The far-field expansion.

Proof of Lemma 9.26. By Fact 2.5 and Fact 2.8, $\widehat{\Xi^2 P(t^2)} = \pi \sum_N d(N) N^{-1/2} \mathfrak{F}_P(2\pi N x)$ with $\mathfrak{F}_P := P^\Xi(-\theta^2)[e^{-y}\tilde{\kappa}]$: indeed $K_0 = (\pi/2y)^{1/2} e^{-y}\tilde{\kappa}$, and $\theta(y^{-1/2}h) = y^{-1/2}(\theta - \frac{1}{2})h$ turns $P(-(\theta + \frac{1}{2})^2)\theta^2(\theta + 1)^2$ into $P^\Xi(-\theta^2)$. Next, $\theta(e^{-y}h) = e^{-y}(\theta - y)h$, so $e^y L(\theta)e^{-y} = L(\theta - y)$ for every polynomial L , and $\theta - y$ satisfies $(\theta - y)(fg) = (\theta f)g + f(\theta - y)g$. Hence $(\theta - y)^m \tilde{\kappa} = \sum_i \binom{m}{i} (\theta^i \tilde{\kappa}) T_{m-i}(-y)$, where $T_n(-y) = e^y \theta^n e^{-y}$ is a polynomial of degree n . Watson's lemma applied to $e^y K_0(y) = \int_0^\infty e^{-yv} (v(v+2))^{-1/2} dv$ [18] gives $\tilde{\kappa} = 1 - 1/(8y) + O(y^{-2})$ and $\theta^i \tilde{\kappa} = O(1/y)$ for $i \geq 1$, with expansions that may be differentiated. So $(\theta - y)^m \tilde{\kappa} = \tilde{\kappa} T_m(-y) + O(y^{m-2})$, which gives the first formula. For $P = u^k$, $Q_{P^\Xi} = \tau_{k+2} + \frac{1}{2}\tau_{k+1} + \frac{1}{16}\tau_k$ with $\tau_j = (-1)^j [y^{2j} - \binom{2j}{2} y^{2j-1} + \dots]$, and multiplying by $\tilde{\kappa}$ gives γ_k . For $N \geq 2$, $|\theta^j K_0(y)| \leq C_j (1+y)^j e^{-y}$ ($y \geq 1$) bounds the terms by $Cy^{C'} \sum_{N \geq 2} d(N) N^{C'} e^{-(N-1)y}$ relative to $N = 1$. The size of the second-order term comes from $\binom{m}{2}$ -type coefficients with $m = 2k + 4$. $\square$

We now prove Theorem 9.27. Fix $\mathbf{y}$ of size J with $(N)_{\mathbf{y}}$ and $p := p_{2J}(\mathbf{y}) \neq 0$, and write y for the variable and Y for the new node. Put $f_0 := e^y \mathcal{F}_{P_{\mathbf{y}}}/\pi$ and $f_m := e^y \mathcal{F}_{E_m}/\pi$ ($m \geq 1$), and $g(k) := \gamma_k = 2k^2 + 7k + 6 + \frac{1}{8}$ (Lemma 9.26), so that $g(k+1) - g(k) = 4k + 9$. With $M := 4J + 4$ and $k := 2J$, Lemma 9.26 gives, with expansions that may be differentiated termwise,

$$\begin{aligned} f_0 &= p y^M (1 + \frac{a_1}{y} + O(y^{-2})), & f_1 &= -y^{M+2} (1 + \frac{b_1}{y} + \dots), \\ f_2 &= y^{M+4} (1 + \frac{c_1}{y} + \dots), & f_3 &= -y^{M+6} (1 + \frac{d_1}{y} + \dots), \end{aligned}$$

with $a_1 = -g(k)$, $b_1 = -g(k+1)$, $c_1 = -g(k+2)$, $d_1 = -g(k+3)$. Indeed $E_m = u^{2J+m} - \Pi_{\mathbf{y}} u^{2J+m}$ and $\Pi_{\mathbf{y}} u^{2J+m}$ has degree $\leq 2J$, so its contribution is $O(y^M)$; in f_0 the term $p_{2J-1} u^{2J-1}$ contributes relatively $O(y^{-2})$. The divisor terms are relatively $O(e^{-y} \operatorname{poly}(y))$ and do not affect any of the expansions below.

Two-term Wronskian rule. If $f = y^m(1 + f_1/y + O(y^{-2}))$ and $h = y^n(1 + h_1/y + O(y^{-2}))$ with differentiable expansions, then $\text{Wr}(f, h) = (n - m)y^{m+n-1} + [(n - m + 1)f_1 + (n - m - 1)h_1]y^{m+n-2} + O(y^{m+n-3})$. Hence

$$\begin{aligned}\text{Wr}(f_1, f_2) &= -[2y^{2M+5} + (3b_1 + c_1)y^{2M+4} + \dots], \\ \text{Wr}(f_0, f_2) &= p[4y^{2M+3} + (5a_1 + 3c_1)y^{2M+2} + \dots], \\ \text{Wr}(f_0, f_1) &= -p[2y^{2M+1} + (3a_1 + b_1)y^{2M} + \dots].\end{aligned}$$

The first shows that $\mathcal{W}(Y) = \pi^2 e^{-2Y} \text{Wr}(f_1, f_2)(Y) \neq 0$ for large Y , which is (N) $_{\mathbf{y} \cup \{Y\}}$ (Proposition 4.6(b)).

(a) and (b). By Lemma 3.4(a) and Proposition 4.6(c), $a = -\text{Wr}(f_0, f_2)/\text{Wr}(f_1, f_2)$ and $b = \text{Wr}(f_0, f_1)/\text{Wr}(f_1, f_2)$ at Y . Hence $a = 2pY^{-2}[1 + (5a_1 + c_1 - 6b_1)/(4Y) + O(Y^{-2})]$ and $b = pY^{-4}[1 + (3a_1 - 2b_1 - c_1)/(2Y) + O(Y^{-2})]$. Now $5a_1 + c_1 - 6b_1 = 6(g(k+1) - g(k)) - (g(k+2) - g(k)) = 6(4k+9) - (8k+22) = 16k+32 = 32(J+1)$ and $3a_1 - 2b_1 - c_1 = 2(4k+9) + (8k+22) = 16k+40 = 32J+40$, which gives (a). Then $\Delta_{\mathbf{y}}(Y) = -\ell_1^* a - \ell_2^* b$, the term $\ell_2^* b$ is $O(Y^{-4})$, and $S_{\mathbf{y}} = -p\ell_1^*$, which gives (b).

(c). Let $f := f_0 + af_1 + bf_2 = e^y \mathcal{F}_{\mathbf{y} \cup \{Y\}}/\pi$, which has a double zero at Y ; there $\mathcal{F}_{\mathbf{y} \cup \{Y\}}''(Y) = \pi e^{-Y} f''(Y)$. By (a), $f''(Y) = pY^{M-2}[M(M-1) - 2(M+2)(M+1) + (M+4)(M+3)](1 + O(1/Y)) = 8pY^{4J+2}(1 + O(1/Y))$. The expansion of $\Delta_{\mathbf{y}}$ may be differentiated, since a and b are ratios of Wronskians of functions with differentiable expansions; so $\Delta_{\mathbf{y}}'(Y) = -4S_{\mathbf{y}}Y^{-3} + O(Y^{-4})$, and Proposition 4.6(b) gives $\mu_Y = -\Delta_{\mathbf{y}}'(Y)/\mathcal{F}_{\mathbf{y} \cup \{Y\}}''(Y)$. The ξ -forms follow from $d/d\xi = 2\pi y d/dy$ at a double zero.

(d). By Lemma 3.4(a), $p_{2J+2}(\mathbf{y} \cup \{Y\}) = b$. As in the proof of Proposition 4.6, $\Pi_{\mathbf{y} \cup \{Y\}} u^{2J+3} = \Pi_{\mathbf{y}} u^{2J+3} + \alpha E_1 + \beta E_2$, where $\alpha f_1 + \beta f_2$ has the value and derivative of f_3 at Y ; so $\alpha = \text{Wr}(f_3, f_2)/\text{Wr}(f_1, f_2) = -Y^4[1 + (d_1 + 2c_1 - 3b_1)/(2Y) + O(Y^{-2})]$ and $\beta = O(Y^2)$. Here $d_1 + 2c_1 - 3b_1 = -(g(k+3) - g(k+1)) - 2(g(k+2) - g(k+1)) = -(16k+56)$, so $\alpha = -Y^4(1 - (16J+28)/Y + O(Y^{-2}))$. Then $\ell_{2J+3}(\mathbf{y} \cup \{Y\}) = \ell_3^* - \alpha\ell_1^* - \beta\ell_2^*$, with $\ell_3^* := \ell_{2J+3}(\mathbf{y})$, and fact (c) after Definition 3.1 gives

$$S_{\mathbf{y} \cup \{Y\}} = -b(\ell_3^* - \alpha\ell_1^* - \beta\ell_2^*) = -p\ell_1^*(1 + \frac{16J+20}{Y})(1 - \frac{16J+28}{Y}) + O(Y^{-2}) = S_{\mathbf{y}}(1 - \frac{8}{Y}) + O(Y^{-2}).$$

(e). If $p > 0$ and $S_{\mathbf{y}} > 0$, then by (b) and (c) $\Delta_{\mathbf{y}}(Y) > 0$, $\mathcal{F}_{\mathbf{y} \cup \{Y\}}''(Y) > 0$ and $\mu_Y > 0$ for large Y ; if $p > 0$ and $S_{\mathbf{y}} < 0$, then $\Delta_{\mathbf{y}}(Y) < 0$ and $\mu_Y < 0$. In the model, $g(k)$ is replaced by $\binom{2k}{2}$, whose first differences are $4k+1$ instead of $4k+9$; the same computation gives $32J$ in place of $32(J+1)$ (Theorem 7.1).

A.7. Proofs for the model.

Proof of Proposition 5.2. (a) Expand $e^{-y} = \sum_j (-y)^j/j!$ and use $\theta(-y)^j = j(-y)^j$. (d) By (a), $F_P = \sum_m b_m \sum_{j \geq m} (-y)^j/(j-m)! = \sum_m b_m (-y)^m e^{-y}$. (b) Integration by parts gives $\int_0^\infty y^{s-1} \theta g dy = -s \int_0^\infty y^{s-1} g dy$ for $g = Qe^{-y}$ and $\text{Re } s > 0$, so the Mellin transform of $P(-\theta^2)e^{-y}$ is $P(-s^2)\Gamma(s)$. Since $\int_0^\infty y^{s-1} y^m e^{-y} dy = \Gamma(s) s(s+1) \cdots (s+m-1)$, the Mellin transform of Qe^{-y} is $\Gamma(s)M_Q(s)$. The rising factorials form a basis of the polynomials, so $Q \mapsto M_Q$ is a bijection preserving degree; $M_{Q_P}(s) = P(-s^2)$ is even, and conversely an even M_Q of degree $\leq 4J$ equals $P(-s^2)$ for a unique P of degree $\leq 2J$, whence $Q = Q_P$. (c) Since $s(s+1) \cdots (s+k-1) = \sum_i |s(k, i)|s^i$, evenness of M_Q means $\sum_{k \geq 2l-1} |s(k, 2l-1)|q_k = 0$ for every l , and $|s(2l-1, 2l-1)| = 1$. $\square$

Proof of Proposition 5.3. For $0 < \text{Re } s < \frac{1}{4}$ all of the following integrals converge absolutely. From $\int_0^\infty J_0(2\sqrt{Y})Y^{s-1} dY = \Gamma(s)/\Gamma(1-s)$ and Fubini, the Mellin transform of $\mathcal{H}_{0g}$ is

$\frac{\Gamma(s)}{\Gamma(1-s)}\mathcal{M}g(1-s)$, where $\mathcal{M}g(s) := \int_0^\infty y^{s-1}g(y)dy$. With $\mathcal{M}F(s) = \Gamma(s)M_Q(s)$ (proof of Proposition 5.2) we get $\mathcal{M}[\mathcal{H}_0F](s) = \Gamma(s)M_Q(1-s)$; and since $-F' = (Q-Q')e^{-y}$ and $M_{Q-Q'}(s) = M_Q(s-1)$ (check on monomials: $s^{(m)} - ms^{(m-1)} = (s-1)^{(m)}$, with $s^{(m)} := s(s+1)\cdots(s+m-1)$), also $\mathcal{M}[-F'](s) = \Gamma(s)M_Q(s-1)$. By injectivity of the Mellin transform, (ii) holds if and only if $M_Q(1-s) = M_Q(s-1)$, i.e. if and only if M_Q is even, which is (i) by Proposition 5.2(b). For (iii): the same computation gives $\mathcal{M}[\mathcal{H}_0(y^m e^{-y})](s) = \Gamma(s)(1-s)(2-s)\cdots(m-s)$, while $\mathcal{M}[m!L_m(y)e^{-y}](s) = \Gamma(s)m!\sum_k \binom{m}{k} \frac{(-1)^k}{k!} s(s+1)\cdots(s+k-1) = \Gamma(s)(1-s)\cdots(m-s)$ by the Chu–Vandermonde identity; hence $\mathcal{H}_0F = (\mathcal{L}Q)e^{-y}$, and $-F' = (Q-Q')e^{-y}$. Finally, for radial $f(v) = g(|v|)$ one has $\hat{f}(\eta) = 2\pi \int_0^\infty g(r)J_0(2\pi r|\eta|)rdr$, and the substitution $w = \pi r^2$ gives $\hat{f}(\eta) = \mathcal{H}_0F(\pi|\eta|^2)$. A double zero of Q at $2\pi n$ means $F(2\pi n) = F'(2\pi n) = 0$, i.e. $f = 0$ on $\pi|v|^2 = 2\pi n$ and, by (ii), $\hat{f} = 0$ on $\pi|\eta|^2 = 2\pi n$. $\square$

Proof of Proposition 5.5. (a) Substituting $t = 1 + v/z$ in $K_0(z) = \int_1^\infty e^{-zt}(t^2 - 1)^{-1/2} dt$ [10, 10.32.8] gives the integral; the integrand increases with z and tends to $e^{-v}v^{-1/2}$, and $\int_0^\infty e^{-v}v^{-1/2}dv = \sqrt{\pi}$. The derivatives of $w \mapsto (1+w)^{-1/2}$ alternate in sign and are monotone on $[0, \infty)$, so $(1+w)^{-1/2} = 1 - \frac{w}{2} + \frac{3w^2}{8} - r(w)$ with $0 \leq r(w) \leq \frac{5w^3}{16}$; with $w = v/(2z)$ and $\int_0^\infty e^{-v}v^{k-1/2}dv = \Gamma(k + \frac{1}{2})$ this gives the expansion.

(b) With the Fourier normalisation of Section 2.1, $\widehat{t^2 F} = -\theta^2 \widehat{F}$, so $\widehat{\Xi^2 P(t^2)} = P(-\theta^2)\Psi$ with $\theta = x \frac{d}{dx}$. The operator θ commutes with dilations and $\theta(\sqrt{x}h) = \sqrt{x}(\theta + \frac{1}{2})h$, so $P(-\theta^2)\Psi = 2\pi\sqrt{x}\sum_N d(N)g_P(2\pi Nx)$ with $g_P := P(-(\theta + \frac{1}{2})^2)\theta^2(\theta + 1)^2K_0$. Write $K_0(z) = \sqrt{\pi/2}z^{-1/2}e^{-z}\tilde{\kappa}(z)$. Since $\theta(z^{-1/2}h) = z^{-1/2}(\theta - \frac{1}{2})h$,

$$\begin{aligned} g_P &= \sqrt{\pi/2}z^{-1/2}P(-\theta^2)(\theta - \frac{1}{2})^2(\theta + \frac{1}{2})^2[e^{-z}\tilde{\kappa}] \\ &= \sqrt{\pi/2}z^{-1/2}\mathfrak{F}_P(z), \end{aligned}$$

because $P(-\theta^2)(\theta^2 - \frac{1}{4})^2 = P^\Xi(-\theta^2)$. Hence

$$\widehat{\Xi^2 P(t^2)}(\xi) = 2\pi\sqrt{x}\sum_N d(N)\sqrt{\pi/2}(2\pi Nx)^{-1/2}\mathfrak{F}_P(Ny),$$

and $2\pi\sqrt{\pi/2}(2\pi)^{-1/2} = \pi$.

(c) By $\int_0^\infty K_0(z)z^{s-1}dz = 2^{s-2}\Gamma(\frac{s}{2})^2$ [10, 10.43.19], the Mellin transform of $e^{-z}\tilde{\kappa}(z) = \sqrt{2/\pi}z^{1/2}K_0(z)$ is

$$\sqrt{2/\pi}2^{s-3/2}\Gamma(\frac{s+1/2}{2})^2,$$

and the duplication formula $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$ shows that this equals $m(s)\Gamma(s)$, the symbol times the Mellin transform of e^{-z} . A Mellin multiplier commutes with θ , so $\mathfrak{F}_P = P^\Xi(-\theta^2)m[e^{-z}] = m[P^\Xi(-\theta^2)e^{-z}] = m[Q_{P^\Xi}e^{-z}]$. Log-convexity of Γ gives $\Gamma(\frac{s+1/2}{2})^2 \leq \Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$, i.e. $m(s) \leq 1$, and Stirling's formula gives $\log m(s) = O(1/s)$. $\square$

Proof of Proposition 5.6. (a) There is an open interval in (z_0, ∞) on which $f > 0$; the integrand is non-negative and positive on a set of positive measure. (b) Substituting $v = z(t-1)$ in Proposition 5.5(a) gives $e^{-z}\tilde{\kappa}(z) = \pi^{-1/2}z^{1/2}\int_1^\infty e^{-zt}(t-1)^{-1/2}(\frac{t+1}{2})^{-1/2}dt = T_k g(z)$. The integrals converge absolutely, locally uniformly in $z > 0$, for g replaced by $w^{1/2}$ times any polynomial times e^{-w} , so we may differentiate under the integral sign; since $z \frac{d}{dz}f(zt) = (\theta f)(zt)$, T_k commutes with θ . Finally $P^\Xi(-\theta^2)[w^{1/2}e^{-w}] = w^{1/2}P^\Xi(-(\theta + \frac{1}{2})^2)e^{-w} = w^{1/2}Q_P^{\text{Ab}}(w)e^{-w}$. (c) Apply (a) to $f = w^{1/2}Q_P^{\text{Ab}}e^{-w}$: since Q_P^{Ab} is a non-zero polynomial, $f \geq 0$ on (z_0, ∞) would force $\mathfrak{F}_P(z_0) > 0$. $\square$

Proof of Proposition 5.8. With $\tau_1 = y - y^2$ and $\tau_2 = y^4 - 6y^3 + 7y^2 - y$, the system $Q(a) = Q'(a) = 0$ for $Q = 1 + p_1\tau_1 + p_2\tau_2$ has determinant $\tau_1(a)\tau_2'(a) - \tau_1'(a)\tau_2(a) = -a^2D(a)$, and Cramer's rule gives p_1, p_2 ; dividing Q by $(y-a)^2$ gives R . The p_1 -rule $\lambda_1\delta_a + \mu_1\delta'_a$ is determined by $\lambda_1\tau_k(a) + \mu_1\tau'_k(a) = \delta_{k1}$ ($k = 1, 2$), which gives $\mu_1 = -\tau_2(a)/(-a^2D(a))$; $Q''(a)$ is computed directly, and $S_{(a)} = -p_2\ell_3$ with $\ell_3 = \lambda_1\tau_3(a) + \mu_1\tau'_3(a)$, $\tau_3 = -y^6 + 15y^5 - 65y^4 + 90y^3 - 31y^2 + y$ (Definition 3.1(iii)). Since $D'(a) = 6(a-1)(a-2)$ and $D(1), D(2) < 0$, D has one real root, and $D(2.6776) < 0 < D(2.6777)$. (a) The constant coefficient of R is $a^{-2} > 0$. If $a > a_D$, then $D > 0$ and (R^+) holds if and only if $a > 3$. If $a < a_D$, then $D < 0$, and either $2a - 1 > 0$ and the leading coefficient is negative, or $2a - 1 \leq 0$ and the middle coefficient $2(2a-1)(a-3)/(a^2D)$ is ≤ 0 . (b) $4a^3 - 12a^2 + 9a - 3$ has a single real root, $2.0979\dots < a_D$. (c) For $a > a_D$, $\mu_1 > 0$ if and only if $a^3 - 6a^2 + 7a - 1 > 0$, whose roots are $0.1657\dots, 1.3433\dots$ and $4.49086\dots$; combine with (b). (d) The expansion is immediate from the formula. For $a > a_D$ the sign of $S_{(a)}$ is that of the degree-7 factor, which has exactly one real root in (a_D, ∞) (Sturm's theorem), and changes sign between 3.2651 and 3.2652 . $\square$

Proof of Proposition 5.14. Let $\pi^*(x)$ be the number of prime powers $n \leq x$, $n \geq 2$. Since every prime $p \geq 5$ is $\equiv \pm 1 \pmod{6}$, $\pi(x) \leq x/3 + 2$ for $x \geq 1$; and $\pi^*(x) = \sum_{k \geq 1} \pi(x^{1/k})$, where the terms with $k > \log_2 x$ vanish, so

$$\pi^*(x) \leq b(x) := \frac{x}{3} + 2 + \sqrt{x} + (\log_2 x) x^{1/3} \quad (x \geq 1).$$

Each term of $b(x)/x$ is non-increasing for $x \geq e^{3/2}$. (Sharper explicit bounds, such as $\pi(x) < 1.25506 x / \log x$ for $x > 1$ [20, (3.6)], would shorten the finite range below; the elementary bound suffices.) (a) Put $c_0 := 8.4/(2\pi)$, so that $2\pi c_0 K = y_K^* + 5.25$. At $x = 120$, $b(120) < 40 + 2 + 11 + 7 \cdot 5 = 88 < 120/c_0 = 89.75\dots$ (using $\log_2 120 < 7$, $120^{1/3} < 5$); hence $\pi^*(x) < x/c_0$ for all $x \geq 120$. If $K \geq 90$ then $x = c_0 K > 120$, so fewer than K prime powers are $\leq c_0 K$, i.e. $n_K > c_0 K$ and $2\pi n_K > y_K^* + 5.25$. For $K \leq 89$ the inequality is checked directly. (b) Put $c_1 := 1.575$; then $2\pi c_1 > 9.896 > 1.2366 \cdot 8$. At $x = 250$, $b(250) < 83.34 + 2 + 15.82 + 8 \cdot 6.3 < 151.6 < 250/c_1$, so as in (a) $n_j > c_1 j$ for $j \geq 159$, and then $2\pi n_j - 1.2366(8j - 5) > 0$. Since $\sum_{j \leq K} (8j - 5) = K(4K - 1)$, the difference $\sum_{j \leq K} (2\pi n_j - 1.2366(8j - 5))$ is increasing for $K \geq 158$, and it suffices to check (b) for $K \leq 158$. The same argument locates the minimum of the ratio for every K : with $r := 124\pi/315 = 1.236694\dots$, the ratio at $K = 9$, we have $8r < 9.8936 < 9.896 < 2\pi c_1$, so $\sum_{j \leq K} (2\pi n_j - r(8j - 5))$ is also increasing for $K \geq 158$, and it is non-negative for all K once it is non-negative for $K \leq 158$. Both finite checks (in fact for all $K \leq 2.17 \cdot 10^5$) are exact integer computations with rational bounds for π (ancillary file `toy/pp/verify_pp_prefix.py` with `toy/pp/params/pp_prefix.json`); they also locate the minima in (a) and (b). $\square$

Proof of the reduction in Remark 5.15. Argue by induction on J . Removing a node maps $\mathcal{U}_J$ into $\mathcal{U}_{J-1}$, so by induction every $\mathbf{y} \setminus y_j$ is regular with positive cofactor. Fix $\mathbf{y} \in \mathcal{U}_J$ and put $\mathbf{y}(\tau) := \mathbf{y}^* + \tau(\mathbf{y} - \mathbf{y}^*)$, $\tau \in [0, 1]$, which stays in $\mathcal{U}_J$ and is coordinatewise non-decreasing in τ . Let S be the set of τ such that $\mathbf{y}(\tau')$ is regular with $\tilde{R} > 0$ for all $\tau' \leq \tau$; $0 \in S$ by $(AP^+)_J$, and S is relatively open. On S , (LOC) gives $\frac{d}{d\tau} \tilde{R}(\mathbf{y}(\tau)) = \sum_j (y_j - y_j^*) \partial_j \tilde{R} \in \mathbb{R}_{\geq 0}[y]$, so $\tilde{R}(\mathbf{y}(\tau)) \geq \tilde{R}(\mathbf{y}^*)$ coefficientwise. Let $\tau_1 := \sup S$, $x := y_J(\tau)$, and let T be the half-step cofactor of $\mathbf{y}(\tau)$ (nodes $y_1, \dots, y_{J-1}$ double, y_J simple), which exists by Lemma 4.1. By Lemmas 4.1 and 4.4, $\mathbf{y}(\tau_1)$ is regular as soon as $T(x) \neq 0$ at τ_1 ; and once it is regular, continuity gives $\tilde{R}(\mathbf{y}(\tau_1)) \geq \tilde{R}(\mathbf{y}^*) > 0$, so $\tau_1 \in S$ and hence $\tau_1 = 1$. Now $T(x) \neq 0$ on S (a chain value, Lemma 4.4) and $T(x) > 0$ at $\tau = 0$ by $(B)_J$, so $T(x) > 0$ on S and $\kappa := \tilde{R}(x)/T(x) > 0$. By the ladder identity (Lemma 4.3), $\partial_{y_J} \tilde{R} = 2(\tilde{R} - \kappa T)/(y - x)$, so (LOC) gives $q_\tau := (\rho_\tau - T)/(y - x) \in \mathbb{R}_{\geq 0}[y]$ with $\rho_\tau := T(x)\tilde{R}/\tilde{R}(x)$. If $T(x) \rightarrow 0$ as $\tau \uparrow \tau_1$, then $\rho_\tau \rightarrow 0$ coefficientwise (the coefficients of $\tilde{R}/\tilde{R}(x)$

lie in $(0, x^{-k}]$, and by synthetic division $[y^{2J-2}]q_\tau = [y^{2J-1}](\rho_\tau - T) + x[y^{2J}](\rho_\tau - T) \rightarrow -1$, since T is monic of degree $2J - 1$; a contradiction. $\square$

Proof of Lemma 6.2. Under (I0), $f(0) > 0$ and $f(z) > 0$, so $Z + 0$ and $Z + z$ are regular (Lemma 4.1) and (6.1) applies; $\lambda = \lambda(z) > 0$ since $g(z) > 0$. Put $N := \lambda f - yg = (z - y)f_{Z+z}$, so $N(z) = 0$ and $N_k = \lambda f_k - g_{k-1}$ (with $f_{K+1} = f_{K+2} = 0$, $g_{K+1} = 1$, $g_{-1} = 0$). Expanding $1/(z - y)$ at $y = 0$,

$$(A.1) \quad [y^k]f_{Z+z} = z^{-k-1} \sum_{l \leq k} N_l z^l.$$

(i) In particular $f_{Z+z}(0) = \lambda f_0/z > 0$, and a short computation from (A.1) gives, for $1 \leq k \leq K + 1$,

$$[y^k]\hat{f}_{Z+z} - [y^k]\hat{f}_Z = \frac{z^{-k}}{\lambda f_0} \sum_{m=0}^{k-1} w_m, \quad w_m := z^m(\lambda f_m - z g_m).$$

For $m \leq K$, $w_m = z^{m+1}f_m(\lambda/z - \varrho_m)$ has the sign of $\lambda/z - \varrho_m$, which is non-increasing in m ; $w_{K+1} = -z^{K+2} < 0$; and $\lambda/z = g(z)/f(z) = (\sum_{k \leq K} \varrho_k f_k z^k + z^{K+1})/f(z) > \varrho_0$, so $w_0 > 0$. Thus the signs of (w_m) change once, from $+$ to $-$, and $\sum_{m=0}^{K+1} w_m = \lambda f(z) - z g(z) = 0$. Hence every partial sum $\sum_{m < k} w_m$, $1 \leq k \leq K + 1$, is positive: it is $\geq w_0$ before the sign change and $\geq -w_{K+1}$ after it. So $\hat{f}_{Z+z}$ exceeds $\hat{f}_Z$ coefficientwise in degrees $1, \dots, K + 1$, and $f_{Z+z} \in \mathbb{R}_{>0}[y]$. (ii) Now $N_0 = \lambda f_0 > 0$, $N_k = f_k(\lambda - \sigma'_{k-1})$ for $1 \leq k \leq K$ has non-increasing sign, $N_{K+1} = -g_K < 0$ and $N_{K+2} = -1$; so (N_l) changes sign once, and the partial sums $\sum_{l \leq k} N_l z^l$ rise and then fall to $N(z) = 0$, the last one before that being $z^{K+2} > 0$. By (A.1) all coefficients of f_{Z+z} are positive. (iii) $g_1 > 0$ and (II') give $\varrho_k > 0$ for $k \geq 1$, so $\sigma'_k = \varrho_k \cdot (f_k/f_{k+1})$ is a product of positive non-decreasing factors for $k \geq 1$ (log-concavity is $f_{k-1}/f_k \leq f_k/f_{k+1}$). If $g_0 \leq 0$ then $\sigma'_0 \leq 0 < \sigma'_1$; if $g_0 > 0$ the product argument applies to $k = 0$ as well. $\square$

Proof of Theorem 6.3. At $K = 0$, $f^{(0)} = 1$ and $g^{(0)} = y - 1$, since the monic element of $\mathcal{A}_1^{\text{pol}}$ vanishing at 0 is $y^2 - y = -\tau_1$; so (I0)₀ holds. If (I0)_K holds and $K < K_0$, Lemma 6.2(i) with $z = z_{K+1}$, (II')_K and (Y0)_K gives regularity of Z_{K+1} , (I0)_{K+1} and the half-step increment. For a doubled chain, $f^{(2J)} = \hat{R}_{\mathbf{y} \leq J}$. Regularity means that the space of P with $\deg P \leq 2J$ and Q_P vanishing doubly at $\mathbf{y} \leq J$ is spanned by one element P^{top} with $Q_{P^{\text{top}}} = A\tilde{R}$; its value at 0 is $Q_{P^{\text{top}}}(0) = A(0)\tilde{R}(0) > 0$, so no non-zero element of the space vanishes at $u = 0$, which is (N); then $P_{\mathbf{y} \leq J} = P^{\text{top}}/(A(0)\tilde{R}(0))$, $p_{2J} = 1/(A(0)\tilde{R}(0)) > 0$ and $R = p_{2J}\tilde{R} \in \mathbb{R}_{>0}[y]$. Finally $\hat{R}_J - \hat{R}_{J-1} = (\hat{f}^{(2J)} - \hat{f}^{(2J-1)}) + (\hat{f}^{(2J-1)} - \hat{f}^{(2J-2)})$ is positive in degrees $1, \dots, 2J$ and $1, \dots, 2J - 1$ respectively. $\square$

Proof of Theorem 7.1. By (7.1), for monic $f = y^n + f_1 y^{n-1} + \dots$ and $g = y^{n'} + g_1 y^{n'-1} + \dots$ one has $\text{Wr}(f, g) = (n' - n)y^{n+n'-1} + [(n' - n + 1)f_1 + (n' - n - 1)g_1]y^{n+n'-2} + \dots$. Hence $\mathcal{W} = -[2y^{4J+5} + (3\kappa_1 + \kappa_2)y^{4J+4} + \dots]$, $\text{Wr}(R_0, R_2) = p_{2J}[4y^{4J+3} + (5\kappa_0 + 3\kappa_2)y^{4J+2} + \dots]$ and $\text{Wr}(R_0, R_1) = -p_{2J}[2y^{4J+1} + \dots]$. By (7.2), $a = \frac{2p_{2J}}{Y^2}(1 + \frac{5\kappa_0 - 6\kappa_1 + \kappa_2}{4Y} + O(Y^{-2}))$ and $b = \frac{p_{2J}}{Y^4}(1 + O(Y^{-1}))$, and $5\kappa_0 - 6\kappa_1 + \kappa_2 = -5\binom{4J}{2} + 6\binom{4J+2}{2} - \binom{4J+4}{2} = 32J$. Since $[u]E_m = -\ell_{2J+m}$, $\Delta_{\mathbf{y}} = -a\ell_1^* - b\ell_2^*$, which gives (2). For (3), $Q_{\mathbf{y} \cup \{Y\}} = A_{\mathbf{y}}\rho_Y$ with $\rho_Y := R_0 + aR_1 + bR_2$, which vanishes doubly at Y , so $Q''_{\mathbf{y} \cup \{Y\}}(Y) = A_{\mathbf{y}}(Y)\rho''_Y(Y)$, and $\rho''_Y(Y) = p_{2J}Y^{2J-2}[2J(2J - 1) - 2(2J + 2)(2J + 1) + (2J + 4)(2J + 3)] + O(Y^{2J-3}) = 8p_{2J}Y^{2J-2} + O(Y^{2J-3})$. A rational function may be differentiated termwise in its expansion at ∞ , which gives Δ'_Y , and $\mu_Y = -\Delta'_Y(Y)/Q''_{\mathbf{y} \cup \{Y\}}(Y)$. $\square$

Proof of Proposition 7.3. For $z > Y_{\mathcal{W}}(\mathbf{y})$, $p_{2J+2}(\mathbf{y} \cup \{z\}) = b$, and the projection of u^{2J+3} for the node set $\mathbf{y} \cup \{z\}$ is $\Pi_{\mathbf{y}} u^{2J+3} + \alpha E_1 + \beta E_2$, where $R_3 - \alpha R_1 - \beta R_2$ vanishes doubly at z . Hence $\ell_{2J+3}(\mathbf{y} \cup \{z\}) = \ell_3^* - \alpha \ell_1^* - \beta \ell_2^*$ (with $\ell_3^* := \ell_{2J+3}$) and $S_{\mathbf{y} \cup \{z\}} = -b(\ell_3^* - \alpha \ell_1^* - \beta \ell_2^*)$. By Cramer's rule $\alpha = \text{Wr}(R_3, R_2)/\mathcal{W}$ and $\beta = \text{Wr}(R_1, R_3)/\mathcal{W}$; by (7.1) and the Wronskian expansion, $\alpha = -z^4(1 - \frac{3\kappa_1 - 2\kappa_2 - \kappa_3}{2z} + O(z^{-2}))$, $\beta = O(z^2)$, and $b = \frac{p_{2J}}{z^4}(1 + \frac{3\kappa_0 - 2\kappa_1 - \kappa_2}{2z} + O(z^{-2}))$. So $S_{\mathbf{y} \cup \{z\}} = -p_{2J}\ell_1^*(1 + \frac{3\kappa_0 - 5\kappa_1 + \kappa_2 + \kappa_3}{2z}) + O(z^{-2})$, and $3\kappa_0 - 5\kappa_1 + \kappa_2 + \kappa_3 = -3\binom{4J}{2} + 5\binom{4J+2}{2} - \binom{4J+4}{2} - \binom{4J+6}{2} = -16$. $\square$

Proof of Proposition 7.4. By Proposition 4.6(a) with two new nodes, $P_{\mathbf{y} \cup \{z, s\}} = P_{\mathbf{y}} + \sum_{m \leq 4} a_m E_m$, where $\rho := R_0 + \sum_{m \leq 4} a_m R_m$ vanishes doubly at z and s , and $p_1(\mathbf{y} \cup \{z, s\}) - p_1(\mathbf{y}) = -\sum_m a_m \ell_{2J+m}$. Write $a_m = (-1)^m p_{2J} c_m$ and $c_0 := 1$. By (7.1), $\rho = p_{2J} y^{2J} \sum_{m=0}^4 c_m y^{2m} (1 + \kappa_m y^{-1} + O(y^{-2}))$. After the scaling $y = L\eta$, and with the double-zero conditions written as divided differences, this is a linear system for (c_m) whose matrix is a polynomial in $1/L$ and equals, at $1/L = 0$, a confluent Vandermonde matrix in η^2 (non-singular uniformly on compacts, the diagonal included). So each c_m has an expansion in powers of $1/L$, uniformly with all derivatives. At leading order, $\sum_m c_m^{(0)} v^m = \varphi(v) := (1 - v/z^2)^2 (1 - v/s^2)^2$ ($v = y^2$), so $c_1^{(0)} = -2(z^{-2} + s^{-2})$ and $c_2^{(0)} = O(L^{-4})$. At the next order, the corrections $c_m^{(1)}$ make $h(v) := \sum_{m \geq 1} c_m^{(1)} v^m$ interpolate $v^{-1/2} \chi(v)$ at the double nodes $Z := z^2$, $S := s^2$, where $\chi(v) := \sum_m \binom{4J+2m}{2} \varphi_m v^m = \mathcal{D}\varphi$, $\mathcal{D} := 2\theta_v^2 + (8J-1)\theta_v + (8J^2-2J)$, $\theta_v = v \frac{d}{dv}$; the part $\sigma\varphi$ of $\sum_m \kappa_m \varphi_m v^m$ vanishes doubly at the nodes and contributes nothing. Write $h = vk$ with $\deg k \leq 3$; since k interpolates $v^{-3/2} \chi$ and $k(\xi)/(\xi \omega(\xi))$, with $\omega(\xi) := (\xi - Z)^2 (\xi - S)^2$, has residue sum 0,

$$c_1^{(1)} = k(0) = -\omega(0) \sum_{\xi_0 \in \{Z, S\}} \text{Res}_{\xi=\xi_0} \frac{\chi(\xi)}{\xi^{5/2} \omega(\xi)} = -\sum_{\xi_0} \text{Res}_{\xi=\xi_0} \xi^{-5/2} \frac{\mathcal{D}\omega}{\omega}(\xi).$$

With $\Lambda := \omega'/\omega = \frac{2}{\xi-Z} + \frac{2}{\xi-S}$ one has $\frac{\mathcal{D}\omega}{\omega} = 2\xi^2(\Lambda' + \Lambda^2) + (8J+1)\xi\Lambda + (8J^2-2J)$, and the residues give $c_1^{(1)} = -16J(z^{-3} + s^{-3}) + 16/(zs(z+s))$. Since $S_{\mathbf{y}} = -p_{2J}\ell_1^*$, $p_1(\mathbf{y} \cup \{z, s\}) - p_1(\mathbf{y}) = -S_{\mathbf{y}}(c_1^{(0)} + c_1^{(1)}) + O(L^{-4})$, which is the formula. The mixed derivative follows from $\partial_z \partial_s (zs(z+s))^{-1} = 2(z^2 + 3zs + s^2)/(z^2 s^2 (z+s)^3)$, the other terms depending on one variable only. $\square$

Proof of Proposition 7.5. Let B be the doubled chain of $\mathbf{y}$; its window- r interpolants are $F_r^B = Q_{V_r} e^{-y}$, which exist since $D_r(\mathbf{y}), D_{r+1}(\mathbf{y}) \neq 0$. Apply Proposition 4.7(H2) at B and then at $B+z$: first $D_r(B+z) = D_r(B)F_r^B(z)$, then $D_r(B+z+z) = D_r(B+z)F_r^{B+z}[z]$, where by (H1) $F_r^{B+z} = F_{r+1}^B - q_r^B(z)F_r^B$ and its confluent value at z (the derivative) is $\text{Wr}(F_r^B, F_{r+1}^B)(z)/F_r^B(z)$. Multiplying, $D_r(B+2z) = D_r(B) \text{Wr}(F_r^B, F_{r+1}^B)(z)$ wherever $F_r^B(z) \neq 0$, and by continuity everywhere. Since $\text{Wr}(F_r^B, F_{r+1}^B) = e^{-2y} \text{Wr}_{r,r+1}$ and $B+z+z$ is the doubled chain of $\mathbf{y} \cup \{z\}$ up to an even permutation of rows, this is the first claim; the factors e^{-2z} cancel in $S = D_0 D_2 / D_1^2$. $\square$

Proof of Lemma 7.9. Let d_k be the data column of τ_k (rows: the nodes of $\mathbf{y}$, then those of Z , where $Z = \{x\}$ or $\{z, x\}$, $r := |Z|$), and let $D_m := \det[d_m, \dots, d_{m+2J-1}]$ on the rows of $\mathbf{y}$ (this differs from $D_m(\mathbf{y})$ by the factor $\prod_j e^{-2y_j}$). By Cramer's rule $p_1(\mathbf{y} \cup Z) = -\det[d_0, d_2, \dots, d_{2J+2r}]/\det[d_1, \dots, d_{2J+2r}]$, $p_{2J} = D_0/D_1$ and $\ell_1^* = -D_2/D_1$, so $S_{\mathbf{y}} = -p_{2J}\ell_1^* = D_0 D_2 / D_1^2$. Eliminate with the block (rows of $\mathbf{y}$) $\times$ (columns 1, $\dots$, $2J$) in the denominator: the Schur complement is the Z -data of $E_1, \dots, E_{2r}$, so the denominator is $D_1[E_1, \dots, E_{2r}](Z)$. In the numerator eliminate with the columns 2, $\dots$, $2J+1$ (determinant D_2 ; moving d_0 past $2J$ columns costs no sign): the reduced columns are $1 - \Pi^{[2]}1$ and

$W_m := u^{2J+m} - \Pi^{[2]}u^{2J+m}$ ($2 \leq m \leq 2r$), where $\Pi^{[2]}$ interpolates at $\mathbf{y}$ in $\text{span}(u^2, \dots, u^{2J+1})$. The elements of $\text{span}(1, u, \dots, u^{2J+1})$ with zero data at $\mathbf{y}$ form the plane spanned by $P_{\mathbf{y}}$ and E_1 ; since $1 - \Pi^{[2]}1$ lies there and has u -coefficient 0, it equals $P_{\mathbf{y}} + (p_1/\ell_1^*)E_1$. Moreover $W_m - E_m$ and $W_m - V_m$ lie in the zero-data part of lower windows, so in all the determinants we may replace E_m and W_m by V_m ($m \geq 2$) and E_1 by V_1 . Hence

$$p_1(\mathbf{y} \cup Z) = -\frac{D_2}{D_1} \left(\frac{[P_{\mathbf{y}}, V_2, \dots, V_{2r}](Z)}{[V_1, \dots, V_{2r}](Z)} + \frac{p_1}{\ell_1^*} \right) = p_1(\mathbf{y}) + \ell_1^* p_{2J} \frac{[V_0, V_2, \dots, V_{2r}](Z)}{[V_1, \dots, V_{2r}](Z)},$$

and $\ell_1^* p_{2J} = -S_{\mathbf{y}}$. For $r = 1$ this is the formula for $\Delta_{\mathbf{y}}(x)$; taking $r = 2$ and subtracting the cases $Z = \{z\}$, $\{x\}$ gives the second. $\square$

Proof of Lemma 7.10. Multiplying all v_r by y^{-d} changes neither Tu nor I , so we may take $d = 0$.

(i) With $e_i = 2i$ and $\varepsilon_i = (-1)^i$, a direct computation gives

$$[ij]_y = \varepsilon_i \varepsilon_j (e_j - e_i) y^{e_i+e_j-1} \left(1 + \frac{t}{y} \frac{(e_j - e_i + 1)\kappa_i + (e_j - e_i - 1)\kappa_j}{e_j - e_i} \right) + O(t^2),$$

and in Tu the prefactors cancel, while the first-order terms add up to $(3\kappa_0 + \kappa_1 + 3\kappa_2 + \kappa_3 - 6\kappa_1 - 2\kappa_2)/(2z)$. (ii) By Cramer's rule, $c(z) := -[02]_z/[12]_z$ is the coefficient c_1 of the unique combination $v_0 + c_1 v_1 + c_2 v_2$ with a double zero at z , and $c(z, x) := -[0234]/[1234](z, x)$ is the coefficient c_1 of the unique $v_0 + \sum_{r=1}^4 c_r v_r$ with double zeros at z and x ; so $I = c(z, x) - c(z) - c(x)$. In the variable $v = y^2$ we have $v_r = (-1)^r (v^r + t\kappa_r v^{r-1/2})$, the systems are non-singular at $t = 0$, and at $t = 0$ the combination is $\varphi_Z(v) := \prod_{\zeta \in Z} (1 - v/\zeta^2)^2$ ($Z = \{z\}$ or $\{z, x\}$), so that $c_1|_{t=0} = 2 \sum_{\zeta \in Z} \zeta^{-2}$ and $I|_{t=0} = 0$. Differentiating at $t = 0$, the polynomial $h(v) := \sum_{r \geq 1} (-1)^r \dot{c}_r v^r$ interpolates $-v^{-1/2}(\mathcal{K}\varphi_Z)(v)$ at the double nodes ζ^2 , where $\mathcal{K} := \alpha + \beta\theta_v + C\theta_v^2$ and $\theta_v = v \frac{d}{dv}$, because $\sum_r \kappa_r [v^r] \varphi_Z v^r = \mathcal{K}\varphi_Z$. Write $h = vk$ with $\deg k \leq 2|Z| - 1$ and $\omega(\xi) := \prod_{\zeta} (\xi - \zeta^2)^2$; as $k(\xi)/(\xi\omega(\xi)) = O(\xi^{-2})$, its residues sum to 0, and since $\varphi_Z = \omega/\omega(0)$,

$$-\dot{c}_1 = [v]h = k(0) = \sum_{\zeta} \text{Res}_{\xi=\zeta^2} \xi^{-5/2} \frac{\mathcal{K}\omega}{\omega}(\xi) = (2\beta + C) \sum_{\zeta \in Z} \zeta^{-3} - 8C \sum_{\{\zeta, \zeta'\} \subset Z} \frac{1}{\zeta \zeta' (\zeta + \zeta')},$$

using $\frac{\mathcal{K}\omega}{\omega} = \alpha + (\beta + C)\xi\Lambda + C\xi^2(\Lambda' + \Lambda^2)$ with $\Lambda = \omega'/\omega = \sum_{\zeta} \frac{2}{\xi - \zeta^2}$. Only the pair term survives in $\dot{c}(z, x) - \dot{c}(z) - \dot{c}(x)$. $\square$

Proof of Proposition 7.11. (1) Each column of W_I is a sum of two monomial columns, so by multilinearity $W_I(y) = y^{N_I-m} p_I(y)$ with $m = |I|$, $N_I \geq m$ and $p_I(y) = a_0 y^m + a_1 y^{m-1} + \dots + a_m$, where $a_0 \neq 0$ is the Wronskian constant of the monomials (a Vandermonde product of distinct exponents) and $|a_j| \leq C_d |\kappa|^j$. The coefficient of t^k in $p_I(c+t)$ is $a_0 \binom{m}{k} c^{m-k} (1 + \sum_{j \geq 1} \frac{a_j}{a_0} c^{-j} \binom{m-j}{k} / \binom{m}{k})$, which has the sign of a_0 once $\sum_j |a_j/a_0| c^{-j} < 1$; multiplying by $(c+t)^{N_I-m}$ keeps the signs. (2) follows from Lemma 7.10 with $t = s$ and $C = 1$, since Tu and I are rational in s and regular at $s = 0$. $\square$

Proof of Proposition 8.5. $[u]E_* = -\ell_1^* \ell_2^* + \ell_2^* \ell_1^* = 0$ and $E_*(0) = 0$, so $E_* \in u^2 \mathbb{R}[u]$; its top coefficient is ℓ_1^* , and the formula for μ_Y is (7.2). (a) On $\mathcal{I} \setminus \mathbf{y}$, $A_{\mathbf{y}} > 0$ and R_* , W have no zeros; at a node in $\mathcal{I}$, μ_Y has a pole without change of sign. So the sign is constant, and as $Y \rightarrow \infty$, $\mu_Y \sim \ell_1^* Y^{2J+4}/(Y^{2J} \cdot (-2)Y^{4J+5})$; finally $S_{\mathbf{y}} = -p_{2J} \ell_1^*$. (b) If $E_* = c u^2 \tilde{h}(u)$, then $Q_{E_*} e^{-y} = c \tilde{h}(-\theta^2)[\theta^4 e^{-y}]$, which vanishes doubly at the nodes. (c) y_j is the inserted node of $\mathbf{y} \setminus y_j$; apply (a) with $\mathcal{I} = [y_j, \infty)$ to that node set (its ℓ_1^* is non-zero since $S/p_{2J-2} \neq 0$). $\square$

APPENDIX B. CERTIFICATES FOR THE MODEL

The arithmetic is that of [15]. *Exact* computations use integers and rationals (FLINT `fmpz`, `fmpq` and their polynomial and matrix types [23]). *Interval* computations use Arb ball arithmetic [14]: a real number is represented by a ball $[m \pm r]$ that is guaranteed to contain it, and every operation returns a ball containing all possible results. Both are accessed through python-flint (version 0.9.0); constants such as π are balls, so the nodes $2\pi n$ are exact. Linear systems are solved by Arb’s ball solver, which either returns balls containing the exact solution of every system in the input balls, and thereby certifies that the exact matrix is invertible, or fails. A sign is accepted only if the ball excludes 0. Positivity of polynomials is certified by coefficient signs.

Lemma B.1 (Positive-coefficient certificates). (a) (Descartes at c .) *If $p \in \mathbb{R}[t]$ and every coefficient of $p(c+t) \in \mathbb{R}[t]$ is non-negative with positive constant term, then $p > 0$ on $[c, \infty)$. If p vanishes to exact order m at c and the coefficients of $t^m, t^{m+1}, \dots$ in $p(c+t)$ are positive, then $p > 0$ on (c, ∞) .*

(b) (Region certificates.) *Let $\mathcal{R} \subseteq \bigcup_i \phi_i([0, \infty)^m)$, where the coordinates of each ϕ_i are rational functions with a common denominator $w_i > 0$ on $[0, \infty)^m$. Let G be a polynomial of total degree d . If each polynomial $w_i^d \cdot (G \circ \phi_i)$ has non-negative coefficients and a positive constant term, then $G > 0$ on $\mathcal{R}$.*

(c) *A polynomial with non-negative coefficients in $v \in [0, \infty)^m$ is non-negative there, and positive at every point at which some monomial with a positive coefficient is positive.*

With ball coefficients, “positive” means that the ball lies in $(0, \infty)$.

Parts (a) and (c) are also in [15].

Proof. All three statements follow from the fact that a polynomial with non-negative coefficients takes non-negative values on the non-negative orthant, with the indicated strictness. In (b), $w_i^d \cdot (G \circ \phi_i)$ is a polynomial because $\deg G = d$, and $w_i > 0$ transfers its sign to $G \circ \phi_i$. $\square$

In the applications the coefficients are exact integers (Propositions 5.10–5.12, 8.3, 8.4) or balls (the Descartes tests in Section 7.1 and Proposition 9.19); Taylor shifts of ball polynomials are computed by Horner’s scheme at a ball containing c , which encloses the exact shifted coefficients.

B.1. The certificates for small J . This subsection describes the computations behind Propositions 5.10–5.12; the logic was given after Proposition 5.12. Throughout, $c = \frac{21}{20}$, $V = \prod_{i < j} (y_i - y_j)$, and all arithmetic is exact.

The polynomials G_i . Let $\mathfrak{M}(y)$ be the $(2J+1) \times (2J+1)$ matrix whose rows are the data rows $(\tau_k(y_j))_{k \leq 2J}$, $(\tau'_k(y_j))_{k \leq 2J}$, $j \leq J$, followed by the row $(\tau_k(y))_{k \leq 2J}$. Its determinant $\mathfrak{Q}(y)$ is a combination of the τ_k , hence admissible, and vanishes doubly at each y_j ; its constant term is $\pm D$, where D is the determinant of the system in $(N)_y$. The verifier computes the minors by Laplace expansion along the pairs of node rows, divides $\mathfrak{Q}$ by A_y exactly, checks that every coefficient N_i of the quotient is divisible by V^4 , and sets $G_i := \pm N_i/V^4$, with the sign chosen so that $G_0 > 0$ at y^* . It then checks the identities (i) and (ii) of the proof of Proposition 5.12 symbolically. For $J = 3$ and $J = 4$ the fingerprints of the G_i are printed in the logs.

Slack coordinates and region covers. Write $y_1 = c(3+A)$ and $y_{k+1} = y_k + cB_k$ with $B_1 = B$, $B_2 = E$, $B_3 = F$. The hypotheses of the propositions become $A \geq 0$, $B, E, F > 0$ and $2A+B \geq 8$, $3A+2B+E \geq 24$, $4A+3B+2E+F \geq 48$ (as far as they apply). All parameters s, t, a, b, e, f below range over $[0, \infty)$.

- $J = 2$: piece I, $A = 4 + s$, $B = t$; piece II, $A = 4s/(1+s)$, $B = 8 - 2A + t$. Every point with $A \geq 4$ lies in I, and every point with $A < 4$ lies in II (take $s = A/(4-A)$).

- $J = 3$: PA, $A = 8s/(1+s)+a$, $B = 12/(1+s)+b$, $E = e$, which covers $3A+2B \geq 24$ with $B > 0$ (given (A, B) , any $s \in [\max(0, 12/B-1), A/(8-A)]$ works, and this interval is non-empty exactly when $3A+2B \geq 24$); PA', $A = 8+a$; and PB1, PB2, in which (A, B) runs over the triangles with vertices $(0, 8), (4, 0), (8, 0)$ and $(0, 8), (8, 0), (0, 12)$ in the projective barycentric form $\lambda = (1, s, t)/(1+s+t)$ with apex $(0, 8)$, and $E = 24 - 3A - 2B + e$. The edges with $\lambda_1 = 0$ lie in $\{B = 0\}$ or in $\{3A + 2B = 24\} \subset \text{PA}$.
- $J = 4$: if $\ell := 4A + 3B + 2E \geq 48$ (which implies $3A + 2B + E \geq 24$), the pieces Ia1 ($B = 16 + b$), Ia' ($A = 12 + a$) and Ia3 (the triangle $(12, 0), (0, 16), (12, 16)$ in (A, B) , with apex $(12, 0)$) cover $4A + 3B \geq 48$, and Ib1, Ib2 (the triangles $(0, 8), (4, 0), (12, 0)$ and $(0, 8), (12, 0), (0, 16)$, both with apex $(0, 8)$, and $E = (48 - 4A - 3B)/2 + e$) cover the rest, each with $F = f$. The edge omitted from Ia3 is $\{B = 16\}$, which lies in Ia1; the edge omitted from Ib1 lies in $\{B = 0\}$, outside the region; the edge omitted from Ib2 is the segment $\{4A + 3B = 48\}$, which is not part of the domain $4A + 3B < 48$ of Ib1, Ib2 and lies in Ia3 (its endpoint $(0, 16)$ in Ia1). If $\ell < 48$, the pieces II1–II6 are the cones from the apex $(0, 8, 8)$ (which is the point y^*) over the six triangles that triangulate the facets $B = 0$, $E = 0$ and $\ell = 48$ of the polytope $\mathcal{P} := \{A, B, E \geq 0, 2A + B \geq 8, 3A + 2B + E \geq 24, \ell \leq 48\}$, with $F = 48 - \ell + f$. The apex lies on the three other facets $A = 0$, $2A + B = 8$ and $3A + 2B + E = 24$, so $\mathcal{P}$ is the union of the cones from the apex over the facets $B = 0$, $E = 0$, $\ell = 48$ (a convex polytope is the union of the cones from one of its vertices over the facets that do not contain it). The parametrisation omits the base face of each cone, which lies in $\{B = 0\} \cup \{E = 0\}$, outside the region, or in $\{\ell = 48\}$, covered by the first part.

What is machine-checked. For every piece, `toy/pt/verify_pt.py` checks that the piece lies in the region (the line `inside` of its logs). The converse inclusion, that the pieces cover the region, is the hand argument above, for $J = 2$, for $J = 3$, and for the part $\ell \geq 48$ of $J = 4$. For the part $\ell < 48$ of $J = 4$ the script `toy/pt/check_pt4_region.py` checks in exact arithmetic that the eight vertices of $\mathcal{P}$ and the six triangles are as stated and that the six cones have total volume $1024/3$, the volume of $\mathcal{P}$ computed independently; together with the convexity argument above this gives the cover. In each piece the coordinates are rational functions of the parameters with a positive common denominator, and the verifier checks that each homogenised $G_i \circ \phi$ has only positive coefficients and a positive constant term (Lemma B.1(b)); the numbers of polynomials and their sizes are in Table 4. No subdivision beyond these pieces is needed: on the whole region the numerators are coefficientwise positive in the slack variables.

B.2. Design of the certificates for Theorem 6.5. *Chain route.* For a chain Z of level K , the coefficients q_m of $\omega_Z f$, for a monic f of degree K , are affine in the K unknown lower coefficients of f . By Proposition 5.2(c), $\omega_Z f$ is admissible if and only if $\sum_m q_m |s(m, 2i-1)| = 0$ for $i = 1, \dots, K$; the unsigned Stirling numbers are exact integers. This gives a $K \times K$ linear system for f_Z (and a $(K+1) \times (K+1)$ system for $g_Z = f_{Z+0}$), whose successful ball solution proves regularity. The script then checks $(\text{IO})_K$, the increments $\varrho_{k+1} - \varrho_k > 0$, and $\mathbf{g}^{(K)}(z_{K+1}) > 0$, at 4000 bits; the worst relative radius over all levels is below 10^{-785} .

Direct route. Here the interpolants are computed in the basis $\tau_0, \dots, \tau_K$ of $\mathcal{A}_K^{\text{pol}}$, with τ_k generated exactly by $\tau_k = (-1)^k (\theta - y)^{2k} 1$ (the conjugate $e^y \theta e^{-y} = \theta - y$); the cofactors are obtained by exact division by ω_Z in ball arithmetic and compared with the expected degrees and leading coefficients. The script checks $(\text{IO})_K$, $(\text{II}')_K$ in the cross-multiplied form $g_{k+1} f_k - g_k f_{k+1} > 0$, and $(\text{Y0})_K$ for every $K \leq 160$; and, for every $J \leq 80$ and without appeal to Theorem 6.3, (N) for the system with $P(0) = 1$, $p_{2J} > 0$, strict TEL_J , positivity of the half-step cofactor, and the consistency relation $p_{2J} \prod_j y_j^2 \tilde{R}_J(0) = 1$ as a ball containment. The level range

is split into two runs ($K \leq 134$ and $135 \leq K \leq 160$). At $K = 160$ about 1150 decimal digits are lost, so 6000 bits leave a wide margin.

The two routes share no code beyond the python-flint library and agree on every level: they certify the same signs, and their margins agree to the printed digits.

TABLE 4. Computer-assisted statements about the model. “Exact”: integer or rational arithmetic; “Arb”: ball arithmetic at the stated precision. Scripts are in `toy/` (Appendix C).

Statement	Arithmetic	Size	Script
Prop. 5.10	exact	G_i : 54, ..., 26 terms; 10 substituted polynomials; second route by Cramer’s rule	<code>pt/verify_pt.py</code> , <code>pt/verify_pt2_cramer.py</code>
Prop. 5.11	exact	G_i : 1304, ..., 636 terms; 4 pieces, 28 polynomials, up to 38 104 terms	<code>pt/verify_pt.py</code>
Prop. 5.12	exact	D : 308 386 terms; G_i : 45 612, ..., 22 604 terms; 11 pieces, 99 polynomials, up to 219 336 terms; region cover checked separately	<code>pt/verify_pt.py</code> , <code>pt/check_pt4_region.py</code>
Prop. 5.16	exact	$K \leq 90$; up to 181 coefficients	<code>ap/verify_ap.py</code> , <code>ap/verify_halfstep.py</code>
Thm. 6.5	Arb, 4000 and 6000 bits	levels $K \leq 160$, $J \leq 80$; two independent routes	<code>pp80/verify_pp80_chain.py</code> , <code>pp80/verify_pp80_tau.py</code>
Prop. 5.14	exact	$K \leq 2.17 \cdot 10^5$	<code>pp/verify_pp_prefix.py</code>
Prop. 8.2	Arb, 1500–7000 bits	up to $J = 50$	<code>w/verify_w_cex.py</code>
Props. 8.3, 8.4	exact	two routes; $J = 3$: up to about 37 000 terms	<code>w/verify_w_small.py</code> , <code>w/verify_w_insertion.py</code>

TABLE 5. Margins in Theorem 6.5 along the prime-power chain (direct route, 6000 bits): certified lower bound for the smallest relative increment $\min_k(\varrho_{k+1} - \varrho_k)/|\varrho_k|$ at level K , and certified upper bound for the largest relative radius of a cofactor coefficient.

K	increment $\geq$	radius $\leq$
20	0.3869948	10^{-1732}
40	0.2650057	10^{-1609}
60	0.1690426	10^{-1464}
80	0.1159777	10^{-1311}
100	0.08632101	10^{-1154}
120	0.06810364	10^{-990}
140	0.05605644	10^{-818}
160	0.04739370	10^{-648}

B.3. Tables. The certified increments decrease with K , roughly like $1/K$: the overall log–log slope between $K = 20$ and $K = 160$ is about 1.01 (local slopes 0.55, 1.11, 1.31, 1.32, 1.30, 1.26, 1.26), and K times the increment falls monotonically from 10.6 ($K = 40$) to 7.58 ($K = 160$). Divided by the scale $K/(k^*(K - k^*))$ of Newton’s inequalities at the position $k^* \approx 0.66K$ of the minimum, they fall from about 2.38 ($K = 40$) to 1.70 ($K = 160$) (computed from the table). So, relative to the natural $1/K$ scale, the margin is still decreasing at $K = 160$; nothing in the data suggests a sign change, but nothing suggests a positive limit either.

TABLE 6. Counterexamples to W (Proposition 8.2), in the units $n = y/2\pi$; values rounded, signs certified.

Node set	Failure	Values
$\{1, \dots, 7\}$	$\mu_7 < 0$	$\mu_7 \approx -2.332 \cdot 10^{-18}$
$\{1, \dots, 10\}$	$\mu_9, \mu_{10} < 0; p_1^{(10)} < p_1^{(9)}$	$\mu_9 \approx -8.924 \cdot 10^{-23}, \mu_{10} \approx -1.403 \cdot 10^{-25},$ $p_1^{(10)} - p_1^{(9)} \approx -7.341 \cdot 10^{-11}$
$\{1, \dots, 5\} \cup \{m\}$	$S < 0; \mu_Y, \Delta < 0$	$S \approx -3.478 \cdot 10^{-6}; \Delta \approx -1.972 \cdot 10^{-9}$ $(m = 10), -1.773 \cdot 10^{-13} \ (m = 1000)$
$(1 + \varepsilon)j, j \leq 10$	$\mu_9, \mu_{10} < 0$	$\varepsilon = 10^{-4}: S_{\leq 5} \approx -3.29 \cdot 10^{-6}; \varepsilon = 10^{-3}: S_{\leq 5} \approx -1.54 \cdot 10^{-6}$
$(1 + \frac{1}{200})j, j \leq 10$	$\mu_{10} < 0$	$\mu_{10} \approx -3.025 \cdot 10^{-26}$
$\frac{11}{10} + j, 0 \leq j < 50$	$\mu_{48}, \mu_{49}, \mu_{50} < 0$	$\mu_{48} \approx -7.050 \cdot 10^{-122}, \mu_{50} \approx -1.720 \cdot 10^{-129}$

APPENDIX C. ANCILLARY FILES

The ancillary files of this paper contain, for every computer-assisted statement, the exact input, a standalone verifier and the log of a fresh run. Their file `README.md` describes each script in detail; this appendix is an index. Paths are relative to the top directory of the ancillary files.

Requirements. Python ≥ 3.10 with python-flint 0.9.0 (FLINT 3.6.0, which contains Arb) and mpmath 1.3.0; nothing else. All commands are run from the top directory of the ancillary files, with PY set to such an interpreter. The wrapper `runlog.sh LOG "$PY" SCRIPT ARGS` writes a log whose header records the command, the date and the software versions, and whose footer records the exit code and the wall time. A run succeeds if it exits with code 0 and its final output line reports success (CERTIFIED: True, ALL CERTIFIED: ... or ALL CHECKS PASSED; the exact line for each script is listed in `README.md`). Every script prints the SHA-256 of each input file; all hashes are listed in `SHA256SUMS`. The ancillary files are self-contained: they include a copy of the shared library of [15] (`lib/`) and copies of the certified enclosures `exact/out/PJ_ball.txt` ($J = 10, 60, 61, 110, 111$) of the exact members, which the certificates of Propositions 9.9, 9.19 and 9.20 read; these enclosures are produced by the verifier of Fact 2.13 in the ancillary files of [15], and the copies are byte-identical. The ancillary files are licensed under the Apache License 2.0 (copyright 2026 Nic Johns); the text is licensed under CC BY 4.0.

Re-running. The script `toy/run_all.sh` repeats all certificates of the model (about 80 CPU-minutes) and `exact/run_all.sh` those of Section 9.3, after which `sha256sum -c SHA256SUMS` checks that inputs and shipped outputs are unchanged. A single certificate is repeated with the command recorded in the first line of its log, for example

```
./runlog.sh toy/pp80/logs/pp80_tau_K0-134.log "$PY" \
toy/pp80/verify_pp80_tau.py toy/pp80/params/pp80.json 0 134
```

The precision, Taylor orders and number of worker processes affect only the running time and the width of the enclosures, not the validity of a successful run. Computed but uncertified data

TABLE 7. Index of the certificates. Directories: `exact/` (members of the true family), `toy/` (the model). Runtimes are wall times of the shipped logs.

Statement	Script and input	Arithmetic	Runtime
Prop. 9.19	<code>exact/verify_bf.py</code> , <code>exact/verify_base.py</code> , <code>exact/verify_r1.py</code>	Arb	up to 464 s
Prop. 9.20	<code>exact/verify_r1.py</code> (Δp_1); <code>exact/verify_increments.py</code> (signs)	Arb	up to 464 s
Prop. 9.9	<code>exact/verify_roots.py</code> with <code>exact/out/PJ_ball.txt</code> , $J = 60, 110, 111$	Arb, 2000 bits	< 5 s
Thm. 6.5	<code>toy/pp80/verify_pp80_tau.py</code> <code>toy/pp80/</code> <code>params/pp80.json</code> (two parts); <code>toy/pp80/verify_pp80_chain.py</code> <code>toy/pp80/</code> <code>params/pp80_chain.json</code>	Arb, 6000 / 4000 bits	658 s; 458 s
Props. 5.10–5.12	<code>toy/pt/verify_pt.py</code> <code>toy/pt/params/pt.J.json</code> ; for $J = 2$ also <code>toy/pt/verify_pt2_cramer.py</code> ; for $J = 4$ also <code>toy/pt/check_pt4_region.py</code>	exact	0.1 s, 3 s, 2190 s
Remark 5.9	<code>toy/pt/verify_pt0_false.py</code> <code>toy/pt/params/</code> <code>pt0_false.json</code>	exact	0.1 s
Prop. 5.14	<code>toy/pp/verify_pp_prefix.py</code> <code>toy/pp/params/</code> <code>pp_prefix.json</code>	exact	0.4 s
Prop. 5.16	<code>toy/ap/verify_ap.py</code> , <code>toy/ap/verify_halfstep.py</code> with <code>toy/ap/params/ap_delta005.json</code>	exact	643 s; 642 s
Props. 7.6, 7.7, 7.8	<code>toy/ft/verify_farbound.py</code> , <code>toy/sm/verify_smset.py</code> , <code>toy/um/verify_um.py</code> , each with its parameter file	Arb	0.2 s; 119 s; 38 s
Prop. 8.2	<code>toy/w/verify_w_cex.py</code> <code>toy/w/params/w_cex.</code> <code>json</code>	Arb, 1500–7000 bits	6 s
Props. 8.3, 8.4	<code>toy/w/verify_w_small.py</code> and <code>toy/w/verify_w_insertion.py</code> with <code>toy/w/params/w_small.json</code> , argument $J = 2$ or 3	exact	< 5 s

TABLE 8. SHA-256 of the input files of the main certificates of the model (all others: SHA256SUMS).

File	SHA-256
<code>toy/pp80/params/pp80.json</code>	9f840fb2e65142ddf7bd47f33f2e95be51ac05dc42a03ef93fea5401cc82f91c
<code>toy/pp80/params/pp80_chain.json</code>	0fedf5dd3f57bd2335abc6678b9451ace7cab7de6fb8e8943dc9dd3bcc90f9ca
<code>toy/pt/params/pt2.json</code>	0db9256e86d5345238649f9544948bb06a39eed7c982391d9e0d146893e529f7
<code>toy/pt/params/pt3.json</code>	2f55a0f3f7792042c1410fd58f36c2cbe5c303fb473ff366ec3b0d28fbc63813
<code>toy/pt/params/pt4.json</code>	03bb73e2e7baab96bb2ccd5a9365a93c7cfdaa323bbb9588e2615a2df9cb2cb6
<code>toy/ap/params/ap_delta005.json</code>	8154e4dd3a63096e6626be846752083bde36b5c5c26267db59506145d4175235

used in numerical observations are listed separately in `README.md`; no statement of this paper depends on them.

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